

Pathophysiology of Proteinuria

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CHAPTER OUTLINE

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MECHANISMS OF PROTEINURIA

One of the most common features of glomerular diseases is an abnormal excretion of plasma proteins in the urine. Proteinuria is the cause and effect of several complications not only at a kidney but also at a systemic level. There are complex changes in the structure and function of the glomerular capillary, as well as in the entire nephron, that are responsible for the final elevation in urine protein concentration in several kidney disorders. In this chapter, before describing the consequences of proteinuria, the pathophysiology of protein excretion is reviewed.

The functional properties of the glomerular filtration barrier, tubular interaction with filtered proteins and the mechanisms of proteinuria have been reviewed in detail,¹ and the characterization of structural molecules relevant to the filtration barrier in glomerular endothelial cells and in particular on podocytes has been reported.^{2,3} There are, in principle, two distinct phenomena that can result in proteinuria. The first is the elevation of glomerular filtration of circulating plasma proteins that are almost completely retained in the circulating plasma in normal physiologic conditions; the second is a defective or incomplete reabsorption of proteins by the proximal tubule. The two phenomena are interrelated, and likely are both present in so-called glomerular proteinuria, when proteins the size of albumin and larger are present in urine. Despite several experimental and clinical observations investigating the structural and molecular alterations involved in kidney diseases resulting in proteinuria,⁴ the precise nature of the functional changes responsible and their quantification remain the subject of numerous ongoing investigations.⁵

STRUCTURE AND FUNCTION OF THE GLOMERULAR CAPILLARY WALL

The function of the glomerular capillary is to allow a large amount of water and small solute filtration while efficiently

restricting glomerular passage of protein macromolecules within blood circulation. This selective function is specific to the glomerular capillary membrane, which is far more permeable to water than any other capillary membrane in the body. With the development of glomerular diseases, the capillary membrane structure at molecular and/or cellular level may be altered, resulting in loss of hydraulic permeability, reduction in surface area available for filtration, and consequent loss of glomerular filtration rate (GFR). Despite the reduction in permeability to water, the capillary membrane often becomes more permeable to circulating macromolecules. Experimental research has elucidated a number of glomerular structural and molecular alterations that are responsible for these functional changes.

GLOMERULAR CAPILLARY WALL ORGANIZATION

Morphologic studies available in the literature describe in detail the complex organization of the glomerular capillary and the capillary membrane (see Chapter 2). However, the interpretation of filtration barrier function has been largely based on major simplifications. Thus although the glomerular capillary is composed of numerous branching segments, the glomerular capillary organization has usually been considered as a simple capillary segment or as a set of several uniform segments in parallel. Similarly, the capillary membrane has been considered as a uniform three-layer structure. As described later, recent investigations allow a better understanding of the functional effects of the geometric and spatial organization of the glomerular capillary, as well as specific features of cell organization and interactions at the capillary membrane level. These aspects have revealed some new insights in the mechanisms responsible for glomerular capillary dysfunction.

Glomerular Capillary Network

According to classic optical microscopy observations of kidney tissue sections, the capillary network is composed of a number of capillary segments connecting afferent and efferent arterioles within a tuft that, in humans, has a mean diameter of 120–150 μm . More realistic and direct visualizations of the

capillary organization are usually derived from scanning electron microscopy, but this technique allows predominantly only views from the outer surface of the capillary. Specific investigations with reconstructions from serial sections⁶ or confocal microscopy⁷ allow investigation of the capillary segment organization and, in particular, calculation of blood flow distribution and water filtration along the network.⁸ Due to the large number of capillary segments, around 200 in the rat, and their apparent uniform size, the blood flow is expected to be uniformly distributed along the network with lower blood velocity as compared with that in the afferent arteriole. This hemodynamic arrangement allows the blood to remain in close contact with the filtration membrane. However, more detailed geometric reconstructions of the glomerular capillary show that the network has some heterogeneity. The size of some capillary segments (with diameters less than 3–4 μm)⁶ would suggest that they are perfused only by plasma, excluding red blood cell transit, and may represent a sort of shunt in the network to decrease overall network pressure. This finely organized geometry is the result of cellular organization and remodeling and seems to be importantly affected by disease processes, resulting in simplification of capillary network, changes in local pressure and flow distribution, ultimately resulting in capillary obliteration in areas of segmental sclerosis.⁹ These local hemodynamic changes affect the filtration function of the capillary network, as elevation of blood flow and hydraulic pressure is expected to occur in some capillary segments leading to abnormal filtration of circulating proteins.¹⁰

Glomerular Capillary Wall

At a smaller scale the organization of the glomerular capillary membrane is rather heterogeneous. The arrangement that is generally described usually refers to the portion of the capillary wall that is considered the filtering surface, characterized by a three-layer composition consisting of endothelial cells, glomerular basement membrane (GBM), and epithelial cells. The structure and function of this highly differentiated arrangement of cells and matrix is presented in Chapter 2. Mechanisms whereby structural and functional changes may result in abnormal protein filtration and ultimately in proteinuria are reviewed later.

According to the classic concept, hydraulic resistance and macromolecule retention are functions of the so-called filtering surface of the glomerular capillary membrane, but recent evidence indicates that the entire structure of the epithelial cells and the relative position of the capillary membrane within the tuft may also affect water and macromolecule filtration. As reported by Neal et al.,¹¹ a large fraction of the filtration membrane is covered by epithelial cell bodies or by the presence of adjacent epithelial cells. The three-dimensional spaces created by these structures have been called subpodocyte (SPS) and interpodocyte spaces (IPS), respectively. Theoretical analysis of the transport of both water and macromolecules through the SPS¹² indicates that the structural organization of this compartment induces significant resistance to water flow from the filtration membrane to the urinary space. This resistance appears to not be insignificant as compared with that of the three-layer membrane structure. Macromolecule transport may also be influenced by the SPS.¹³

Specific evaluations of the structural changes that characterize SPS and IPS, and their functional consequences in experimental models of kidney disease, or in patients with

renal dysfunction, are not yet available. The difficulties in obtaining such quantitative evaluations derive from the heterogeneous nature of these three-dimensional structures. In addition, they can be visualized only using electron microscopy, both transmission electron microscopy and scanning electron microscopy (SEM), but their quantification is not easy because they are in the inner portion of the glomerular capillary tuft.¹⁴

ULTRASTRUCTURE OF THE GLOMERULAR CAPILLARY MEMBRANE

Endothelial Cell Layer

Glomerular endothelial cells are the most fenestrated in the circulation, with a pore area in peripheral zone that occupies from 20% to 50% of the cell surface.¹⁵ The surface of endothelial cells has been considered to have negative electric charge due to the presence of electrical charges of glycoproteins, glycosaminoglycans, and membrane-associated proteoglycans (glycocalyx).¹⁶ These negative charges are expected to act as an electrostatic barrier to the transmural passage of anionic circulating proteins, such as albumin. Thus even if endothelial fenestrae are much larger than albumin (about 60 nm in diameter as compared with a radius of 3.6 nm for albumin), negatively charged circulating macromolecules stay away from the endothelial surface due to electrical repulsion and remain within the circulation. It is now evident¹ that the first restriction to albumin filtration across the glomerular membrane consists of the endothelial surface layer and its role is to substantially decrease protein concentration in the fluid that enters the GBM layer. Endothelial cell glycocalyx expression is importantly affected by fluid shear stress.¹⁷ Increased shear stress is associated with glycocalyx formation and reorganization on the cell surface in contact with fluid flow, with lower expression in static conditions. Thus pathologic conditions in which changes in glomerular capillary flow (i.e., reduced flow) may occur are expected to decrease glycocalyx formation and consequently reduce the retention of anionic proteins within the bloodstream. It has been demonstrated¹⁸ that disruption of the endothelial glycocalyx increases glomerular albumin filtration even in the presence of only minor changes in both the GBM and glomerular epithelial cells. The role of the glomerular polysaccharide-rich endothelial surface layer (ESL) to act as a filtration barrier for large molecules such as albumin has been recently confirmed in C57Bl/6 mice given long-term infusion of hyaluronidase, a hyaluronan degrading enzyme that disrupts the endothelial glycocalyx proteoglycans.^{19,20} A new electron microscopy technique that allows visualization of the ESL and albumin transport within the entire glomerular section at nanometer resolution was used in this set of experiments.²¹ It was shown that glomerular fenestrae are filled with dense negatively charged polysaccharide structures that are largely removed in the presence of circulating hyaluronidase, leaving the polysaccharide surface of other glomerular cells intact.¹⁹ Both retention of cationic ferritin in the glomerular basement membrane and systemic blood pressure were unaltered. Hyaluronidase treatment, however, induced albumin passage across the endothelium in 90% of glomeruli, whereas this could not be observed in untreated control animals. Nevertheless, there was no net albuminuria due to binding and uptake of filtered albumin by the podocytes and parietal epithelium. The ESL structure and function completely recovered after

cessation of hyaluronidase infusion. Thus the polyanionic ESL component hyaluronan is a key component of the glomerular endothelial permeability barrier whose reduction facilitates albumin passage across the endothelial layer and the GBM toward the epithelial compartment.

Glomerular Basement Membrane Organization

The basement membrane layer that characterizes the capillary wall (see Chapter 2) has been shown to exert an important contribution to protein retention by the capillary wall. The molecular composition and organization of this basement membrane would suggest a sieving function due to both size and charge.²² Structural proteins such as collagen type IV and laminin, as well as heparansulfate proteoglycans, represent not only a steric hindrance but also a charge effect on the filtration of circulating molecules. Both *vivo* and *in vitro* studies²³ indicate that small neutral and charged solutes are freely filtered across this extracellular matrix layer, but an important restriction is observed for macromolecules the size of albumin or larger. Thus changes in composition and/or organization of GBM molecules are expected to reduce water filtration and retention of circulating macromolecules.²⁴ The role of GBM in glomerular permselectivity is highlighted by the discovery of mutations affecting genes encoding GBM components in humans and mouse models.²⁵ Mutations in the *COL4A3*, *COL4A4*, or *COL4A5* genes that encode collagen type IV $\alpha 3$, $\alpha 4$, and $\alpha 5$ chains, respectively, cause Alport syndrome, a hereditary glomerular, auditory, and ocular disease.²⁶ Mutations in the gene encoding laminin $\beta 2$ (*LAMB2*) cause Pierson syndrome, a congenital nephrotic syndrome with associated extrarenal manifestation.²⁷ Studies using knockout mouse models of Alport and Pierson syndromes have documented that GBM, defective of these specific components, is more permeable to ferritin or albumin than is the normal GBM, indicating it has a role in glomerular permselectivity.^{28,29}

Tight adherence to the basement membrane is required to prevent podocyte detachment into the Bowman capsule. On a molecular level, a multitude of adhesion receptors including heterodimeric integrins mediate interaction of cells with the surrounding basement membrane.^{30,31} One common form of integrin-mediated adhesion is focal adhesions (FAs), which have been extensively studied in cultured cells.³¹ Recently, FERM-domain protein EPB41L5 has been identified as a highly enriched podocyte-specific FA component.³² This provided the clue to document that genetic deletion of the related *Epb41l5* gene resulted in severe proteinuria, detachment of podocytes, and development of focal and segmental glomerulosclerosis.³²

Epithelial Filtration Slits

A large amount of experimental and clinical research has been generated in the past few decades on the molecular and structural composition of the epithelial junctional complex, known as filtration slits. The characterization of several molecular components of this structure³³ has allowed detailed definition of the proteins that compose the filtration slits (see Fig. 30.1); however, detailed information on the ultrastructure of this intracellular junction is still under investigation.⁵ The original observations by Rodewald and Karnowsky³⁴ suggested a zipper-like structure of the epithelial filtration slit, with rectangular openings of 4 by 14 nm. These

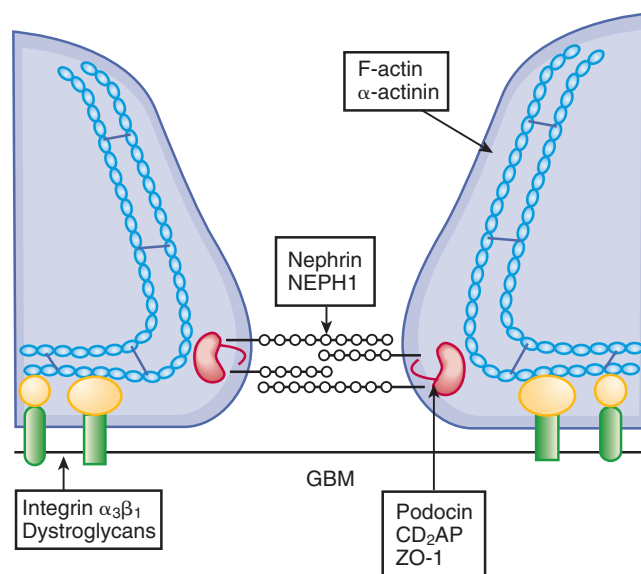


Fig. 30.1 Hypothetical model of the podocyte slit diaphragm. See text for discussion. (From Jalanko H. Pathogenesis of proteinuria: Lessons learned from nephrin and podocin. *Pediatr Nephrol.* 2003;18:487–491, with permission.)

dimensions are in contrast with the observation that a limited amount of albumin can traverse the filtration barrier in physiologic conditions,¹ because the mean molecular radius of albumin is 3.6 nm. Observations with high-resolution SEM and three-dimensional electron microscopy reconstruction³⁵ suggest that the filtration slits are perforated by larger openings of the size of albumin, with more complex geometry. The morphology of the filtration slit has been further imaged with high-resolution SEM, providing evidence of a new ultrastructure composed of circular pores of different sizes with an average radius of 12 nm^{36,37} (Fig. 30.2). Despite the small size of filtration slit openings, a large amount of plasma water is filtered because of the high filtration slit length per unit surface area. As mentioned earlier, under physiologic conditions about 20% of peripheral capillary filtering surface is directly in communication with the Bowman space, and the epithelial slits are the last resistance encountered by water and filtered solutes.¹¹ In the remaining portion of the glomerular membrane, water and solutes, after passing through the filtration slits, must traverse the SPS and the IPS before arriving in the Bowman capsule.¹¹

It should also be considered that the flow of ultrafiltrate over the cell surface directly causes shear stress (SS) on the podocyte membrane associated with the filtration slit as well as to cell body. In animal models of solitary kidney where filtrate flow is increased, SS on the podocyte surface increases 1.5- to 2-fold.³⁸ These forces are highest within the central portion of the filtration slit diaphragm. The forces acting parallel to the GBM are balanced by the mechanical resistance produced by the slit diaphragm complex, opposing foot processes, preventing widening of the slit.³⁹ On the other side, SS acting perpendicular to the GBM plane, tends to detach the foot process from the GBM. However, these shear forces are balanced by the tight junctions between podocyte cell membrane and GBM proteins that allow transmission of this mechanical load into the cytoskeleton apparatus. These

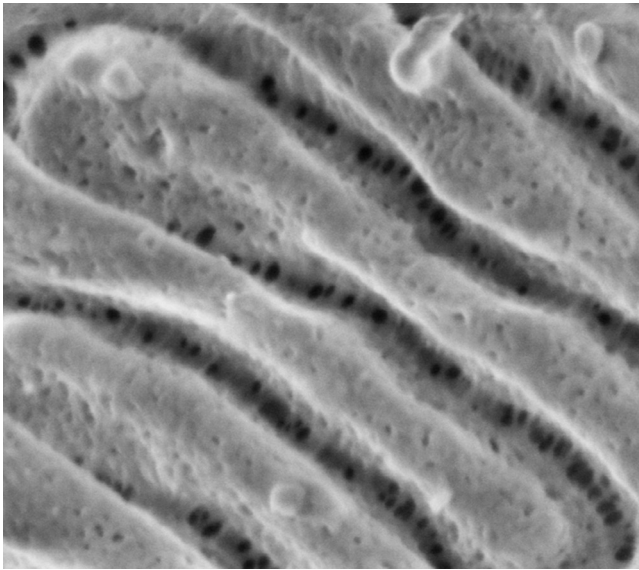


Fig. 30.2 Visualization of epithelial filtration slits obtained using scanning electron microscopy and an in-lens detector to enhance electron detection. Sample was obtained from a Wistar rat, dehydrated with a critical point dryer. The ultrastructure of the filtration slit appears different from the model conventionally proposed in the literature.³⁴ The radius of the circular pores averages 12 nm. (From Gagliardini E, Conti S, Benigni A, et al. Imaging of the porous ultrastructure of the glomerular epithelial filtration slit. *J Am Soc Nephrol.* 2010;21:2081–2089.)

complex mechanical challenges present in physiologic conditions within the podocyte, induced by glomerular ultrafiltrate flow, make mechanotransduction an important function of podocytes. In addition, SS acting on the podocyte membrane adjacent to the filtration slit, as well as to the entire cell body, seems to play a central role in the process of podocyte damage that results in foot process effacement as well as in podocyte detachment from the GBM.³⁹

THEORETICAL MODELS OF GLOMERULAR PERMSELECTIVITY

In addition to structural investigation, functional evaluations of the glomerular capillary wall permselectivity have been extensively used to characterize physiologic conditions and to quantify the effects of pathologic changes. These studies are based on the estimation of filtration of endogenous plasma molecules, such as albumin, IgG, and other proteins, or on the use of test macromolecules of different size, either neutral or electrically charged. Macromolecule filtration depends on convective and diffusive transport that is influenced by glomerular hemodynamic conditions (glomerular plasma flow and pressure) and water filtration. As described later, several investigators have developed theoretical models to derive intrinsic sieving properties of the capillary wall from estimation of macromolecule filtration in experimental and human studies.

Heteroporous Models of Glomerular Size Selectivity

The most-used theoretical models of glomerular size-selective function are based on the assumption of water-filled pores of different sizes as functional equivalents of the glomerular membrane. The passage of water is calculated along the network taking into account the balance between hydraulic and oncotic pressure, as well as membrane hydraulic

permeability.^{9,40} For the calculation of solute filtration, convective and diffusive transport are taken into account, whereas pore resistance to solute filtration is based on steric and hydrodynamic hindrance.^{10,40} The use of these models indicated that glomerular hypothetical pores have mean radius of 4.5–5.0 nm in humans, and a lognormal statistical distribution of pore size around the mean. However, the best simulation of experimental measurements has been obtained assuming that in parallel to restrictive pores, there is a nonselective shunt pathway.⁴¹

The application of these theoretical models clearly showed that in several proteinuric conditions the increased glomerular filtration of the largest neutral test macromolecules is not associated with important changes in the size of restrictive pores, but rather with changes in the nonselective shunt pathway.⁴² This suggests that in normal conditions the small amount of albumin present in the urine may be the result of a small amount of protein filtration that takes place in some focal areas of the epithelial junction, while the majority of the filtration slits retain the protein.

Fiber Models of Glomerular Size Selectivity

Fiber models of solute filtration across the glomerular membrane have also been developed and used.¹ Similar to porous models, the fiber models allow separation of the effect of glomerular hemodynamic changes from those related to intrinsic changes of glomerular membrane selective properties. The advantage of the fiber model is that other than steric hindrance, the effect of membrane and protein electrical charge can be embedded in the model, allowing the estimation of changes in membrane properties both in terms of molecular structural organization and electric charge.⁴³ This modeling approach indicates that filtration of albumin is importantly affected by electrical charge, whereas on the basis of size selectivity alone the molecule could easily escape the capillary membrane.

Multilayer Membrane Models. The structural complexity of the glomerular capillary wall suggested a need to develop more complex theoretical models to simulate more reliably the resistance to water and solute movement across the membrane. These models have been developed and tested by Deen et al.,²² with the aim to estimate the role of individual layers on the filtration of water and solutes. In these models the resistance of endothelial cells, GBM, and epithelial cells is assumed to act in series. In normal conditions, hydraulic resistance of the endothelial layer is negligible, while GBM and epithelial resistance are comparable. The contribution of the three layers to solute hindrance has been considered and the major contribution to membrane selectivity is exerted by the filtration slit.⁴⁴ Although these models describe in detail the physical interaction of water and macromolecules with the membrane structure, their application is difficult because they require extensive measurement of ultrastructural parameters.

Models of Glomerular Charge Selectivity

As mentioned previously, the fact that negative electrical charges are present in the glomerular membrane (in the glycocalyx of endothelial cells, the negatively charged heparansulfate of the GBM and the glycoproteins of the cell membrane of podocytes) strongly suggests that circulating negatively charged proteins, like albumin, are restricted within

the circulation not only for their size but also for electrical charges. The use of theoretical models for the simulation of charge-selective function of the glomerular membrane allowed estimation of the amount of electrical charge present within the membrane.¹⁰ These studies indicated that electrical charge is an important component of glomerular permselective function, and changes in membrane electrical charge can explain abnormal albumin filtration even without changes in membrane structural parameters, such as pore size or fiber size and length per unit volume. However, the use of these models to investigate glomerular membrane charge is limited by the difficulties in measuring glomerular filtration of charged test solutes that interfere with circulating macromolecules and are not filtered simply on the basis to their molecular shape and electric charge alone.⁴⁵

PROTEIN REABSORPTION BY THE PROXIMAL TUBULE

PROXIMAL TUBULE STRUCTURE AND FUNCTION

The glomerular ultrafiltrate, once flowing inside the proximal tubule, undergoes important changes in composition, due to processes of water and solute reabsorption. In addition to small solutes and electrolytes, proteins such as albumin are also reabsorbed.⁴⁶ Thus final urinary excretion of proteins depends largely on the interaction of proteins with proximal tubular epithelial cells. These cells form a compact epithelial layer with a basal side in contact with the tubular basement membrane, an intercellular junction, and a luminal surface in contact with tubular fluid. They are characterized by a large number of mitochondria, an index of important metabolic activity, and a prominent layer of microvilli (Fig. 30.3) that results in the extension of the luminal cell surface area.

The microvillar membrane is the site for receptor mediated endocytosis of low-density lipoprotein and negatively charged proteins. Albumin in the proximal tubule undergoes specific

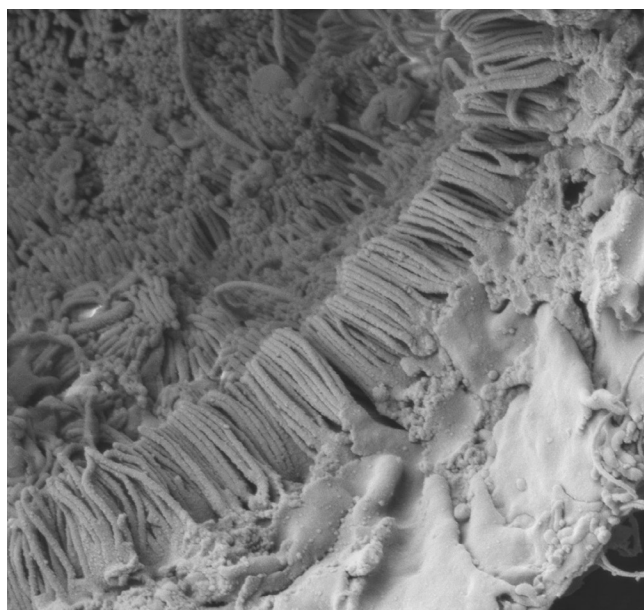


Fig. 30.3 Scanning electron microscopy of proximal tubule cells. Inner cell membrane is covered by microvilli of the brush border. In order to be taken up by cell receptors, albumin must diffuse through the dense layer of microvilli.

binding to the extracellular domain of a membrane receptor complex, megalin–cubilin receptor⁴⁶ (Fig. 30.4), which is followed by the internalization of the protein by membrane vesicles. These vesicles are then processed for protein degradation, amino acid transport to basal membrane of tubule cells, release into interstitial space, and ultimately into peritubular capillaries. Receptors are recycled into the luminal cell membrane by dense apical tubules.

The amount of albumin filtered at glomerular level and that reabsorbed by proximal tubule cells is not easy to quantify. Ideally one would have to sample the early proximal tubule and quantify albumin concentration in these microsamples. Despite technical difficulties, micropuncture techniques have been used to avoid sample contamination with plasma present close to the puncture site (in interstitial space and peritubular capillaries). The protein concentration in the urinary space was estimated to range from 10 to 25 $\mu\text{g}/\text{mL}$.^{47,48} More recently, direct in vivo imaging of fluorescent albumin by two-photon microscopy has been used to directly estimate albumin concentration in the Bowman capsule fluid.⁴⁹ It has been recently demonstrated that reliable measurements of albumin fractional clearance, the ratio between urinary space and plasma albumin concentration, allow estimation of an albumin concentration of about 60 $\mu\text{g}/\text{mL}$ in the Bowman capsule in normal conditions in the rat,⁵⁰ corresponding to a fractional clearance of 0.002. Once filtered at glomerular level, albumin and smaller proteins are almost entirely reabsorbed at the proximal tubular level. In pathologic conditions, when the filtered load overwhelms the reabsorptive capacity, proteins are detected in the urine.

THEORETICAL MODELS FOR TUBULAR REABSORPTION

The process of albumin reabsorption by proximal tubule cells has been modeled⁵¹ to allow a quantitative assessment of the relationship between albumin ultrafiltration at the glomerular level, proximal tubular uptake, and final excretion in the urine. In this model, the process of diffusion of albumin across the microvillar space and the uptake of the protein by tubule cell receptors are taken into consideration. The amount of albumin that is reabsorbed during the proximal tubule passage is simulated assuming the presence of a high-affinity site for binding and internalization of albumin at the base of tubule cell microvilli. According to in vitro and ex vivo data, these receptors are assumed to be half-saturated at concentrations similar to those mentioned for albumin in the Bowman capsule (20–30 $\mu\text{g}/\text{mL}$). The effect of the assumption of different values for the maximum absorptive capacity (V_{max}) on the albumin concentration along the proximal tubule is reported in Fig. 30.5. This modeling approach showed that the transport of albumin across the microvillar space has a modest effect on the value of V_{max} needed to fit micropuncture data.

The two most important parameters that determine the fraction of reabsorption of albumin are the single-nephron glomerular filtration rate (SNGFR) and the albumin concentration in the filtrate fluid (Cb). A 50% increase in SNGFR is predicted to cause a four- to fivefold increase in albumin excretion in rats and humans. For large increases in Cb, such as those measured by micropuncture,⁵² there is a threshold above which the reabsorption of albumin is overwhelmed and the protein appears in the urine. According to theoretical analysis⁵¹ this value corresponds to an albumin

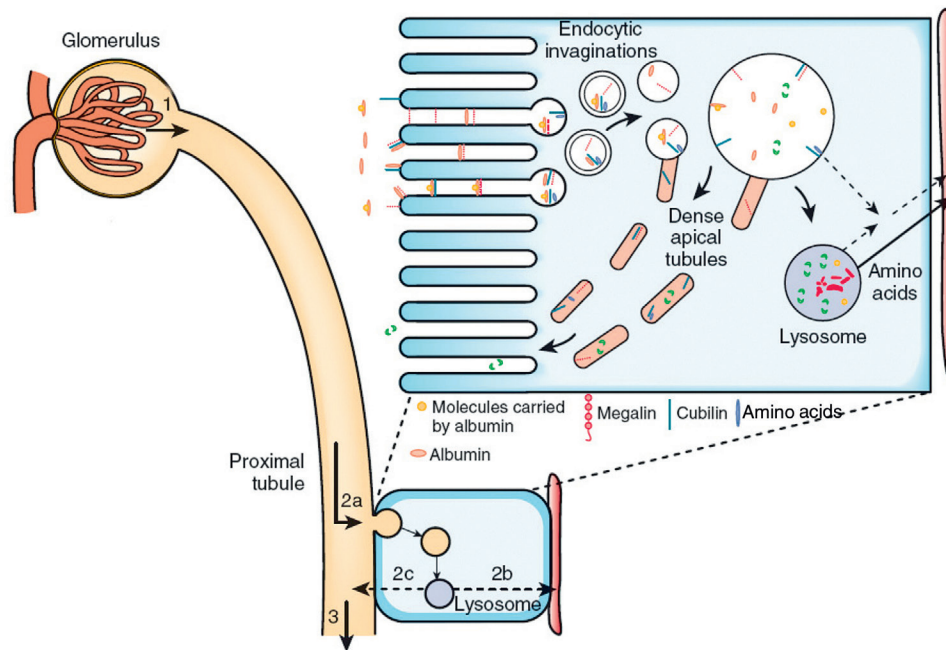


Fig. 30.4 Pathways of albumin degradation in the proximal tubule. Albumin is filtered in the glomeruli (1) and reabsorbed by the proximal tubule cells by receptor-mediated endocytosis (2a). Internalization by endocytosis is followed by transport into lysosomes for degradation. Some intact albumin may escape tubular reabsorption (3), the amount being greater as the glomerular filtration fraction of albumin increases or tubular function is compromised. The *upper right* shows a schematic representation of the intracellular pathways following endocytic uptake of albumin and possible associated substances. Following binding to the receptors, cubilin, or megalin, the receptor–albumin complex is directed into coated pits for endocytosis. The complex dissociates following vesicular acidification, most likely also leading to the release of any bound substances. Albumin is transferred to the lysosomal compartment for degradation. Some albumin may be degraded within a late endocytic compartment and recycled as fragments to be released at the luminal surface. Alternatively, albumin fragments may be recycled from the lysosomal compartment by a yet unknown route. Receptors recycle through dense apical tubules, whereas released substances carried by albumin may be released into the cytosol or transported across the tubular cell. (From Birn H, Christensen EI. Renal albumin absorption in physiology and pathology. *Kidney Int.* 2006;69:440–449, with permission.)

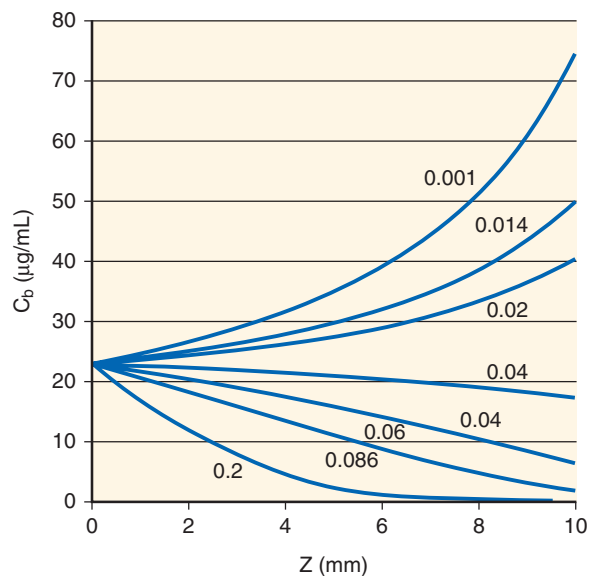


Fig. 30.5 Theoretical calculation of albumin concentration along the proximal tubule. Predictions of bulk albumin concentration (C_b) vs. axial position (z) in rats for V_{max} ranging from 0.001 to 0.2 ng/sec per millimeter squared. The curve that most closely corresponds to normal rats is that for $V_{max} = 0.086$ ng/sec per millimeter squared. (From Lazzara MJ, Deen WM. Model of albumin reabsorption in the proximal tubule. *Am J Physiol Renal Physiol.* 2007;292:F430–F439, with permission.)

fractional clearance of approximately 0.001. The combination of increased SNGFR and elevation in filtrate albumin concentration is shown to have important additive effects.

PROTEINURIA OF GLOMERULAR ORIGIN

Changes in glomerular protein filtration and/or defects in tubular reabsorption cause the appearance of proteins in the urine. At values exceeding 300 mg/day, or 200 mg/L the condition is termed *proteinuria*. Smaller amounts of proteins may appear in the urine in the early stages of progressive diseases, such as diabetic nephropathy. In this case an albumin excretion between 30 and 300 mg/day (20–200 mg/L) is termed *microalbuminuria*.⁵³ Proteinuria is considered severe or in the “nephrotic range” when protein excretion is greater than 3.5 g/day.¹ When proteins in the urine have a large molecular weight, they are considered to have glomerular origin. If the molecular weight is low, there is evidence that the defect causing proteinuria is likely related to abnormal proximal tubular reabsorption, often related to toxic damage of tubule cells.⁵⁴ Proteinuria associated with progressive kidney disease is predominantly of glomerular origin and mainly composed of plasma albumin. Mechanisms responsible for glomerular proteinuria are discussed in the next section.

GLOMERULAR PERMSELECTIVE DYSFUNCTION

As mentioned earlier, abnormal plasma protein filtration at glomerular level may be caused by a defect in both size and

charge selectivity. Glomerular size-selective dysfunction has been extensively investigated in several kidney diseases using neutral test macromolecules, usually neutral dextrans.^{42,55} These studies demonstrated that in most cases there are statistically significant changes in fractional clearance of the largest test macromolecules, whereas the fractional clearance (sieving coefficient) of molecules the size of albumin is unaltered. These data consistently indicate that permselectivity defects responsible for albumin filtration must be focal and likely due to changes in glomerular cell components, most likely the podocytes. These proteinuric conditions are frequently associated with podocyte foot process effacement and simplification, and likely with defective intercellular junctions.⁵⁶

The quantification of the contribution of charge-selectivity defects to proteinuria is more difficult. A few experimental and clinical investigations clearly indicate that proteinuria is indeed associated with abnormal filtration of charged macromolecules,^{57,58} but these data have been questioned because electrically charged test or endogenous macromolecules in the circulation are expected to interfere with other circulating charged solutes and cell membranes.¹ This would result in measured fractional clearance that does not represent effective transport of probe macromolecules. However, even without direct evidence that glomerular membrane charge distribution is altered in proteinuric conditions, the evidence that glomerular albumin filtration is increased without important changes in the fractional clearance of test macromolecules of the same size strongly suggests a role for a charge-selectivity defect in proteinuria of glomerular origin.^{42,58}

Experimental and clinical research has allowed the identification of molecular defects underlying some genetic disorders associated with nephrotic syndrome. In Finnish-type nephropathy a defect in the nephrin gene *NPHS1* is responsible for glomerular dysfunction, proteinuria, and end-stage renal disease.^{3,59} Similarly, defects in other genes (*NPHS2*, *LMX1B* and several others) have been shown to result in defective structure and function of filtration slit proteins or glomerular epithelial cells.⁶⁰

Another condition in which proteinuria is manifest is ischemia-reperfusion injury.⁶¹ Studies in kidney transplantation and in experimental settings suggest that abnormal elevation of protein in the urine in this condition occurs without major changes in glomerular capillary membrane structure. The evidence indicates⁶² that ischemia per se is responsible for loss of glomerular endothelial glycocalyx, and the previously mentioned effect of fluid shear stress on endothelial cell glycocalyx would reinforce this evidence. Thus, abnormal elevation of glomerular protein filtration may derive from selective changes in ultrastructure and function of membrane components.

TUBULAR HANDLING OF EXCESSIVE FILTERED PROTEINS

EFFECTS OF PROTEIN FILTRATION ON PROXIMAL TUBULE CELLS

An excessive increase in albumin and other plasma protein filtration may result from a defective glomerular capillary membrane and/or an increase in SNGFR. Both conditions, and their combination, result in elevated protein concentration in the ultrafiltrate. These filtered proteins are expected

not to be entirely reabsorbed by proximal tubule cells, because protein reabsorption is believed to operate near maximum under physiologic conditions.

The presence of high protein concentration within the renal tubule may influence the progression of disease processes. There are at least two phenomena that are expected to occur. The first is related to the fact that if albumin is still present in tubular fluid at the end of the proximal tubule, its concentration increases substantially along the remaining portion of the nephron because of water reabsorption. Thus protein concentration in the distal tubule and collecting duct can reach very high values even for a small amount of proteins filtered at glomerular level, with the possibility for these proteins to precipitate and form protein casts.⁶³ Tubular obstruction may then occur and the entire nephron function lost, with complete loss of glomerular water filtration.

In addition, structural changes are expected to occur with tubular atrophy, disconnection of the tubule from the Bowman capsule and glomerular capillary tuft structural changes. This condition is frequently observed in proteinuric kidney diseases at experimental and clinical level.⁶⁴

Even before tubular obstruction, important functional changes are expected to occur in proximal tubule cells exposed to abnormal protein concentrations. The protein overload of these cells exposes them to increased workload and this can lead to loss of reabsorptive capacity due to loss of receptor activity.⁶⁵ In this condition, the elevated protein concentration along the proximal tubule further increases due to the lower level of absorption and the concomitant water reabsorption. Thus a vicious circle develops, inducing further damage in tubule cells and progressively higher protein concentration along the entire nephron. The consequences of this abnormal glomerular filtration of plasma proteins at both the organ and systemic level are discussed in the following sections.

RENAL CONSEQUENCES OF PROTEINURIA

GLOMERULAR DAMAGE

Data from animal models have shown that a wide variety of insults result in a common pathway of glomerular capillary hypertension, increased permeability with excess passage of proteins across the glomerular capillary wall, and progressive glomerular injury⁶⁶ (Fig. 30.6). The key glomerular lesion is sclerosis, characterized by accumulation of extracellular matrix and obliteration of the capillary tuft leading to the loss of renal function.

PODOCYTES: CHANGES IN FUNCTION AND CELL NUMBER

Podocytes show a fairly uniform pattern of response to damage. The intercellular junction and cytoskeletal structure of the foot processes are altered, and the cell shows a simplified, effaced phenotype.^{67,68} These alterations result in the disappearance of the typical slit diaphragm structures and the development of proteinuria. Although podocyte effacement is a hallmark of podocyte disease and nephrotic syndrome, damage to these cells may present as very subtle changes that are difficult to quantify.⁶⁹ Major advances in the field of live imaging have allowed the investigation of

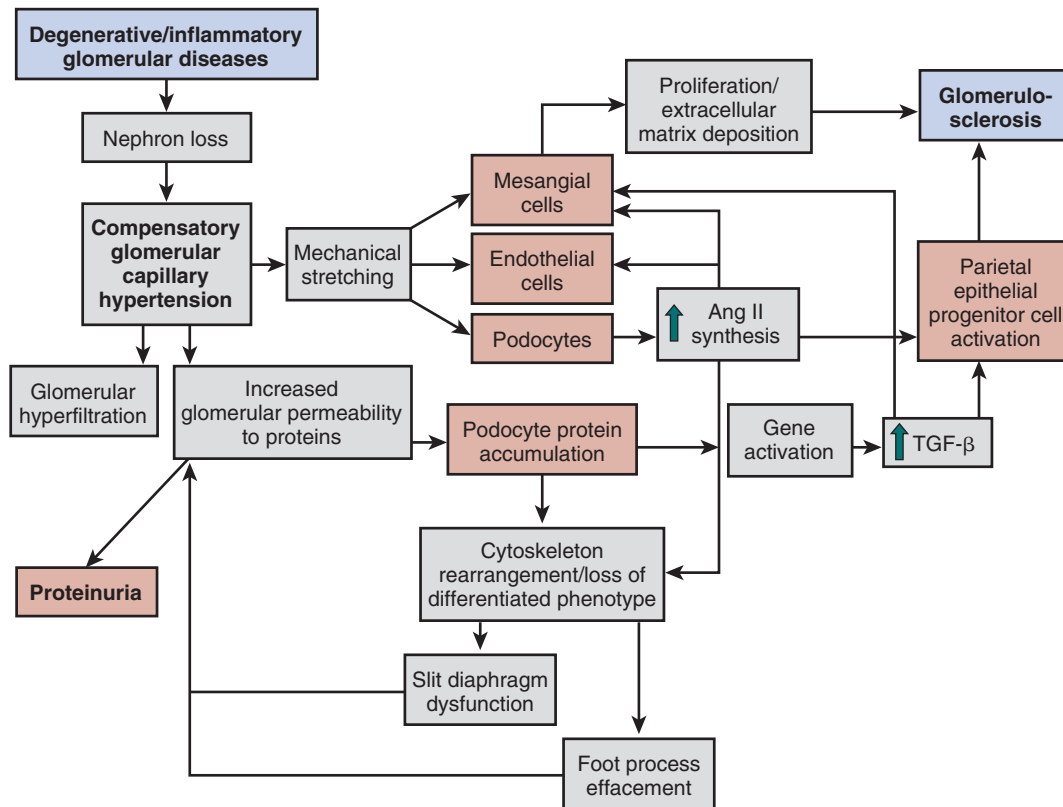


Fig. 30.6 Mechanisms of progressive glomerular injury. A reduction in the number of nephrons as a consequence of various glomerular diseases results in compensatory glomerular hemodynamic changes that are ultimately detrimental. In particular, by mechanical stretching, the increased glomerular capillary pressure directly injures glomerular cells. Glomerular hypertension also impairs the glomerular capillary size-selective function, which causes excessive protein ultrafiltration and eventually podocyte injury and proteinuria.

podocyte biology in unprecedented detail.⁷⁰⁻⁷² These new tools are likely to facilitate the next stage in gaining insights to podocyte responses to injury. So far there is evidence that experimental models of chronic proteinuria as well as their human counterparts (that is, minimal change glomerulopathy, focal and segmental sclerosis, diabetic nephropathy, and membranous nephropathy) have in common ultrastructural findings of severe glomerular epithelial cell damage that include vacuolization, fusion of foot processes and focal detachment of epithelial cells from the underlying basement membrane.⁷³ These changes appear to be the consequence mainly of persistent abnormalities in intraglomerular capillary hemodynamics. Increased capillary hydraulic pressure and flow and activation of local tissue renin-angiotensin system in podocytes⁷⁴ eventually impairs the size-selective function of the glomerular capillary wall, allowing excess plasma protein to move into the urinary space.⁷⁵

In addition to being affected by mechanical stress, podocytes are also damaged by excessive protein load resulting from alterations of glomerular permeability to macromolecules. Protein uptake by podocytes may occur through binding to megalin, a receptor for albumin and immunoglobulin (Ig) light chains, which is endocytosed after ligand binding, as shown in cultured murine podocytes.⁷⁶ Mice with protein overload proteinuria induced by repeated injection of bovine serum albumin developed podocyte injury followed by glomerulosclerosis.⁷⁷⁻⁷⁹ Evidence of a causal link between podocyte protein overload and podocyte damage is provided

by studies showing that in rats with renal mass reduction protein accumulation in podocytes preceded cell dedifferentiation and injury, as characterized by loss of synaptopodin and an increase in desmin expression.⁸⁰

Podocyte abnormalities were accompanied by upregulation of transforming growth factor- β (TGF- β) messenger ribonucleic acid (mRNA) and enhanced production of the related protein.⁸⁰ In vitro, albumin loading of immortalized mouse podocytes promoted actin-cytoskeleton rearrangement and upregulation of intracellular transduction signals, such as activating protein-1 (AP-1), which is a known stimulus of TGF- β 1 synthesis.⁸¹

Podocytes possess a complex contractile structure composed of F-actin microfilaments, most abundant in the foot process, connected with adaptor molecules that anchor the slit diaphragm proteins and α 3 β 1 integrins, transmembrane proteins that form focal adhesion complexes and mediate podocyte-GBM matrix interaction.^{82,83} In vitro, actin filament disorganization, as occurs after albumin loading of mouse podocytes,⁸¹ is closely associated with podocyte shape changes that affect cell adhesion to the extracellular matrix. Podocyte detachment from the GBM likely underlies the decrease in podocyte number in proteinuric glomerular diseases, which has been shown in many experimental and clinical studies.^{84,85}

Apoptosis is considered an additional cause of podocyte loss in proteinuric glomerulopathies. Once detached from the GBM, podocytes become extremely susceptible to apoptosis.⁶⁷ Furthermore, apoptosis may be promoted by locally

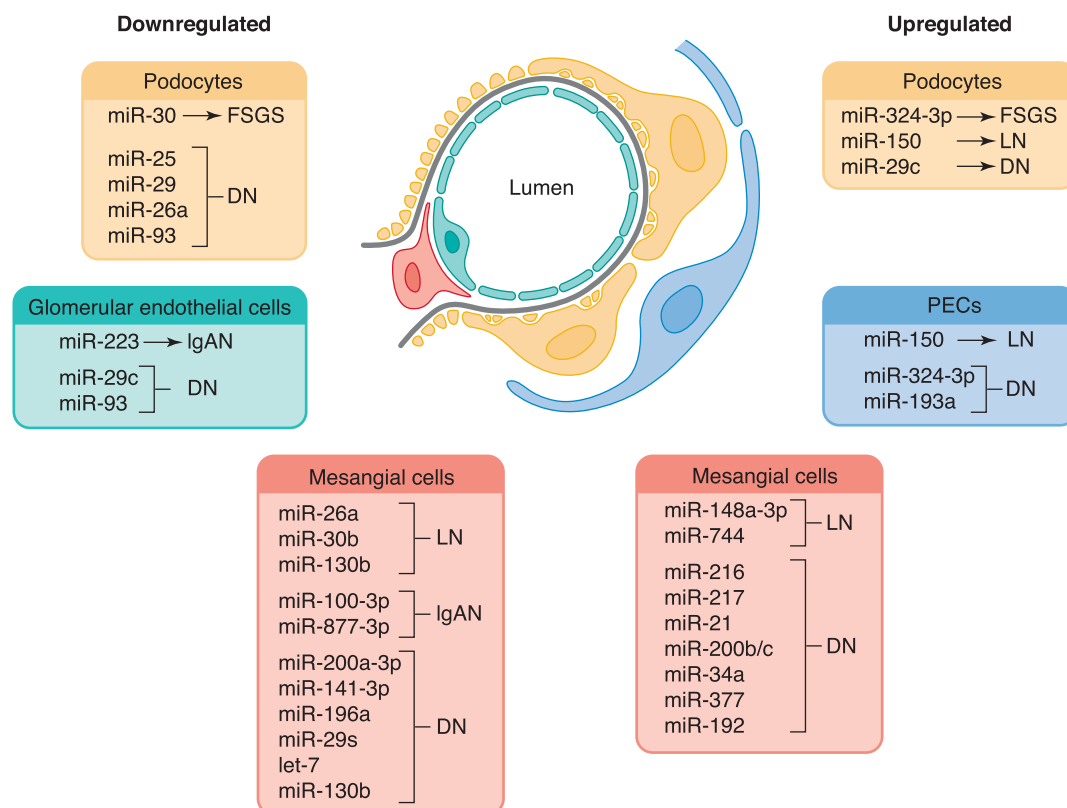


Fig. 30.7 MicroRNA (miRNA) dysregulation in glomerular disease. Changes in miRNAs in different populations of glomerular cells (podocytes, parietal epithelial cells, glomerular endothelial cells, and mesangial cells) occurring in focal and segmental glomerulosclerosis (FSGS), lupus nephritis, IgA nephropathy, and diabetic nephropathy. DN, Diabetic nephropathy; IgAN, IgA nephropathy. (From Trionfini P, Benigni A. MicroRNAs as master regulators of glomerular function in health and disease. *J Am Soc Nephrol*. 2017;28:1686–1696.)

produced proapoptotic factors. Studies have demonstrated that exogenous TGF- β 1 induced apoptosis in cultured podocytes via the p38 mitogen-activated protein kinase (MAPK) and classic caspase-3 pathways.⁸⁶ This effect occurred only in wild-type, not in p21-null cultured podocytes, indicating that the cyclin-dependent kinase (CDK) inhibitor p21 is required for TGF- β 1-induced apoptosis.⁸⁷ Of note, like TGF- β 1, p21 is increased in podocytes in experimental models of membranous nephropathy⁸⁸ and diabetic nephropathy.⁸⁹ In summary, protein accumulation in podocytes induces TGF- β 1 production, leading to podocyte apoptosis.

Recent evidence indicates that angiotensin II (Ang II) contributes to perpetuate podocyte injury in proteinuric nephropathies eventually promoting progression to end-stage kidney disease.⁹⁰ Mechanical strain increases Ang II production and expression of Ang II type 1 (AT₁) receptors in podocytes⁷⁴ potentially contributing to further sustaining the glomerular hypertension-induced damage in chronic kidney disease. However, there is also evidence that, independent of its hemodynamic effect, Ang II may directly impair the glomerular barrier sieving function, possibly through inhibition of nephrin expression by podocyte, the essential protein component of the glomerular slit diaphragm.^{91,92} This observation has been confirmed in studies in diabetic animals showing that blockade of Ang II synthesis/activity preserved the expression of nephrin in the glomeruli and prevented overt proteinuria.^{93,94} Thus at least in diabetes, a pathogenetic relationship between Ang II and early proteinuria via

functional podocyte alteration through modulation of nephrin protein level has been suggested. Moreover, in the setting of diabetes, after the initial insult of hyperglycemia and intraglomerular hypertension, Ang II plays a relevant role to sustain glomerular injury via persistent activation of Notch1 and Snail signaling in the podocyte, eventually resulting in persistent downregulation of nephrin expression.⁹⁵ The consistency of these findings in diabetic ZDF rats with overt nephropathy and in type 2 diabetic patients with established nephropathy provides robust insight to implicate an important role for the Ang II Notch1/Snail axis in perpetuating podocyte damage.

MicroRNAs (miRNAs) are a class of short (21–24 nucleotides) noncoding RNAs that regulate gene expression through posttranslational and epigenetic mechanisms, thereby affecting several cellular processes from development to disease conditions.⁹⁶ miRNAs are critical players in podocyte homeostasis (Fig. 30.7), because targeted deletion of the helicase with RNase motif Dicer or the class II ribonuclease III enzyme Drosha in these cells leads to proteinuria and glomerulosclerosis.^{97–100} Moreover, several miRNAs have been found to be dysregulated in podocyte injury, a pathologic mechanism causing glomerular injury and sclerosis. Because mature podocytes must withstand fluctuating pressures and potentially harmful molecules contained in the primary filtrate, they are unlikely to be static structures.¹⁰¹ Remodeling of the actin cytoskeleton plays a primary role in the structural adaptations made by podocytes to preserve their

glomerular filtration properties. The podocyte cytoskeleton is finely regulated by a set of miRNAs that are expressed mainly in the adult kidney, such as miR-30, miR-132, miR-134, and miR-29a. miR-30 family members are highly expressed in human podocytes and, in addition to protecting them against apoptosis,¹⁰² they promote podocyte actin fiber stability by controlling calcium/calcieneurin signaling through the inhibition of several components of this pathway.^{103,104} Dysregulation of calcium/calcieneurin signaling leads to podocyte cytoskeletal damage, which is a key feature of various glomerular diseases, such as FSGS, characterized by the early onset of podocyte injury.¹⁰³ It is notable that miR-30s are significantly downregulated in the podocytes of patients with FSGS and in the rat model of FSGS induced by the podocyte toxin puromycin aminonucleoside. Furthermore, fibrogenic factors, such as TGF- β , reduce miR-30 expression both in vivo and in cultured podocytes.^{102,105} A recent study underlines the role of miRNAs in regulating podocyte cytoskeletal dynamics. The study showed that brain-derived neurotrophic factor can repair podocyte damage both in vitro and in a mouse model of FSGS by inducing miR-132 and inhibiting miR-134 expression upon binding to its receptor on podocytes. Brain-derived neurotrophic factor-induced modulation of miR-132 and miR-134 has been found to be essential for increasing actin polymerization, favoring foot process elongation that contrasts the cell flattening induced by proteinuric conditions.¹⁰⁶

A crosstalk with proximal tubule cells has been recently suggested to contribute to podocyte function through the release of nicotinamide mononucleotide (NMN) that might have implications for persistent podocyte dysfunction in proteinuric diseases.¹⁰⁷ In a mouse model, diabetes induced downregulation of Sirtuin1 (Sirt1), a highly conserved protein deacetylase in proximal tubules.¹⁰⁸ The low Sirt1 expression reduces the release of nicotinamide mononucleotide by tubule cells, eventually decreasing local NMN concentrations. In vitro studies have shown that in the absence of NMN the expression of the tight junction protein claudin-1 in podocytes is no longer silenced.¹⁰⁷ Claudin-1 is reported to activate the intracellular β -catenin–Snail pathway¹⁰⁹ that eventually leads to glomerular barrier dysfunction through downregulating synaptopodin or podocin expression in podocytes.¹¹⁰

MESANGIAL CELLS: PROLIFERATION AND DEPOSITION OF EXTRACELLULAR MATRIX

Because it is close to the capillary lumen, the mesangium may be exposed to macromolecules crossing the endothelial layer, although under normal conditions they do not accumulate.¹¹¹ However, in rats having undergone unilateral nephrectomy¹¹² or in puromycin-aminoglycoside-induced nephrosis,¹¹³ intravenous infusion of colloidal carbon leads to the accumulation of the macromolecular tracer in the mesangial space. To prevent accumulation of proteins, mechanisms exist for their effective removal. These include transport along the mesangial stalk in cleft-like spaces as well as phagocytosis and degradation by mesangial cells.¹¹⁴

It has been shown that IgG and IgA can be taken up by both receptor-independent and receptor-mediated processes.¹¹⁵ Another important factor for clearance of Igs from the mesangium may be complement factor D, a serine protease essential for activation of the complement system through the alternative pathway, which is constitutively expressed within

the glomerulus.¹¹⁶ Interestingly, mice deficient in complement factor D spontaneously develop a mesangial immune-complex deposition disease associated with albuminuria.¹¹⁷ This indicates that complement factor D is necessary to prevent mesangial accumulation of immunoglobulin deposits.

Whether abnormal local accumulation of proteins promotes mesangial cell proliferation and mesangial matrix deposition remains, however, ill-defined. Nevertheless, the significance of the protein-clearing function of the mesangial cells has been illustrated by the deleterious consequences of mesangial immune complex accumulation leading to complement activation and generation of mediators of inflammation, such as reactive oxygen species, prostanooids, and cytokines, such as tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6).¹¹⁸

The mesangial cell is a critical part of the glomerular functional unit interacting closely with endothelial cells and podocytes.¹¹⁶ Alterations in one cell type can produce changes in the others. As such, key survival factors for mesangial cells, including platelet-derived growth factor-B (PDGF-B) are generated by endothelial cells, and mesangiolysis has been shown in knockout mice lacking endothelial PDGF-B.¹¹⁶ Whether cytokines generated by podocytes also influence mesangial cells has yet to be clearly defined, but the observation that podocyte injury frequently results in mesangial cell proliferation supports the existence of such cytokine crosstalk.¹¹⁶

Besides PDGF-B, other growth factors shown to influence mesangial cell proliferation and mesangial matrix accumulation include PDGF-C, fibroblast growth factor, hepatocyte growth factor, epidermal growth factor (EGF), connective tissue growth factor, and TGF- β .¹¹⁶ Effects of vasoactive hormones such as Ang II on mesangial cell proliferation may be mediated indirectly through the generation of growth factors such as EGF.¹¹⁹ In rats with renal mass ablation, glomerular TGF- β 1 upregulation is associated with phenotypic transformation of mesangial cells.⁸⁰ In vitro exposure of cultured murine mesangial cells to TGF- β induced a sclerosing phenotype as shown by α -smooth muscle actin (SMA) expression, which was blocked by anti-TGF- β 1 antibodies.⁸⁰ Moreover, the transfection of the TGF- β 1 gene into normal kidneys in rats or the transgenic TGF- β expression in mice increased extracellular matrix accumulation in the mesangial space.¹²⁰ At least in experimental diabetes, increased matrix production in mesangial cells is induced by a fine balance between the upregulation of miR-192, miR-200b/c, miR-216a, and miR-377 and the downregulation of miR-29s and let-7. High glucose, via TGF- β , upregulates miR-377, which suppresses the expression of p21-activated kinase and superoxide dismutases, leading to enhanced susceptibility to oxidant stress and the accumulation of fibronectin.¹²¹ In mesangial cells, TGF- β lowers the expression of miR-29s¹²² and let-7,¹²³ antifibrotic miRNAs targeting different isoforms of collagen. Moreover, decreased miR-29a expression attenuates the Dickkopf-1/Wnt/ β -catenin signaling, contributing to apoptosis and extracellular matrix deposition in streptozotocin-treated mice.¹²⁴ Notably, in mesangial cells TGF- β induces signaling loops that amplify and create a chronic state of profibrotic pathway activation, modulating the expression of miR-192, miR-200s, miR-21, and miR-130b.¹²⁵⁻¹²⁷

ENDOTHELIAL CELLS: APOPTOSIS

Glomerular endothelial injury is a common feature of many human diseases, such as diabetic nephropathy, hypertension,

thrombotic microangiopathy and preeclampsia. Evidence has been provided that there is close crosstalk of podocytes with glomerular endothelial cells, a key interaction for the normal function of the glomerular capillary barrier.¹²⁸ The final steps in glomerular endothelial cell differentiation involve the formation of fenestrae, plasma membrane-lined circular pores that perforate the flattened glomerular endothelium.¹²⁹ The fenestrated phenotype of glomerular endothelial cells is induced by vascular endothelial growth factor (VEGF), a molecule constitutively expressed and secreted by podocytes.^{128,129} That VEGF regulates fenestrae formation was suggested by the observation that mature fenestrated endothelium is typically located adjacent to podocytes expressing high levels of VEGF mRNA.¹³⁰ There is also in vitro evidence that VEGF type A and C regulate glycosaminoglycan synthesis, charge, and shedding of glomerular microvascular endothelial cell glycocalyx.¹³¹ Moreover, deletion of the podocyte-specific transcription factor LMX1B in mice, which results in the loss of many features of podocyte differentiation, including VEGF-A expression, is associated with failure of glomerular endothelium to differentiate and develop fenestrae.¹³² Thus, loss of podocytes secondary to protein-induced cell injury may lead to reduced VEGF production influencing glomerular endothelial fenestrae formation and eventually leading to endothelial cell apoptosis.¹³³ However, how VEGF reaches endothelial cells against the urine flow is not yet known. Conversely, recent in vitro evidence has shown that blockade of VEGF in glomerular endothelial cells enhanced the release of endothelin-1 (ET-1), which induced nephrin shedding from podocytes,¹³⁴ leading to further glomerular protein permeability dysfunction.

Moreover, endothelin-1 activates podocytes to release heparanase.¹³⁵ In mice, podocyte-specific deletion of the endothelin receptor prevented the diabetes-induced increase in glomerular heparanase expression and consequent reduction in heparan sulfate expression, endothelial glycocalyx thickness, and development of proteinuria that was observed in wild-type mice.¹³⁵ Heparanase-deficient mice were resistant to the development of proteinuria and renal damage upon induction of type 1 diabetes mellitus.¹³⁶ In addition, proteinuria was also reduced and renal function improved by treatment with the experimental heparanase inhibitor SST0001.¹³⁶

Taken together, these studies indicate that the toxic effects of excess ultrafiltered plasma proteins on podocytes may alter podocyte–endothelial interaction, thereby further enhancing glomerular permeability to proteins through a complex interplay of molecular signaling.

PARIETAL EPITHELIAL CELLS: ACTIVATION

Changes in glomerular permselective function, as it occurs in proteinuric glomerulopathies, elevate the filtered load of plasma albumin and consequently its concentration in the Bowman space (Fig. 30.8). Evidence has been provided that the abnormally filtered albumin impairs the mechanism underlying regeneration of protein-induced damage to podocytes.¹³⁷ Glomerular injury caused by multiple etiologies can lead to activation and accumulation of parietal epithelial cells (PECs) within the Bowman space as a common response to damage,¹³⁸ as shown in several human proliferative glomerulonephritides. Although extracapillary proliferation is a relatively straight forward pathologic change to recognize,

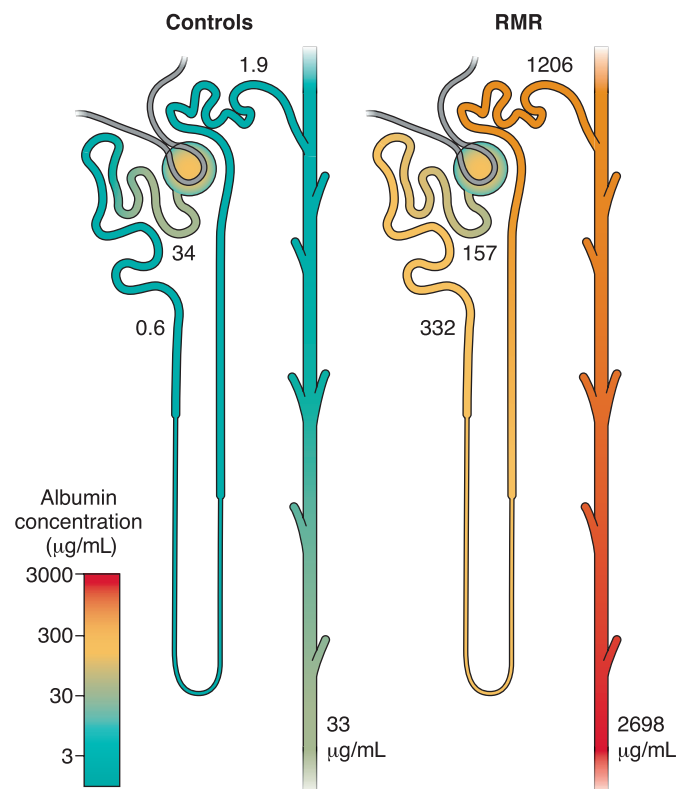


Fig. 30.8 Estimated albumin concentrations along the nephron. Color-coded graphical representation of estimated albumin concentration along the entire nephron in the two animal groups (control group and renal mass reduction group). Numbers represent local group average albumin concentration in $\mu\text{g/mL}$. RMR, Renal mass reduction. (Modified from Sangalli F, Carrara F, Gaspari F, et al. Effect of ACE inhibition on glomerular permselectivity and tubular albumin concentration in the renal ablation model. *Am J Physiol Renal Physiol*. 2011;300:F1291–1300.)

determining its cellular components has been more controversial. The traditional concepts, largely from immunohistochemical studies, indicate that the multilayered cellular lesions are a mixture of glomerular parietal epithelial cells, macrophages, and myofibroblasts,^{139–141} the proportion of such cells in the lesion being variable. In both animal models and human tissues, PECs predominate when the Bowman capsule is intact. Recently, a heterogeneous population of renal progenitor cells, previously identified in the normal human Bowman capsule,¹⁴² has been documented in hyperplastic lesions of human crescentic glomerulonephritis.¹⁴³ The extracapillary lesions could therefore be the result of dysregulated proliferation of renal progenitor cells in response to the injured podocytes.¹⁴³ This possibility is supported by findings in Munich Wistar Frömter (MWF) rats, which are genetically programmed to develop renal damage characterized by excessive progenitor cell migration and proliferation, leading to their accumulation into cellular lesions and glomerulosclerosis.¹⁴⁴ Among the factors that influence the maladaptive PEC response, ultrafiltered albumin has been regarded as a critical player that impairs podocyte regeneration.¹⁴⁵ Notably, albumin prevents PEC differentiation into podocytes by sequestering retinoic acid and impairing retinoic acid response element-mediated transcription of podocyte specific genes.¹⁴⁵

Circulating components of the complement system are also lost in the urine in proteinuric conditions and become activated at the glomerular level, favoring progression of the lesions.^{80,146-148} Thus abnormal fixation of ultrafiltered complement C3 is detected in podocytes showing signs of dedifferentiation and injury during the early stage of proteinuric disease in rats with remnant kidney,⁸⁰ and in mice with protein overload proteinuria.¹⁴⁸ C3-deficient mice with protein overload are protected against podocyte structural damage and sclerosis, indicating that C3 might increase susceptibility to injury. Recently, we demonstrated that complement activation via alternative pathway is a pivotal trigger for podocyte loss and PEC activation, leading to glomerulosclerosis in the model of protein-overload proteinuria.¹⁴⁹ Factor H (*Cfh*^{-/-}) or factor B-deficient mice were studied in comparison with wild-type littermates. Wild-type mice with protein overload induced by bovine serum albumin showed podocyte depletion accompanied by glomerular complement C3 and C3a deposits, PEC migration to capillary tuft, proliferation, and glomerulosclerosis. These changes were more prominent in *Cfh*^{-/-} mice with protein overload induced by bovine serum albumin. The pathogenic role of the alternative pathway was documented by data that factor B deficiency preserved glomerular integrity. In protein-overload mice, PEC dysregulation was associated with upregulation of CXCR4 and GDNF/c-Ret axis. In vitro studies provided additional evidence of a direct action of C3a on proliferation and CXCR4-related migration of PECs. These effects were enhanced by podocyte-derived GDNF. In patients with proteinuric nephropathy, glomerular C3/C3a paralleled PEC activation, CXCR4, and GDNF upregulation. These results indicate that mechanistically uncontrolled alternative pathway complement activation is not dispensable for podocyte-dependent PEC activation resulting in glomerulosclerosis.¹⁴⁹

Parietal epithelial cells expressing the progenitor cell marker CD133⁺CD24⁺ have also been reported to proliferate and accumulate into the multilayered cellular lesions in patients with glomerulonephritides characterized by extracapillary proliferation but not in nonproliferative nephropathies such as membranous or diabetic nephropathies.¹⁵⁰ Upregulation of the CXCR4 chemokine receptor on these progenitor cells was accompanied by high expression of its ligand, SDF-1, in podocytes.¹⁵⁰ Moreover, parietal epithelial cell proliferation was associated with increased expression of the Ang II subtype 1 receptor. Renin-angiotensin system blockade normalized CXCR4 and Ang II subtype 1 receptor expression on parietal progenitor cells concomitant with regression of crescentic lesions. Together these findings suggest that the glomerular hyperplastic lesions derive from the proliferation and migration of renal progenitors in response to injured podocytes, and that the Ang II/AT₁ receptor pathway may contribute, together with SDF-1/CXCR4 axis, to the dysregulated response of parietal epithelial cell precursors.

Parietal epithelial cell activation is increasingly recognized and seems to be present also in most forms of FSGS, characterized by nephrotic syndrome often leading to a progressive decline in renal function.¹⁵¹ Using a lineage tagging approach in mice, it has been documented that activated parietal epithelial cells invade the affected segment of the capillary tuft and initiate glomerular and epithelial basement membrane adhesion and glomerulosclerosis.¹⁵² The Notch signaling pathway has been proposed to play a role in orchestrating

parietal epithelial cell phenotypic changes in FSGS.¹⁵³ This possibility rests on in vitro evidence that in cultured mouse parietal epithelial cells TGF- β enhanced Notch mRNA expression, which resulted in a significant upregulation of target genes associated with mesenchymal cell phenotype, such as α -smooth muscle actin, vimentin, and Snail.¹⁵³ Concurrent inhibition of Notch signaling with the γ -secretase inhibitor DBZ¹⁵⁴ blocked both epithelial-to-mesenchymal transition gene expression changes and cell migration in response to TGF- β , demonstrating a dependence on Notch signaling for induction of mesenchymal gene marker activation in the parietal epithelial cell line. Moreover, in LMB2 antibody-treated NEP25 transgenic mice, a model of collapsing FSGS,¹⁵⁵ Notch inhibition in vivo significantly decreased parietal epithelial cell lesions, pointing to the role of Notch-mediated cell activation in the formation of such lesions.¹⁵³

miRNAs have emerged as important regulators of gene expression in parietal epithelial cells.⁹⁶ Indeed, robust miR-150 expression in PECs and podocytes of patients with lupus nephritis correlated positively with disease chronicity scores and the expression of profibrotic proteins. Increased miR-150 would foster the production of profibrotic molecules through the downregulation of its predicted target, SOCS1. The latter protein acts as a negative regulator of the JAK/STAT signaling pathway, which promotes the transcription of genes involved in cell proliferation, inflammation, and fibrosis.^{156,157} Similar to miR-150, we identified miR-324-3p increased expression in PECs as well as podocytes in an FSGS rat model. Increased expression of miR-324-3p was associated with the downregulation of its target propyl endopeptidase—involved in the formation of the antifibrotic peptide Ac-SDKD—in fibrotic areas of the kidneys of diseased rats.¹⁵⁸ Human PECs isolated from naïve Bowman capsules express significant levels of miR-193a, which works as a suppressor of podocyte differentiation by inhibiting the expression of the transcription factor Wilms tumor protein (WT1).^{159,160} WT1 is essential for the development and maintenance of podocytes and glomeruli by inducing the expression of genes governing podocyte architecture.¹⁶¹ The downregulation of miR-193a is associated with PEC transdifferentiation toward a podocyte phenotype, whereas its overexpression leads to their abnormal activation, a prerequisite for the formation of crescents in proliferative glomerulonephritis.¹⁶⁰ In line with this, isolated glomeruli from individuals with FSGS are characterized by increased expression of miR-193a, compared with normal kidneys or kidneys affected by other glomerular diseases.¹⁵⁹ The involvement of miR-193a in the pathogenesis of FSGS was also supported by data from miR-193a knock in mice, which develop FSGS with extensive podocyte foot process effacement.¹⁵⁹ All of the previous evidence, together with findings that the number of crescents was reduced by anti-miR-193a in a mouse model of nephrotoxic nephritis,¹⁶⁰ concur to indicate that miR-193a is a promising therapeutic target for FSGS. The mouse model of nephrotoxic nephritis was also instrumental for identifying the key role of miRNAs in orchestrating T-cell-mediated crescentic glomerulonephritis. Mice with miRNA-deficient CD4⁺ T cells develop less severe glomerulonephritis upon toxin injection. Moreover, the kidneys of patients with ANCA-associated crescentic glomerulonephritis and mice with nephrotoxic nephritis are characterized by the upregulation of miR-155 that drives the Th17 immune response and tissue injury.¹⁶²

LOSS OF GLOMERULAR CAPILLARIES: POSTGLOMERULAR HYPOXIA

Irrespective of the underlying process leading to glomerular endothelial damage, such as increased intracapillary pressure and/or podocyte loss, the net result is rarefaction of glomerular capillaries. Loss of glomerular capillary loops translates into diminished postglomerular blood flow from affected glomeruli and downstream injury of the peritubular capillary network. Microvascular dysfunction causes progressive scarring of renal tissue by creating hypoxic environment that triggers a fibrotic response in tubulointerstitial cells.¹⁶³ This, in turn, has an impact on adjacent unaffected capillaries and glomeruli, further extending the hypoxic area, and leading to a vicious cycle of progressive destruction of the kidney and decline of renal function to end-stage organ failure.

Indeed, in animal models of proteinuric chronic kidney disease, including anti-Thy1 glomerulonephritis, 5/6 remnant kidney, diabetic nephropathy, and adriamycin-induced nephrosis, the immunohistochemical detection of hypoxia-dependent pimonidazole protein adducts has revealed that renal tissue hypoxia is present early in the course of the disease.¹⁶⁴ Moreover, blood oxygen-dependent magnetic resonance imaging has shown hypoxia in diabetic nephropathy.¹⁶⁵ Although data from animal models provide a compelling argument for postglomerular hypoxia in proteinuric diseases as a primary mediator of progressive renal scarring, data in

humans are scarce. Nevertheless, that hypoxia-related injury also applies to humans may be deduced by the finding that there is increased expression of hypoxia-inducible factor (HIF)—a key regulator of the adaptive response to hypoxia controlling expression of hundreds of genes¹⁶⁶—in biopsies of patients with diabetic nephropathy, IgA nephropathy, and chronic allograft nephropathy.¹⁶⁷

TUBULAR DAMAGE

Glomerular ultrafiltration of excessive amounts of plasma protein-associated factors incites tubulointerstitial damage and further promotes the effects of glomerular disease on the tubular compartment. The noxious substances in the proteinuric ultrafiltrate may set off tubular epithelial injury with tubular apoptosis, secondary generation of inflammatory mediators, and peritubular inflammation.⁴ The mechanisms whereby increased urinary protein concentration leads to nephrotoxic injury are multifactorial and involve complex interaction between numerous pathways of cellular damage (Fig. 30.9).

TUBULAR CELLS: APOPTOSIS AND TUBULO GLOMERULAR DISCONNECTION

Emerging evidence suggests that proteinuria causes tubule cell apoptosis. In cultured proximal tubule cells, delipidated albumin-induced apoptosis in a dose- and time-dependent manner,¹⁶⁸ as characterized by internucleosomal DNA

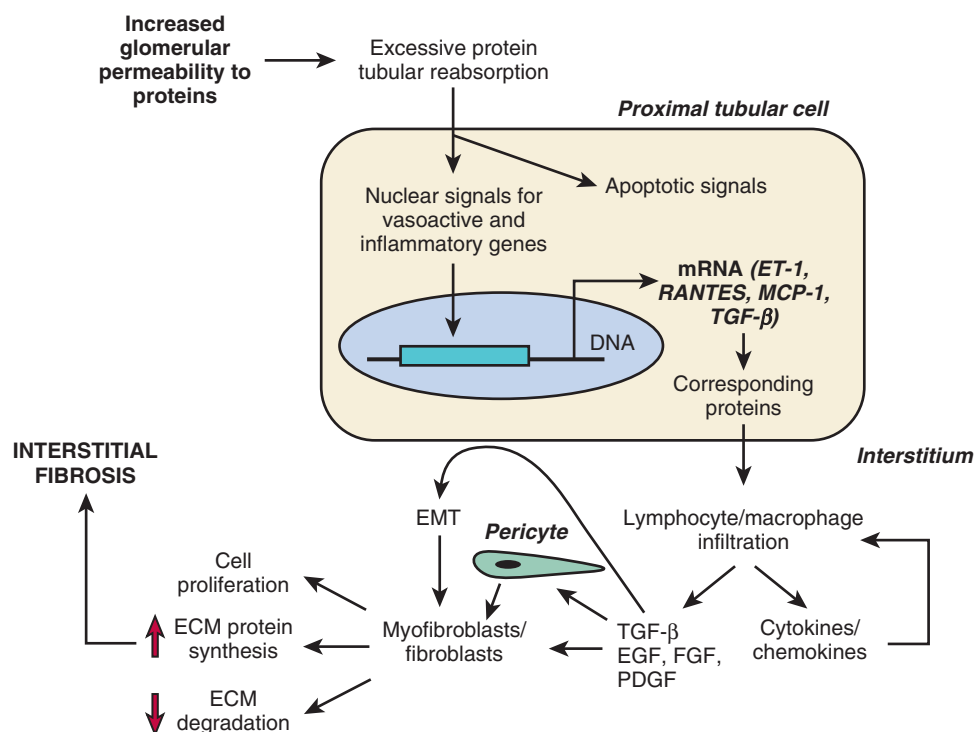


Fig. 30.9 Mechanisms of tubulointerstitial damage induced by proteins. Protein overload of proximal tubule cells as a consequence of increased glomerular permeability to proteins activates intracellular signals that promote cell apoptosis or cause increased production of inflammatory and vasoactive mediators and growth factors. These substances are released into the interstitium, inducing progressive inflammation and injury. *ECM*, Extracellular matrix; *EGF*, epidermal growth factor; *EMT*, epithelial-to-mesenchymal transdifferentiation; *ET-1*, endothelin-1, *FGF*, fibroblast growth factor; *MCP-1*, monocyte chemoattractant protein-1; *PDGF*, platelet-derived growth factor; *TGF-β*, transforming growth factor-β.

fragmentation, morphologic changes including cell shrinkage and nuclear condensation, and plasma membrane alterations.¹⁶⁸ Kidneys of rats with albumin-overload proteinuria¹⁶⁹ or with passive Heymann nephritis⁶⁴ showed increased numbers of terminal dUTP nick-end labeling-positive apoptotic cells in the tubulointerstitial compartment. In tubules, most of the positive cells expressed Ang II subtype 2 (AT2) receptors.¹⁶⁹ Findings of reduced phosphorylation of extracellular signal-regulated kinase (ERK) and Bcl-2 suggested an AT2 receptor-mediated mechanism underlying tubular cell apoptosis.¹⁶⁹

Apoptotic cells expressing both proximal and distal tubular phenotypes were detected in biopsy specimens from patients with primary FSGS.¹⁷⁰ A strong positive correlation was found between proteinuria and incidence of tubular cell apoptosis.¹⁷⁰

Renal proximal tubule cells have a remarkable ability to reabsorb large quantities of albumin through clathrin- and megalin receptor-mediated endocytosis.⁶⁵ Megalin is the sensor that determines whether cells will be protected from or injured by albumin. It has been shown that megalin binds the serine/threonine kinase PKB, crucial for the phosphorylation of Bad, the Bcl2-associated death promoter.¹⁷¹ Low concentrations of albumin lead to activated PKB and phosphorylation of the Bad protein, which inhibits apoptosis.¹⁷² On the other hand, overload of albumin leads to decreased megalin expression on the plasma membrane of proximal tubule cells that is associated with reduction of PKB activity and Bad phosphorylation.¹⁷³ The result is albumin-induced apoptosis.

In cultured proximal tubule cells, albumin repletion with fatty acids and its association with linoleic acid induced more apoptosis than the exposure to defatted albumin alone.¹⁷⁴ Furthermore, another study showed that nondelipidated albumin or albumin conjugated with palmitate, but not fatty acid-free albumin, altered both tubule mitochondrial variability and membrane potential and caused cytochrome *c* release.¹⁷⁵ In concert with the decline of mitochondrial parameters, fatty acid overload led to a redox imbalance, which deactivated the antioxidant protein peroxiredoxin 2 and caused a peroxide-mediated apoptosis through the redox-sensitive pJNK/caspase-3 pathway. These data were taken to suggest that attempts at lowering circulating fatty acid levels may be important in both preserving redox balance and limiting tubule cell damage.¹⁷⁵ A novel biochemical mechanism has been proposed linking lipotoxicity to tubule apoptosis in proteinuric conditions.¹⁷⁶ The study focused on the Na⁺/H⁺ exchanger NHE1, a regulator for proximal tubule cell survival through interaction with the membrane phosphoinositide phosphatidylinositol 4,5-bisphosphate (PI[4,5]P2), which initiates formation of a signaling complex that culminates in Akt activation and opposition to apoptotic stress. Starting from the concept that diseased glomeruli with impaired permselectivity allow filtration and proximal tubule reabsorption of nonesterified fatty acids (NEFA) bound to albumin, it was shown that an accumulation of metabolites of NEFA, long-chain acyl-CoA could stimulate lipoapoptosis by competing with the structurally similar PI(4,5)P2 for NHE1 binding, thus interrupting PI(4,5)P2 prosurvival activity.¹⁷⁶

Proximal tubule cell apoptosis was found to contribute to tubulo glomerular disconnection and atrophy in response to proteinuria in animal models of proteinuric nephropathies.⁶⁴ Injured or dying cells release molecules that serve as

danger signals.¹⁷⁷ Danger molecules trigger inflammation by engaging pattern recognition receptors such as Toll-like receptors (TLRs)¹⁷⁸ and nucleotide-binding domains, leucine-rich repeat-containing proteins (NLRs),¹⁷⁹ and are thus referred as danger-associated molecular patterns (DAMPs).¹⁸⁰ Through TLRs, DAMPs signal cytokine and chemokine production and upregulate the expression of cell adhesion molecules. When DAMPs interact with NLRs, they stimulate NLRs to complex with apoptosis-associated speck-like proteins to form macromolecular complexes, the inflammasomes that cleave proinflammatory cytokines to their mature forms.¹⁸¹ Thus besides promoting tubulo glomerular disconnection, proximal tubular cell apoptosis contributes to create a local proinflammatory microenvironment.

TUBULE CELLS: ACTIVATION

Receptor-mediated endocytosis of excessive proteins at the apical pole of the proximal tubule cells is also associated with phenotypic changes characteristic of an activated state.

Insights into specific mechanisms linking protein uptake to cell activation have come from in vitro studies using polarized proximal tubule cells to assess the effect of apical exposure to proteins. Collectively, they show that protein overload induces a proinflammatory phenotype.¹⁸²⁻¹⁸⁵ Indeed, upregulation of inflammatory and fibrogenic genes and production of related proteins have been reported following a challenge of proximal tubule cells with plasma proteins. They include cytokines and chemokines, such as monocyte chemoattractant protein-1 (MCP-1), RANTES (regulated upon activation, normal T cell expressed and secreted), interleukin-8 (IL-8), and fractalkine.¹⁸²⁻¹⁸⁵ Moreover, levels of the profibrogenic cytokine TGF- β and its type I receptor,¹⁸⁶ tissue inhibitors of metalloproteinase (TIMP)-1 and TIMP-2, as well as membrane surface expression of the α v β 5 integrin¹⁸⁷ were also highly increased in vitro upon stimulation by plasma proteins.

Investigations of the molecular mechanisms underlying chemokine and growth factor upregulation in proximal tubule cells on protein challenge have focused on the activation of transcription factor NF- κ B.¹⁸² Other studies confirmed this pathway^{188,189} and revealed reactive oxygen as a second messenger.^{184,190}

Extrapolation from such in vitro data to the human situation may be difficult considering the conflicting data observed with different proteins in different cell systems,¹⁹¹ as well as the reported changes in the expression of several genes of unknown function.¹⁹² However, the in vitro observations have recently been confirmed through transcription analysis by cDNA microarray of renal proximal tubule epithelial cells isolated by laser capture microdissection from patients with proteinuric nephropathies.¹⁹³ More than 160 genes—including those encoding for signal transduction, transcription, and translation, and apoptotic and inflammatory proteins—were identified as being regulated differently from those in proximal tubule cells from control subjects.

Evidence implicates megalin as a central element of the signaling pathway linking protein reabsorption and gene regulation in proximal tubule cells.¹⁹⁴ Megalin is subjected to regulated intramembrane proteolysis (RIP), an evolutionarily conserved process linking receptor function with transcriptional regulation.¹⁹⁵ Through RIP megalin is subjected to protein kinase C regulated, metalloprotease-mediated ectodomain shedding producing a membrane-associated

C-terminal fragment (MCTF).¹⁹⁴ The MCTF in turn forms the substrate for γ -secretase, which releases the C-terminal, cytosolic domain. The latter translocates to the nucleus where it interacts with other proteins to regulate expression of specific genes. This function may explain the phenotypic change of proximal tubules in proteinuric kidney disease.

That megalin contributes to the early activation of proximal tubule cells in nonselective proteinuria has been documented in megalin knockout/*NEP25* mice given immunotoxin LMB2, a model for nephrotic syndrome, focal segmental glomerulosclerosis, and tubulointerstitial injury.¹⁹⁶ Megalin-deficient proximal tubule cells reabsorbed less proteins in vivo and expressed less tubular injury markers, such as MCP-1 and heme-oxygenase 1.¹⁹⁶

The proteinuric ultrafiltrate may also activate TLRs and promote an innate inflammatory immune response.¹⁹⁷ Besides being expressed by cells of the immune system such as macrophages, dendritic cells, neutrophils, B cells, and natural killer cells, TLRs are also expressed by nonimmune cells, including renal tubule epithelial cells.¹⁹⁷ The cellular effects of TLRs include the production of proinflammatory cytokines and chemokines that contribute to local inflammation and leukocyte accumulation. It has been demonstrated that proximal tubule epithelial cells were sites of robust expression of TLR-9 mRNA and protein—a receptor for CpG DNA—in NZBxNZW lupus mice with overt nephropathy.¹⁹⁸ Upregulation of TLR-9 expression was accompanied by the development of proteinuria and correlated with tubulointerstitial damage. Furthermore, abundant TLR-9 staining in proximal tubule cells of lupus patients correlated with tubulointerstitial damage. Thus tubular TLR activation may occur due to filtration of plasma proteins, which include immune complexes containing DNA enriched in CG motifs.^{199,200}

The proximal tubule bears other receptors for ultrafiltered proteins, such as cytokines and growth factors.⁴ Usually these molecules are present in high-molecular-weight precursor forms or bound to specific binding proteins that regulate their biological activity. They can be found in nephrotic tubular fluid. In experimental proteinuria in rats, there is translocation of insulin-like growth factor I (IGF-I) from plasma into tubular fluid (primarily as the 50-kDa complex).²⁰¹ Similarly, hepatocyte growth factor (HGF) is present in early proximal tubular fluid from rats with streptozotocin-induced diabetic nephropathy, and it is excreted in the urine of diabetic animals.²⁰² Under physiologic conditions the high molecular weight of TGF- β complexes prevents glomerular ultrafiltration of this pluripotent cytokine. However, in proteinuric glomerular diseases, TGF- β is present in early proximal tubular fluid and at least a portion is bioactive.²⁰² IGF-I, HGF and TGF- β are also present in the urine of patients with proteinuric diseases.²⁰³

Collectively the tubular response to these growth factors can be described as activation or as a moderate change toward a cell phenotype resembling cell injury, which includes a moderate increase in collagen type I and IV production in response to IGF-I²⁰¹ and upregulation of the expression of fibronectin by HGF.²⁰⁴ TGF- β also increases the transcription in proximal tubule cells of the genes encoding collagen α_1 III (Col3A1) and collagen α_2 I (Col1A2) as well as fibronectin.

With proteinuria, a putative key factor in tubule cell activation and damage is the excess glomerular filtration of serum-derived complement C3, the central molecule in the

complement system that exerts proinflammatory potential.²⁰⁵ Renal tubule epithelial cells appear most susceptible to luminal attack by the C5b-9 membrane attack complex because of the relative lack of membrane bound complement regulatory proteins, such as membrane cofactor protein (CD46), decay-accelerating factor, or CD55 and CD59 on the apical surface.²⁰⁶ In rats with severely reduced renal mass^{207,208} or with protein overload proteinuria,¹⁴⁸ C3 colocalized with proximal tubule cells engaged in high protein uptake. By limiting the transglomerular passage of proteins, treatment with angiotensin-converting enzyme (ACE) inhibitors was an effective maneuver to reduce C3 load of tubule cells in remnant kidneys.²⁰⁷ C3 and other complement proteins are also found in proximal tubules in renal biopsy material from nephrotic patients.^{205,209} Furthermore, proximal tubule cells are able to synthesize C3 and other complement factors²¹⁰ and to upregulate C3 in response to serum proteins in vitro.²¹¹

The injurious role of plasma-derived C3, as opposed to tubular cell-derived C3, has been documented in C3-deficient kidneys transplanted into wild-type mice.¹⁴⁸ Protein overload led to the development of glomerular injury, accumulation of C3 in proximal tubules, and tubulointerstitial changes. Conversely, when wild-type kidneys were transplanted into C3-deficient mice, protein overload led to a milder disease and abnormal C3 deposition was not observed. Thus ultrafiltered C3 contributes more to tubulointerstitial damage induced by protein overload than locally synthesized C3.

INTERSTITIAL INFLAMMATION AND INJURY

In proteinuric kidney disease, progressive inflammation and injury to the renal interstitium are secondary events following glomerular or vascular injury. Tubule epithelial cells synthesize cytokines and chemokines and accumulate complement components that recruit inflammatory cells and lymphocytes into the interstitium causing progressive fibrosis.

RESIDENT MONOCYTE/DENDRITIC CELLS

The interstitium of normal kidneys contains numerous resident monocytic myelocytes,²¹² which express dendritic cell (DC) markers and can indeed present antigens.²¹² DCs have recently been described to form an immune sentinel network through the entire kidney, where they probe the environment in search of antigens.²¹³ An inflammatory environment converts the tolerogenic status of resident DCs into an immunogenic one, favoring recruitment of T cells. It is known that cross-presentation by DCs is a major mechanism for the immune surveillance of tissue against foreign antigens.²¹⁴ In this process professional antigen-presenting cells, such as DCs, acquire proteins from other tissue cells through endocytic mechanisms, especially phagocytosis or macropinocytosis. The internalized antigen can then be processed and presented on MHC class I molecules to the extracellular environment.²¹⁵ The outcome of cross-presentation regarding immunity depends on the expression of immunostimulatory signals after the uptake of the antigen.²¹⁴

Until recently, the role of resident DCs that accumulate in the renal parenchyma of nonimmune-mediated proteinuric nephropathies remained poorly understood. Recent studies, however, have provided new insights into the activation of DCs in the setting of proteinuria. Administration of ovalbumin—which is freely filtered by the glomerulus—to

normal mice leads to concentration of the protein principally in proximal tubules and to its transfer to DCs in the kidney and renal lymph nodes.²¹⁶ Here, ovalbumin is presented to CD8⁺ T cells, thereby inducing proliferation of these cells.

The importance of kidney DC activation to renal injury has been recently demonstrated by the fact that in transgenic NOH mice (that selectively express the model antigens ovalbumin and hen egg lysozyme in podocytes) DC depletion resolved established periglomerular mononuclear infiltrates.²¹⁷ In vitro experiments have also shown that exposure of rat proximal tubule cells to excess autologous albumin, as in the case of proteinuric nephropathies, results in the formation of the N-terminal 24-residue fragment of albumin (ALB₁₋₂₄).²¹⁸ This peptide is taken up by DCs, where it is further processed by proteasomes into antigen peptides. These peptides were shown to have the binding motif for MHC class I and to be capable of activating CD8⁺ T cells. Moreover, in vivo, in the rat 5/6 nephrectomy model, accumulation of DCs in the renal parenchyma peaked 1 week after surgery and decreased thereafter, concomitant with their appearance in the renal draining lymph nodes. DCs from renal lymph nodes loaded with the albumin peptide ALB₁₋₂₄ activated syngeneic CD8⁺ T cells in primary culture.²¹⁸ Thus inflammatory stimuli released from damaged tubules after protein overload may represent danger signals that, in the presence of albumin peptides, alert DCs to promote local immunity via CD8⁺ T cells that are activated in regional lymph nodes and recruited in the renal interstitium.

MACROPHAGES AND LYMPHOCYTES

The interstitial infiltrate of most human chronic renal diseases consists of a number of different effector cells, including macrophages, CD4⁺ and CD8⁺ T cells.²¹⁹ In animal models, macrophages are the dominant infiltrating cells both in the early and in later stages of chronic renal injury. More specifically, tubulointerstitial macrophage accumulation in chronic nephropathies correlates with the severity of the glomerular and interstitial lesions and the degree of renal dysfunction.²¹⁹ Direct damage to resident cells is caused through the generation by macrophages of reactive oxygen species (ROS), nitric oxide (NO), complement factors, and proinflammatory cytokines.²²⁰ Macrophages can also affect the supporting matrix and vasculature through the expression of metalloproteinases and vasoactive peptides.

Macrophages are only one component of the cellular infiltrate that characterizes inflammation in the renal interstitium. Recent models of overload proteinuria have emphasized the importance of tubulointerstitial infiltration with mononuclear cells. Indeed, T-helper cells and cytotoxic T cells, as well as macrophages, are observed in the tubulointerstitial infiltrate 2 weeks after protein overload.²²¹ T-cell depletion with intraperitoneal anti-T cell monoclonal antibody administration did not modify macrophage infiltration, indicating that the influx of these cells was independent of lymphocytes,²²¹ and more likely resulted from local tubule cell expression of osteopontin, MCP-1 and the adhesion molecules VCAM and ICAM.

T lymphocytes are also abundantly present in the tubulointerstitial infiltrate early after renal mass ablation in rats and remain there in significant numbers for the following weeks.²²² Although the infiltration of macrophages is part of a nonspecific inflammatory reaction, the presence of

lymphocytes within lesions indicates that their recruitment and activation is mediated by an antigen-specific immune response. Their role is to maintain and amplify the inflammatory response in the renal interstitium.

Because B cells are considered to be important mostly in lymph nodes, spleen, and in humoral immune responses, little attention has been paid to their potential role as intrarenal infiltrating cells.²²³ However, a prominent accumulation of CD20⁺ B cells has been described in membranous nephropathy.²²⁴ Furthermore, CD20⁺ B cells formed a prominent part of the infiltrating cells in renal biopsies from patients with IgA nephropathy and chronic interstitial nephritis.²²⁵ Together with CD3⁺ T cells, CD20⁺ B cells formed large nodular structures, like tertiary lymphatic organs in inflamed tissues.²²⁶

The level of mRNA expression of the chemokine CXCL13 was increased and correlated with CD20⁺ mRNA in the tubulointerstitial space. The localization of enhanced CXCL13 immunoreactivity to the nodular infiltrates and that of the corresponding receptor CXCR5 to B cells in the infiltrates point toward a role of CXCL13-CXCR5 for B-cell recruitment into the lymphoid follicle-like structures. In the interstitium, B cells may release proinflammatory cytokines and chemokines, present antigens, and activate T cells, as well as play a role in the development of tissue fibrosis.²²⁶

BONE MARROW-DERIVED FIBROCYTES

In proteinuric renal disease, chemokines generated in the inflammatory milieu may contribute to the recruitment of bone marrow-derived fibrocytes to the renal interstitium.²²⁷ Fibrocytes are circulating connective tissue cell progenitors with a high capacity for collagen I synthesis. In progressive kidney fibrosis induced by unilateral ureteral obstruction in mice, fibrocytes infiltrated the interstitium, and the number of these cells increased with the progression of fibrosis.²²⁸ In addition, the number of infiltrating fibrocytes correlates well with the extent of interstitial fibrosis in several human kidney diseases.²²⁷ Although fibrocytes isolated from mice and humans express chemokine receptors including CCR2, CCR3, CCR5, CCR7, and CXCR4,²²⁷ the specific chemokine and receptor pair involved into the recruitment of these cells in the damaged tubulointerstitium remains uncertain.

FIBROBLASTS: ACTIVATION AND DEPOSITION OF EXTRACELLULAR MATRIX

The process of tubulointerstitial fibrosis involves the loss of renal tubules and the accumulation of myofibroblasts and extracellular matrix (ECM) proteins.²²⁹ Resident interstitial fibroblasts and myofibroblasts proliferate in response to macrophage-derived profibrogenic cytokines, and their number correlates with the subsequent formation of a scar.²³⁰ These cells may be derived from transdifferentiated tubular epithelial cells or pericytes of peritubular capillaries, a process promoted by profibrogenic cytokines, including TGF- β expressed by macrophages.^{231,232}

During the developmental stage embryonic epithelia of different organs, including the collecting duct epithelium of the kidney, may give rise to mesenchymal cells, a process known as epithelial to mesenchymal transition (EMT).²³³ EMT has been suggested as a process that contributes to interstitial fibrosis in chronic kidney diseases by transformation of injured renal tubule cells into mesenchymal cells.²³³

However, the evidence for EMT in adult kidneys and chronic renal disease is controversial and there are no solid data supporting EMT as an *in vivo* process in kidney fibrosis. The most supportive data have come from a study in a model of unilateral ureteral obstruction.²³⁴ With the use of genetically tagged proximal tubule epithelial cells, it has been demonstrated that up to 36% of all matrix-producing cells within the tubulointerstitial space may be of tubular origin.²³⁴ However, the contribution of EMT to the formation of myofibroblasts is less in other models.^{235,236} Moreover, using cell fate–tracing techniques in the unilateral ureteral obstruction model, other investigators did not find any evidence for a contribution of EMT to renal fibrosis.²³⁷ This is in line with the finding that in the remnant kidney model in rats, after the onset of proteinuria α -smooth muscle actinin, a marker of transdifferentiation to myofibroblasts was initially expressed by nonepithelial cells in the peritubular compartment. Recently peritubular pericytes have been identified as the source of myofibroblasts in this transdifferentiation process.²³⁷

Activated renal fibroblasts may secrete chemokines which, in turn, may further attract macrophages and perpetuate tubulointerstitial injury.¹⁹⁷ Eventually activated fibroblasts produce interstitial matrix components contributing to interstitial collagen deposition and fibrosis. Increased tubulointerstitial fibrosis is a common feature of kidney injury and results from accumulation of ECM structural proteins. It is maintained by continuous remodeling through the proteolytic action of matrix metalloproteinases (MMPs) and the synthesis of new proteins.

MMPs are inhibited by tissue inhibitors of matrix metalloproteinases (TIMPs). Therefore the balance between TIMPs and MMPs determines the ECM integrity. Among the four members of the TIMP family, TIMP3 is unique in that it is ECM bound and is highly expressed in the kidney.²³⁸ TIMP3^{-/-} mice had more interstitial fibrosis, increased synthesis and deposition of type I collagen, increased activation of fibroblasts, and greater activation of MMP2 after unilateral obstruction than wild-type mice.²³⁹ TIMP3 levels are upregulated in patients with diabetic and chronic allograft nephropathy.²³⁹

Recent studies link fibrosis to changes in miRNAs.^{96,240,241} A number of miRNAs have been shown to be relevant to fibrotic processes in diabetic nephropathy, including miR-29 and miR-200 families, miR-192, and miR-21.²⁴¹⁻²⁴⁴ These miRNAs are regulated by TGF- β in renal cells, and normalization of their expression ameliorated fibrosis in *in vitro* and *in vivo* models of diabetes.²⁴³

More recently, miR-184 has been shown to be a downstream effector of albuminuria driving renal fibrosis in rats with diabetic nephropathy.²⁴⁵ Indeed, in Zucker diabetic fatty (ZDF) rats, miR-184 showed the strongest differential upregulation compared with lean rats (18-fold). Tubular localization of miR-184 was associated with reduced expression of lipid phosphate phosphatase 3 (LPP3) and collagen accumulation. Transfection of NRK-52E cells with miR-184 mimicked reduced LPP3, promoting a profibrotic phenotype. Albumin was a major trigger for miR-184 expression. Interestingly, anti-miR-184 counteracted albumin-induced LPP3 downregulation and overexpression of plasminogen activator inhibitor-1. In ZDF rats, ACE inhibitor treatment limited albuminuria and reduced miR-184, with tubular LPP3

preservation and tubulointerstitial fibrosis amelioration. Albumin-induced miR-184 expression in tubule cells was epigenetically regulated through DNA demethylation and histone lysine acetylation and was accompanied by binding to NF- κ B p65 subunit to miR-184 promoter. These results suggest that miR-184 may act as a downstream effector of albuminuria through LPP3 to promote tubulointerstitial fibrosis and offer the rationale to investigate whether targeting miR-184 in association with albuminuria-lowering drugs may be a new strategy to achieve fully antifibrotic effects, at least in diabetic nephropathy.²⁴⁵

CHRONIC HYPOXIA

One of the most important contributors to the development of tubulointerstitial fibrosis is chronic ischemia.²⁴⁶ Production of Ang II and inhibition of production of NO underlie chronic vasoconstriction, which may contribute to tissue ischemia and hypoxia.²⁴⁷ In that regard, histologic studies on biopsies from animal models and human kidneys have documented that there is often a loss of peritubular capillaries in areas of tubulointerstitial fibrosis.²³³ Downregulation of VEGF may be functionally implicated in the progressive attrition of peritubular capillaries and tissue hypoxia, as shown in mouse folic acid nephropathy.²⁴⁸

Pericytes play a critical role in the stabilization and proliferation of peritubular capillaries via interaction with endothelial cells.²⁴⁹⁻²⁵¹ This process is mediated by several angioregulatory factors, including angiopoietin-1, produced by pericytes, and angiopoietin-2, produced by activated endothelial cells.²⁵¹⁻²⁵³ Renal ischemia, as it occurs in chronic kidney disease due to microvascular rarefaction,²⁵⁴ promotes an imbalance in angiopoietins that besides leading to proliferation of pericytes may induce interstitial fibrosis in the long term.²⁵⁵

Moreover, given that the size of the interstitial compartment determines the diffusion distance between peritubular capillaries and tubule cells, interstitial fibrosis further impairs tubular oxygen supply. Focal reduction of capillary blood flow leading to starvation of tubules may underlie tubular atrophy and loss. Under these conditions, the remaining tubules are subjected to functional hypermetabolism with increased oxygen consumption, which in turn creates an even more severely hypoxic environment in the renal interstitium. *In vitro*, such hypoxia stimulates fibroblast proliferation and ECM production by tubular epithelial cells.²⁵⁶

ENDOGENOUS SYSTEMS OF TISSUE REPAIR

PROTECTIVE MACROPHAGES

Much remains to be learned about macrophages in tubulointerstitial injury. The role of interstitial macrophages was elucidated in mice with progressive adriamycin-induced nephropathy.²⁵⁷ By treating mice with the monoclonal antibody ED7 directed against the CD11b/CD18 integrin, which is expressed by macrophages, renal cortical macrophages (ED1-positive cells) were reduced by almost 50%, whether ED7 was administered before or after adriamycin administration.²⁵⁸ However, ED7 reduced renal structural and functional injury only when treatment was started prior to adriamycin administration.²⁵⁸

Among several possible explanations for these observations is a temporal change in the predominant macrophage phenotype. If pathogenic macrophages predominated early

and protective macrophages later in the course of the disease, then only early antimacrophage treatment would be expected to protect against progression. Indeed, macrophages can exhibit distinctly different functional phenotypes and can be polarized toward proinflammatory (M1 macrophages) or tissue-reparative (M2 macrophages) phenotype.²⁵⁹ In the peritubular interstitium, macrophages have been shown to mediate tissue repair in response to acute kidney injury by adopting an M2 phenotype and producing a cytokine environment that supports tubular repair and proliferation rather than inflammation.²⁶⁰ Colony-stimulating factor-1 (CSF-1) signaling mediated M2 macrophage-induced recovery from renal injury, because pharmacologic blockade of CSF-1 decreased M2 polarization and eventually inhibited tissue repair.²⁶¹

Recent observations also support the importance of macrophage phenotype. For example, in mice with unilateral ureteric obstruction reconstituted with bone marrow of Ang II subtype 1 receptor gene knockout or wild-type mice, infiltrating macrophages were shown to play a beneficial antifibrotic role.²⁶²

Other studies have demonstrated marked macrophage heterogeneity and context specificity, depending on the nature of the injury and location within the kidney.²⁶³ Evidence is available that macrophages perform both injury-inducing and repair-promoting tasks in different models of inflammation. This has been shown in a reversible model of liver injury, in which the injury and recovery phase are distinct.²⁶⁴ Macrophage depletion when liver fibrosis was advanced resulted in reduced scarring and fewer myofibroblasts. Macrophage depletion during recovery, by contrast, led to a failure of matrix degradation.²⁶⁴ These findings provide clear evidence that a functionally distinct subpopulation of macrophages exists in the same tissue.

Further studies on possible temporal variations in the phenotype, activation status, and net effect on injury of macrophages should give a better understanding of the complex role of macrophages in tubulointerstitial injury and repair of chronic renal disease, particularly in the proteinuric setting.

REGULATORY T CELLS

CD4⁺ T cells constitute a critical component of the adaptive immune system and are typified by their capacity to help both humoral and cell-mediated responses. However, there is a substantial functional diversity among CD4⁺ T cells, and certain subpopulations hinder rather than help immune response. The most well-characterized example of an inhibitory subpopulation is the CD4⁺CD25⁺, which appears to play an active role in downregulating pathogenic autoimmune responses.²⁶⁵ CD4⁺CD25⁺ T cells are potent immunoregulatory cells that suppress T-cell proliferation in vitro and have the capacity to suppress immune responses to auto- and alloantigens, tumor antigens, and infectious antigens in vivo.²⁶⁶

The regulatory activity of these cells in the setting of chronic renal diseases is highlighted by studies in severe combined immunodeficient (SCID) mice reconstituted with CD4⁺CD25⁺ T cells after induction of adriamycin nephrosis.²⁵⁸ Mice reconstituted with these regulatory cells had significantly reduced glomerulosclerosis, tubular injury, and interstitial expansion compared with unreconstituted mice with adriamycin-induced nephrosis.

A study using the green fluorescence protein (GFP)-*Foxp3* mouse suggests that *Foxp3* expression identifies the regulatory T-cell population.²⁶⁷ In the murine model of adriamycin nephropathy, the adoptive transfer of *Foxp3*-transduced T cells protected against renal injury. Urinary protein excretion and serum creatinine were reduced, and there was significantly less glomerulosclerosis, tubular damage, and interstitial infiltrates.²⁶⁸

KIDNEY-DERIVED PROGENITOR CELLS

In chronic proteinuric renal disease, regression of glomerular structural changes is associated with remodeling of the glomerular architecture.²⁶⁹ Instrumental to this discovery were three-dimensional reconstruction studies of the glomerular capillary tuft, which allowed the quantification of sclerosis volume reduction and of capillary regeneration upon treatment.²⁶⁹ The reversal of early glomerular damage in animal models and humans²⁷⁰ argues for the existence of a regenerative mechanism that promotes glomerular repair. However, mature podocytes are postmitotic cells with limited capacity to divide in situ and therefore unable to regenerate.²⁷⁰ A potential mechanism for podocyte replacement by bone marrow-derived stem cells has been described in the Alport mouse model as well as in kidney transplants.^{271,272} Nevertheless, most studies concluded that regeneration occurs predominantly from resident renal progenitors,^{56,67} although the source of these cells remains ill-defined.

Recently, a study using a triple-transgenic mouse model that allowed permanent marking of glomerular parietal cells and their progeny upon administration of doxycycline showed that parietal epithelial cells of the Bowman capsule possess the capability to migrate into the glomerular tuft via the vascular stalk, where they differentiate into podocytes.²⁷³ Similarly, in the adult human kidney, cells localized between the urinary pole and vascular pole of the Bowman capsule—that expressed both progenitor and podocyte markers (CD24⁺CD133⁺PDX⁺ cells)—can differentiate into podocytes by losing stem cell markers and expressing markers indicative of a podocyte phenotype while progressing from the urinary pole to the surface of the glomerular tuft.²⁷⁴

Experimental evidence indicates that intravenous injection of human progenitor cells harvested from the Bowman capsule into SCID mice with adriamycin-induced nephropathy reduced proteinuria and mitigated chronic glomerular damage.²⁷⁴ Even more intriguing from the clinical perspective is the finding that ACE inhibition induces glomerular repair in MWF rats, a model of spontaneous glomerular injury.²⁷⁵ In these proteinuric animals, besides halting age-related podocyte loss, lisinopril increased the number of glomerular podocytes above baseline, which was associated with an increased number of proliferating WT1-positive cells, loss of cycling-dependent kinase inhibitor p27 expression, and increased number of parietal podocytes. This indicates that remodeling of the Bowman capsule epithelial cells contributes to the ACE inhibitor-induced restructuring of the damaged glomerular capillary, primarily by restoring the podocyte population.

Similarly, glomerular repair is augmented when glucocorticoid treatment is given to mice with experimental FSGS at a time when podocyte number was already decreased.²⁷⁶ Prednisone increased podocyte number, which correlated with reduced proteinuria and decreased glomerulosclerosis. This could be the result of direct biological effects on both

glomerular epithelial cells, by reducing podocyte apoptosis, and by enhancing podocyte regeneration via increasing the number of parietal epithelial cell precursors.

In addition to ACE inhibitors^{144,150} and prednisone,²⁷⁶ Notch inhibitors,²⁷⁷ blockers of chemokine stromal-derived factor-1²⁷⁸ and retinoids,²⁷⁹ can be added to the list of agents that improve podocyte regeneration by augmenting the number of parietal epithelial cells progenitors. Indeed, *in vitro* exposure of human renal progenitor cells to human serum albumin inhibited their differentiation into podocytes by sequestering retinoic acid and preventing retinoic acid–response element (RARE)-mediated transcription of podocyte-specific genes.¹⁴⁵ Similarly, in mice with adriamycin-induced nephropathy *in vivo*, a model of human FSGS, blocking endogenous retinoic acid synthesis, increased proteinuria and exacerbated glomerulosclerosis.¹⁴⁵ This effect was related to a reduction in podocyte number. In RARE-lacZ transgenic mice, albuminuria reduced retinoic acid bioavailability and impaired RARE activation in renal progenitors, inhibiting their differentiation into podocytes.¹⁴⁵ Treatment with retinoic acid restored RARE activity and induced the expression of podocyte markers in renal progenitors, decreasing proteinuria and increasing podocyte number, as demonstrated in serial biopsy specimens.¹⁴⁵

Together these experimental studies suggest that restoring the capacity of parietal epithelial progenitor cells to differentiate into podocytes could promote the regeneration of podocytes and potentially result in the regression of glomerular disease.

SYSTEMIC CONSEQUENCES OF NEPHROTIC-RANGE PROTEINURIA

Nephrotic-range proteinuria is accompanied by a cluster of abnormalities known collectively as the nephrotic syndrome. It is characterized by systemic complications that result from profound alterations in the composition of the body protein pool, a state of sodium retention, dyslipidemia, abnormalities of coagulation factors, and a variable degree of renal insufficiency.

HYPOALBUMINEMIA

Clinical manifestations of the nephrotic syndrome become evident in patients with levels of proteinuria in excess of 3.5 g/day. However, proteinuria in overtly nephrotic subjects usually exceeds this lower bound by a factor of 2 to 3. Immunochemical analysis shows that albumin accounts for more than 80% of the excreted proteins.²⁸⁰ The second most copiously excreted protein is immunoglobulin, which, after albumin, is the next most abundant protein in plasma. One of the most common systemic abnormalities associated with nephrotic proteinuria is hypoalbuminemia, which develops in most patients.

PATHOGENESIS OF HYPOALBUMINEMIA

Under normal conditions, albumin production by the liver is 12–14 g/day (130–200 mg/kg). Production equals the amount catabolized, predominantly in extrarenal locations.²⁸¹ However, about 10%, is catabolized in the proximal tubule of the kidney after reabsorption of filtered albumin.²⁸¹ In patients with the nephrotic syndrome, hypoalbuminemia

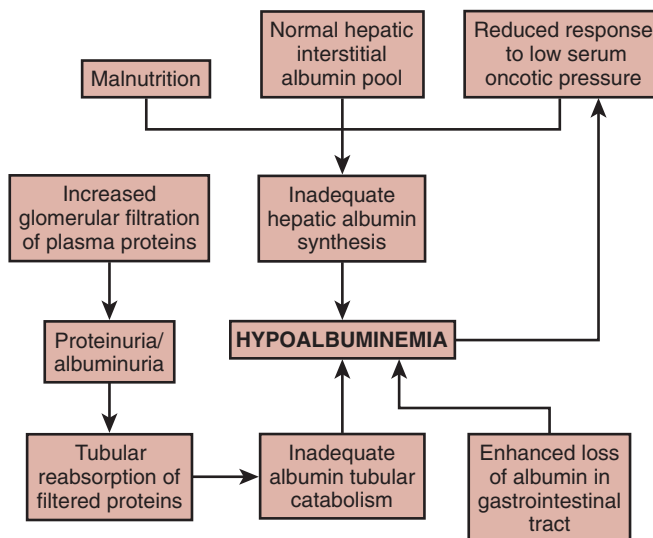


Fig. 30.10 Schematic representation of mechanisms leading to nephrotic hypoalbuminemia. Compensatory mechanisms, such as increase in albumin synthesis and decrease in albumin catabolism, are insufficient to correct the hypoalbuminemia.

results from excessive urinary loss, decreased hepatic synthesis, and increased rates of albumin catabolism (Fig. 30.10).

Urinary albumin loss is an important contributor to the development of hypoalbuminemia. However, it is not a sufficient cause in most patients with the nephrotic syndrome, as the rate of hepatic albumin synthesis can increase by at least threefold, thereby compensating for urinary albumin loss.²⁸¹ Enhanced loss of albumin in the gastrointestinal tract has also been proposed to contribute to hypoalbuminemia, but there is little evidence for this hypothesis.²⁸² Therefore for hypoalbuminemia to develop there must be either an insufficient increase in hepatic synthetic rate or an increase in albumin catabolism.

Normally the rate of hepatic albumin synthesis may increase by as much as 300%. However, studies of the nephrotic syndrome in animal models and in humans with hypoalbuminemia demonstrate that the rate of albumin synthesis is at or only slightly above the upper limit of normal as long as dietary protein is adequate.²⁸³ This indicates an inadequate synthetic response to hypoalbuminemia by the liver.

Oncotic pressure of the plasma perfusing the liver is one major regulator of protein synthesis.²¹⁵ Experimental evidence in rats that are genetically deficient in circulating albumin showed a twofold increase in the hepatic transcription rate of the albumin gene compared with normal rats.²¹⁵ However, in these rats the increase in hepatic albumin synthesis was inadequate to compensate for the degree of hypoalbuminemia, which indicates an impaired synthetic response.²¹⁵ Similarly, in nephrotic patients, reduced oncotic pressure is unable to enhance the albumin synthetic rate of the liver to the extent required to restore plasma albumin concentration.²⁸³ There is also evidence in normal subjects that the hepatic interstitial albumin regulates albumin synthesis.²⁸⁴ Because the hepatic interstitial albumin pool is not depleted in the nephrotic syndrome, the albumin synthetic response is normal or slightly increased, but it remains inadequate relative to the level of hypoalbuminemia.²⁸⁴

Dietary protein-intake further contributes to the synthesis of albumin. Hepatic albumin mRNA and albumin synthesis was not increased in nephrotic rats when fed a low-protein diet but increased with a high-protein diet.²⁸⁵ However, serum albumin levels were not altered because hyperfiltration resulting from increased protein intake led to increased albuminuria.

The contribution of renal albumin catabolism to hypoalbuminemia in the nephrotic syndrome is controversial. Some have argued that the renal tubular albumin transport capacity is already saturated at physiologic levels of filtered albumin and that any increase in filtered protein, instead of being absorbed and catabolized, is simply excreted in the urine.²⁸⁶ Studies in isolated perfused proximal tubules in rabbits, however, demonstrated a dual transport system for albumin uptake.²⁸⁷ In addition to a low-capacity system that became saturated when the protein load exceeded physiologic levels, a high-capacity low-affinity system was also present and allowed the tubular absorptive rate for albumin to increase as the filtered load rose. Thus an increase in the fractional catabolic rate may occur in the nephrotic syndrome.

This hypothesis is supported by the positive correlation between fractional albumin catabolism and albuminuria in rats with puromycin aminonucleoside-induced nephrosis.²⁸⁸ Nevertheless, because total body albumin stores are substantially decreased in the nephrotic syndrome, absolute catabolic rates may be normal or even reduced.²⁸² This outcome is affected by nutritional state, as documented by the fact that in nephrotic rats nourished with a low-protein diet, absolute albumin catabolism was reduced but not in those with normal dietary protein intake.²⁸⁹

In summary, hypoalbuminemia in the nephrotic syndrome results from multiple alterations in albumin homeostasis that are not sufficiently compensated for by hepatic albumin synthesis and by decreased renal tubular albumin catabolism.

CONSEQUENCES OF HYPOALBUMINEMIA

Impairment of kidney function is the rule in patients with nephrotic hypoalbuminemia and it usually manifests in two ways. One is the inability of the kidney to maintain sodium and fluid homeostasis. The other is loss of intrinsic ultrafiltration capacity of glomerular capillary walls, a phenomenon that leads, in turn, to fall in GFR.²⁹⁰

When viewed in physiologic terms, the GFR can be defined as the net rate of water flux across the walls of the capillaries in the glomerular tufts of the kidney. It is determined by the product of the net pressure for ultrafiltration and the ultrafiltration coefficient— K_f —a measure of intrinsic ultrafiltration capacity, derived from the product of the available filtering surface area(s) and the hydraulic permeability of the glomerular capillary wall (k). By estimating GFR and its determinants in humans, it has been shown that a reduced GFR in some forms of the nephrotic syndrome (minimal change and membranous nephropathy) is exclusively a consequence of profoundly lowered hydraulic permeability.²⁹¹ In the nephrotic syndrome associated with lupus nephritis, idiopathic focal and segmental glomerulosclerosis, and diabetic nephropathy, both reduction of the surface area available for filtration and impaired hydraulic permeability contribute to K_f depression.^{292, 293}

The principal cause of impaired hydraulic permeability in nephrotic disorders is broadening and effacement of

epithelial foot processes.²⁹¹ This lowers the frequency of interpodocyte slit diaphragms through which water must pass to gain access to the Bowman space, thereby increasing the resistance to water flow. The low K_f is partially offset by an increase in net ultrafiltration pressure, which is largely due to a substantial lowering of the intraglomerular capillary oncotic pressure. As a result, the fall in GFR is not proportional to the decrease in K_f . This compensatory elevation in net ultrafiltration pressure explains why reduced values of single-nephron GFR are not consistently observed in all experimental nephrotic models.²⁹⁴

The low ultrafiltration capacity induced by glomerular disease and protein depletion makes the nephrotic patient particularly vulnerable to acute exacerbations of hypofiltration and renal insufficiency.²⁹⁵ As the prevailing level of GFR depends heavily on ultrafiltration pressure in the presence of a low K_f , any maneuver lowering the glomerular capillary perfusion pressure can, therefore, cause a precipitous fall in the GFR. The susceptibility of nephrotic patients to episodes of acute kidney injury should thus be borne in mind when prescribing drugs, such as diuretics, cyclooxygenase inhibitors, and cyclosporine, that can compromise the ultrafiltration pressure.

An additional consequence of hypoalbuminemia is the potential for enhanced drug toxicity.²⁹⁶ Indeed, many drugs are bound to albumin. Hypoalbuminemia reduces the number of available binding sites and increases the proportion of circulating free drugs, but in a steady state this is counterbalanced by faster metabolism. Furthermore, because protein binding may enhance tubule drug secretion, diminished protein binding in the nephrotic syndrome may delay renal excretion of some drugs.²⁹⁷ Although the clinical consequences of altered protein binding may be difficult to predict, higher levels of free drugs may be toxic, as shown with prednisolone.²⁹⁸

The case of diuretics is intriguing. Resistance to loop diuretics, which often occurs in patients with the nephrotic syndrome, may be due to reduced delivery of the diuretic to its site of action, secondary to hypoalbuminemia. Anecdotal reports suggest that the administration of furosemide with small amounts of albumin (6–20 g) can enhance the response to furosemide in nephrotic patients.²⁹⁹ These observations are not conclusive, because others did not show a difference in the excretion of intravenous furosemide in the urine of nephrotic patients compared with normal controls.³⁰⁰ On the other hand, excessive amounts of filtered albumin in the tubule may bind furosemide and make it less effective.³⁰¹

In animals, the inhibition of fractional loop Cl^- reabsorption by furosemide is blunted by the presence of albumin in the proximal tubule, and prevention of albumin–furosemide binding with warfarin and sulfisoxazole partially restored the response to the diuretic.³⁰¹ However, these findings were not confirmed in nephrotic patients given sulfisoxazole, raising doubt about the importance of excessive albumin-bound furosemide at the active tubular site in resistance to diuretics.³⁰² Sodium-retaining mechanisms, such as low effective arterial blood volume and activation of neurohumoral factors, may be relatively more important.

Many binding proteins are lost in the urine in the nephrotic syndrome.³⁰³ Consequently, in patients with the nephrotic syndrome, the plasma levels of many ions (iron, copper, and zinc), vitamins (vitamin D metabolites), and hormones

(thyroid and steroid hormones) are low because the level of protein-bound ligands is reduced. Urinary loss of protein-bound ligands can theoretically cause depletion, but there is little convincing clinical evidence for this, with the possible exception of vitamin D.³⁰⁴ Indeed, one of the proteins lost in the urine of patients with the nephrotic syndrome is cholecalciferol binding globulin (also known as vitamin D binding protein DBP), which is a 59-kDa protein easily filtered by nephrotic glomeruli.³⁰⁵ Because 25-hydroxycholecalciferol (25-[OH]D₃) circulates as a complex with DBP, there is also an associated urinary loss of 25-(OH)D₃ in the nephrotic syndrome.³⁰⁶ The oral administration of ³H-labeled cholecalciferol indicated that the serum half-life of 25-(OH)D₃ was reduced and urinary excretion increased in the nephrotic syndrome.³⁰⁷ Nevertheless, in general, nephrotic patients have normal to decreased plasma levels of 1,25-dihydroxycholecalciferol (1,25-[OH]₂D₃).³⁰⁶

Although the hypocalcemia of the nephrotic syndrome was once attributed solely to the reduction in protein-bound calcium secondary to hypoalbuminemia, a subset of patients has been noted with hypocalcemia out of proportion to the hypoalbuminemia. In these patients ionized serum calcium is decreased.³⁰⁸ Secondary hyperparathyroidism is seen in some patients, even in the absence of renal failure, as are changes in bone histology consistent with mixed osteomalacia and osteitis fibrosa cystica bone disease.³⁰⁹ Not all investigators, however, have observed abnormalities in calcium homeostasis in the nephrotic syndrome.³¹⁰ Why only a subset of patients is predisposed to alterations in calcium, vitamin D, and parathyroid hormone homeostasis has not been determined, but it has been suggested that factors such as age, duration of disease, renal function, degree of proteinuria, serum albumin concentration, and corticosteroid therapy might be involved.³⁰⁴

Finally, hypoalbuminemia may play a role in platelet hyperaggregability.³¹¹ Because albumin normally binds arachidonic acid, thus limiting its conversion to thromboxane A₂ by platelets, hypoalbuminemia might allow increased platelet arachidonate metabolism to take place, and platelet hyperreactivity may result.³¹¹

EDEMA FORMATION

The clinical manifestation that most frequently brings the nephrotic patient to medical attention is the formation of edema. This represents an increase in the size of the interstitial fluid compartment. The interstitial fluid accumulates most readily in dependent areas where tissue pressure is low. It thus manifests as periorbital edema upon awakening in the morning, and pedal edema at the end of the day. Even when edema is generalized and massive, a condition referred to as anasarca, it remains most marked in the lower extremities. Not infrequently, anasarca is also accompanied by large effusions into the peritoneal, pleural, and pericardial spaces. The mechanisms responsible for extravascular fluid accumulation in nephrotic patients are complex and only partially understood.

REDUCED PLASMA ONCOTIC PRESSURE

Low colloid oncotic pressure as a result of hypoalbuminemia favors the movement of water from the intravascular to the interstitial space. Under normal conditions, edema formation

is halted by expansion and proliferation of lymphatics that increase lymphatic flow, and by reduction of interstitial oncotic pressure due to protein-free fluid accumulation. In addition, the increased hydraulic pressure in the interstitium because of fluid accumulation lowers the transcapillary pressure gradient, further reducing the transudation of plasma fluid into the interstitial space. However, there is no clear evidence of alterations in these normal defense mechanisms against edema formation in nephrotic patients.³¹² For example, comparable changes in interstitial and plasma colloid oncotic pressure have been documented during relapse and remission phases in patients with nephrotic syndrome.³¹² Moreover, the capillary hydraulic conductivity is elevated in nephrotic patients,³¹³ possibly due to disruption of the intercellular macromolecular complex between endothelial cells, which enhances capillary filtration capacity and may lead to sustained edema formation.³¹⁴ These observations suggest that hypoalbuminemia per se may not be the primary determinant of the severity of edema formation and that intrarenal mechanisms may have a prominent contributory role.

ALTERATIONS IN BLOOD VOLUME

According to the traditional view, lowering of the plasma albumin concentration eventually induces renal sodium and fluid retention in the nephrotic syndrome by causing hypovolemia, the so-called underfill mechanism (Fig. 30.11). Indeed, hypovolemia as a consequence of reduced plasma colloid oncotic pressure triggers a cascade of events that signal the kidney to retain the filtered sodium and water.³¹⁵ Thus hypovolemia is the afferent stimulus of a complex

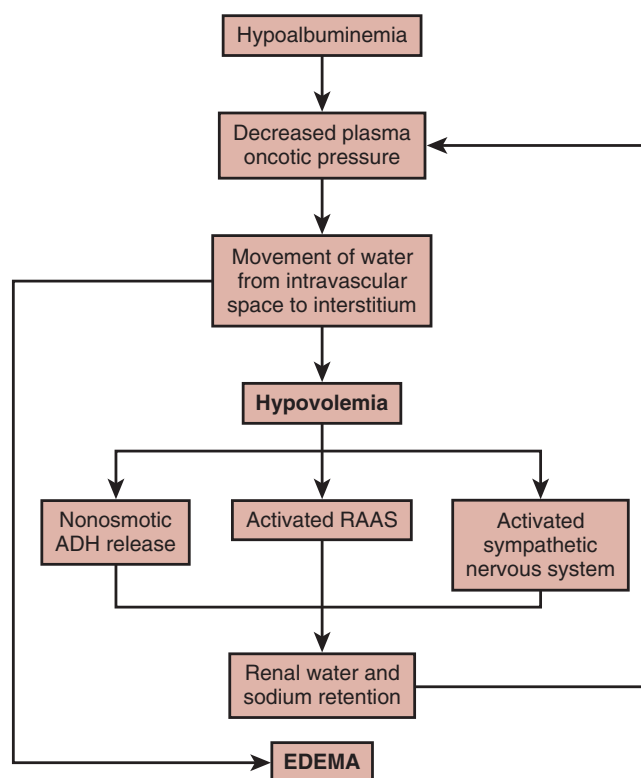


Fig. 30.11 The “underfill” mechanism of edema formation. Hypovolemia, as a consequence of reduced plasma oncotic pressure, is the key event that signals the kidney to retain the filtered sodium and water. *ADH*, Vasopressin; *RAAS*, renin–angiotensin–aldosterone system.

pathway of responses mediated by low- and high-pressure baroreceptors in the cardiac atria, carotid arteries, and aorta that activate the sympathetic nervous system and the renin-angiotensin system. Moreover, hypovolemia also promotes excessive nonosmotic secretion of arginine vasopressin (AVP), which further contributes to water retention by the kidneys.³¹⁵

The homeostatic response of renal sodium and water retention that serves to restore intravascular volume also exacerbates hypoalbuminemia, thereby sustaining transudation of plasma fluid into the interstitial space. The fact that salt retention may be the consequence of an underfilled circulation is consistent with the finding that head-out water immersion, a maneuver that increases plasma volume, is followed by a natriuretic and diuretic response in some nephrotic patients.³¹⁶

This mechanistic scenario of edema formation in the nephrotic syndrome would also imply consistently reduced plasma volume,³¹⁷ as well as an elevated plasma renin activity (PRA),³¹⁸ and increased plasma and urinary levels of catecholamines.³¹⁹ However, only a minority of nephrotic patients have a low plasma volume³¹⁹; in fact, approximately 70% of patients had normal or even high values in some studies.³²⁰ In some cases, plasma volume was lower during remission than during the acute phase of the disease.^{319,320} However, methodologic issues have been raised about the measurement of plasma volume in nephrotic patients that may limit the interpretation of these studies.³²¹

Measurement of vasoactive hormones, which are responsive to low plasma volume and can be taken as surrogate markers of the intravascular volume, also documented that only 50% of nephrotic patients have higher than normal PRA and plasma as well as urinary aldosterone levels.³²² Moreover, pharmacologic blockade of the renin-angiotensin-aldosterone

system in nephrotic patients with a high PRA does not change sodium excretion.³²³ Similarly, plasma levels of norepinephrine, arginine vasopressin, and atrial natriuretic peptide (ANP) are near normal or inconsistently changed.³²⁴ The diuretic and natriuretic response to hyperoncotic plasma or albumin infusions,³²⁵ or to central volume expansion with head-out water immersion, also varies widely from patient to patient.³²⁵ Evidence that PRA often increases rather than decreases after steroid-induced remission of the nephrotic syndrome is additional, albeit indirect, evidence that argues against a key role for hypovolemia in edema formation in most nephrotic patients.³²²

INTRARENAL MECHANISMS

Alternatively, the overfill theory hypothesizes that there is a dominant mechanism by which the kidneys retain sodium independently of circulating plasma volume, leading to hypervolemia (Fig. 30.12).³¹⁵ Examination of the edema-forming, nephrotic patient during consumption of a known amount of sodium reveals a positive sodium balance. This results in increased blood volume, which by altering Starling forces across the capillary wall, leads to plasma leakage into the interstitium and overflow edema. This mechanism has been illustrated in a unilateral model of puromycin aminonucleoside (PAN)-induced nephrosis in rats.³²⁶ In such a model, in which albumin concentration in the systemic circulation is normal, only the proteinuric kidney (not the contralateral intact one) retained excessive amounts of sodium and water. This indicates that abnormal sodium retention by the proteinuric kidney is brought about by intrarenal rather than circulating or systemic factors.

These findings can be partly explained by a lowered filtered sodium load, a consequence of the diminished GFR that

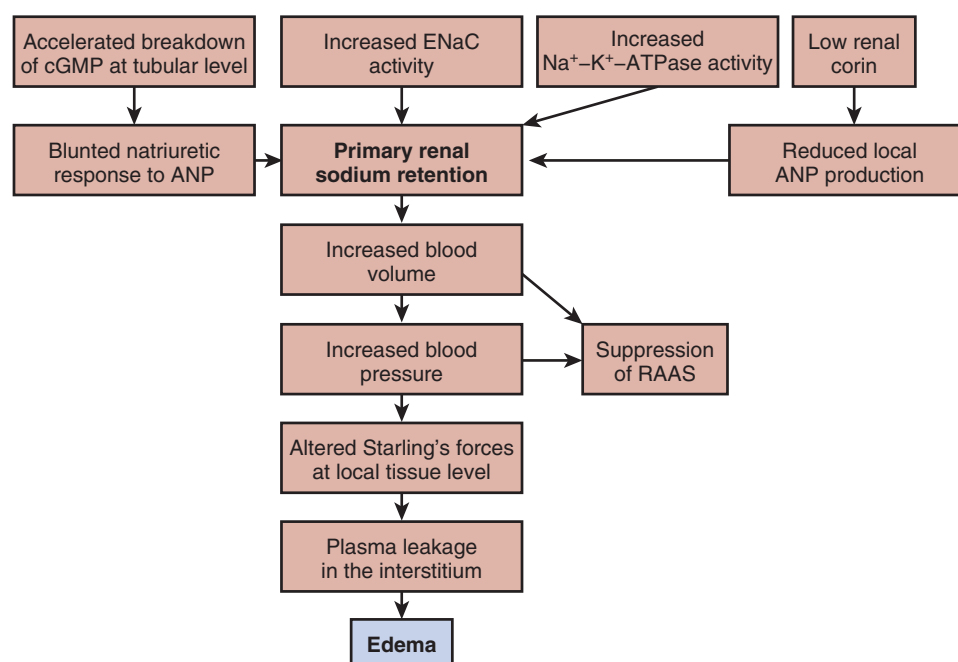


Fig. 30.12 The “overfill” mechanism of edema formation. The abnormal renal sodium retention is the consequence of blunted natriuretic response to atrial natriuretic peptide, increased epithelial sodium channel (ENaC) activity, and $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity, and is the key event of the process. The resulting hypervolemia alters Starling forces across the capillary wall at the local tissue level, leading to overflow edema. cGMP, Cyclic guanosine monophosphate; RAAS, renin-angiotensin aldosterone system.

frequently accompanies nephrotic range proteinuria. However, because the fractional sodium excretion is low, enhanced tubular sodium reabsorption appears to be the predominant cause of sodium retention in the nephrotic syndrome. Analysis of segmental sodium transport in nephrotic rats has identified the collecting duct as the major site of enhanced sodium reabsorption.³²⁶ Refractoriness to the natriuretic action of ANP (which increases urinary sodium and water excretion) in experimental³²⁷ and clinical³²⁸ studies further indicates the distal segments of the nephron as the likely site of sodium retention in nephrotic syndrome. Indeed, the inner medullary segment of the collecting duct is the tubule segment most richly endowed with receptors for ANP.³²⁹

A crucial observation was that natriuretic and diuretic responses to intravenously infused atrial extract (from normal or nephrotic rats) or synthetic ANP were markedly reduced in nephrotic as compared with normal rats.³³⁰ In a rat model of unilateral glomerulopathy, the blunted natriuretic and diuretic response to ANP was confined to the “nephrotic” kidney as opposed to the contralateral normal kidney, despite a comparable increase in GFR.³²⁷ Moreover, both enhanced release of endogenous ANP during water immersion and infusion of exogenous ANP failed to promote an appropriate natriuretic response in nephrotic patients.³²⁸

Taken together, these findings support a role for ANP in intrarenal sodium retention in the nephrotic syndrome. In addition, it can be inferred that alterations in the intrinsic transport properties of the collecting duct render this tubule segment unresponsive to the natriuretic action of ANP.

In some studies, increased activity of efferent sympathetic nerve has been related to the blunted ANP natriuretic response.³³¹ More consistent evidence indicates enhanced phosphodiesterase activity in collecting duct cells from nephrotic animals, leading to accelerated breakdown of normally produced cyclic guanosine monophosphate (cGMP), which is important for intracellular signaling after ANP binding to its specific receptors.³³²

With the discovery of corin, a 1042-amino acid transmembrane serine protease that converts proANP and proBNP into the active forms ANP and BNP,^{333,334} the pathogenesis of edema formation in nephrotic syndrome has been revised. In addition to being initially localized in the heart,³³³ more recently, corin has been shown to also be expressed in renal tissue.³³⁵ Using immunohistochemical analysis, colocalization of corin and ANP in renal tissue has been documented.³³⁵ It is noteworthy that kidneys of corin^{-/-} mice displayed increased amounts of renal β -epithelial Na⁺ channel (ENaC), phosphodiesterase 5 (PDE5), and protein kinase G II, as compared with wild-type mice. Induction of nephrotic syndrome by puromycin aminonucleoside or glomerulonephritis by anti-Thy1 induced concomitant increase in proANP and decrease in ANP in the kidney in association with low renal immunoreactive levels of corin.^{335,336} Upregulation of PDE5 and kinase G II resulted in reduced cGMP in the collecting duct, and subsequently in increased ENaC abundance seen in nephrotic syndrome and glomerulonephritis.³³⁶ These findings suggest that corin deficiency by lowering locally produced ANP might be involved in primary salt retention seen in edematous glomerular diseases.³³⁷ In this regard, reduced urinary corin levels were reported in patients with chronic kidney disease.³³⁸

In the kidney, the ultimate regulation of sodium reabsorption occurs in the collecting duct through the low-conductance epithelial sodium channel (ENaC),³³⁹ located on the apical membrane of principal cells. Evidence is also available that the proteolytic removal of an inhibitory domain from the γ -subunit of ENaC by the serine protease plasmin can activate ENaC.³⁴⁰ Plasmin, present in the urine of nephrotic rats and humans, has been shown to activate ENaC via this mechanism.³⁴¹ Additionally, urokinase-type plasminogen activator present in the rat and human kidney can convert inactive plasminogen (which is filtered by the nephrotic kidney) to the active form plasmin.³⁴¹ In the rat PAN nephrosis model, amiloride increased urine sodium excretion and reduced ascites volume. This effect was attributed to the ability of amiloride to inhibit both ENaC and urokinase-type plasminogen activator, and thus to reduce the amount of active plasmin present.³⁴¹

ENaC is also regulated by aldosterone.³³⁹ In rat models of the nephrotic syndrome, activation of ENaC together with elevated plasma aldosterone levels has been reported.³⁴² Nevertheless, in puromycin-induced nephrosis in rats with clamped aldosterone plasma levels, sodium retention persisted even when ENaC recruitment to the apical membrane was inhibited.³⁴³ Conversely, the transport activity of sodium–potassium–adenosine triphosphatase (Na⁺-K⁺-ATPase), the ubiquitous sodium pump localized exclusively on the basolateral membrane, was increased.³⁴³ These findings indicate that increased Na⁺-K⁺-ATPase activity is the driving force behind enhanced sodium reabsorption in the nephrotic syndrome, an observation confirmed by several studies in the cortical collecting ducts in nephrotic rats.³⁴⁴

Because the Na⁺-K⁺-ATPase pump in the basolateral membrane promotes secondary passive sodium entry from the lumen through the ENaC, Na retention in the collecting duct of nephrotic rats can result from the coordinated overactivity of these tubular sodium transporters. A role for the proximal tubule in the avid sodium retention of nephrotic syndrome has been proposed based on the observation that in PAN-nephrotic rats increased sodium reabsorption was associated with a shift of the apical Na⁺/H⁺ exchanger isoform 3 (NHE-3) from an inactive to an active pool.³⁴⁵ The increase in NHE-3-specific activity may be a response to the increased albumin load presented to the proximal tubule, as indicated by the correlation between albumin exposure and enhanced NHE-3 abundance and activity in opossum kidney (OKP) cells.³⁴⁶

An additional hypothesis concerning intrarenal mechanisms of nephrotic edema proposes that interstitial inflammation of the kidney plays a major role in the pathogenesis of primary sodium retention.³⁴⁷ The generation of vasoconstrictor and the reduction of vasodilator substances in the interstitium, driven by the inflammatory cell infiltrate, can lead to a reduction in Kf and single-nephron GFR. These glomerular hemodynamic changes that reduce filtered sodium load combine with the increased net tubular sodium reabsorption induced by mediators released from the inflammatory cell infiltrate leading to primary sodium retention, an overfilled intravascular volume, and increased capillary hydrostatic pressure. The decrease in plasma oncotic pressure again promotes fluid movement out of the vascular compartment, thereby buffering the changes in blood volume induced by primary sodium retention. A renal inflammatory infiltrate is, however, minimal or absent

in most children with minimal-change nephrotic syndrome. Thus the nephrotic edema may derive from a combination of primary sodium retention and relative arterial underfilling. The predominance of one or the other mechanism is perhaps in accordance with the pathogenesis of nephrotic syndrome or the stage of the disease.

Deranged renal water handling is also a cardinal feature of the nephrotic syndrome. Defects in both urinary diluting ability³⁴⁸ and concentrating capacity³⁴⁸ have been documented in nephrotic patients. The cause of the concentrating defect has been explored in experimental models of the nephrotic syndrome. The extensive downregulation of the expression of the water channels aquaporin 1, 2, and 3 in the collecting duct,³⁴⁹ and of the urea transporter,³⁵⁰ as well as a marked decrease in the abundance of thick ascending limb Na⁺ transporters represent an appropriate renal response to the extracellular volume expansion observed in the nephrotic syndrome despite increased circulating vasopressin, but may occur at the expense of decreased urinary concentrating capacity.

HYPERLIPIDEMIA

Both quantitative and qualitative changes in lipid metabolism occur in the nephrotic syndrome, with virtually all plasma lipid and lipoprotein fractions being elevated.³⁵¹ Blood levels of cholesterol are almost always increased and continue to rise as the severity of the nephrotic syndrome increases.³⁵¹ Total cholesterol and cholesterol esters are all increased.³⁵² Levels of triglycerides are more variable and in many patients do not increase at all, except when the nephrotic state is very severe.³⁵¹ Plasma levels of free fatty acids are within normal limits in the nephrotic syndrome, although a smaller than normal fraction is bound to plasma albumin.³⁵¹ Levels of very low-density lipoprotein (VLDL), intermediate-density lipoprotein (IDL), and low-density lipoprotein (LDL) increased early in the nephrotic syndrome^{351,353}; data on high-density lipoproteins (HDLs) are less clear. Plasma levels

are usually normal but may decrease due to HDL excretion in the urine in severely proteinuric patients.³⁵¹

The composition of the lipoprotein molecules is also abnormal. Greater than usual amounts of cholesterol and triglycerides are present in VLDL, IDL, and LDL. Moreover, an alteration in the specific type and quantity of various apoprotein moieties in the lipoprotein molecules has been described, with reduced apo C despite elevations in apo B, C-II, and E and an increased ratio of apo C-III to apo C-II.³⁵⁴ These abnormalities return to normal promptly when the nephrotic syndrome remits.

PATHOGENESIS OF NEPHROTIC HYPERLIPIDEMIA

Two mechanisms contribute to nephrotic dyslipidemia: overproduction and impaired catabolism/clearance of serum lipids and lipoproteins (Fig. 30.13). There is general agreement that hepatic lipid and apolipoprotein synthesis are both increased and that the clearance of chylomicrons (CM) and VLDL³⁵⁵ is reduced in the nephrotic syndrome. Cholesterol synthesis has been shown to increase both in animals and humans in response to the hypoalbuminemia associated with the nephrotic syndrome.³⁵⁶ Hepatic activity of hydroxymethylglutaryl CoA reductase, the rate-limiting step for hepatic synthesis of cholesterol, is elevated.³²⁰ In general, serum cholesterol levels are inversely proportional to serum albumin levels,³²⁰ and cholesterol levels generally normalize upon remission. Conversely, triglyceride synthesis does not appear to be increased.

It has been suggested that lipoprotein synthesis increases in parallel with albumin synthesis because they share a common secretory pathway.³⁵⁷ This hypothesis was supported by studies showing that infusion of albumin partially corrected nephrotic hyperlipidemia. It has also been reported that apolipoprotein B (apo B) secretion by cultured hepatocytes can be reduced by increasing the oncotic pressure of the culture medium.

Most evidence still indicates that reduced extracellular albumin concentration and/or reduced extracellular oncotic

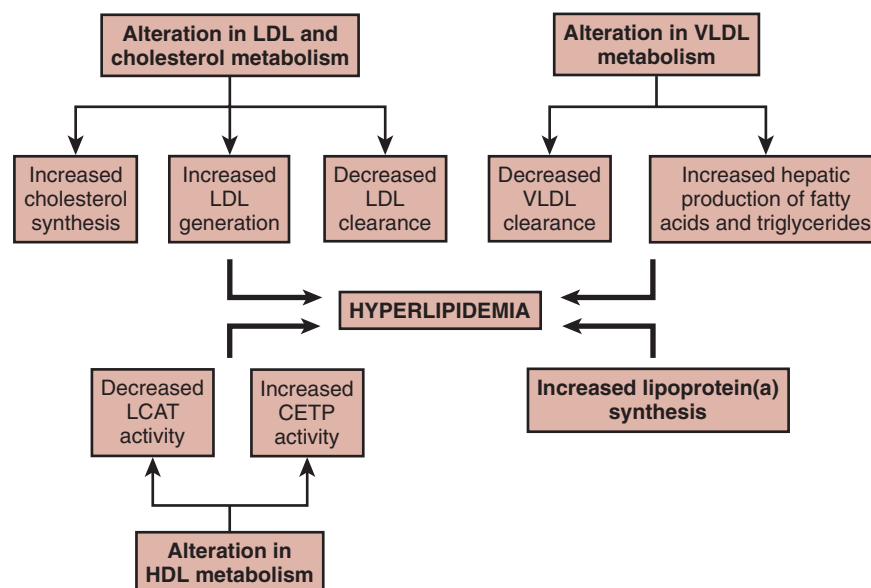


Fig. 30.13 Pathophysiology of nephrotic hyperlipidemia. All abnormalities of lipid profile originate from alteration in low-density lipoprotein (LDL), very-low-density lipoprotein (VLDL), high-density lipoprotein (HDL), and cholesterol metabolism and increased synthesis of lipoprotein(a).

pressure in some way regulate apolipoprotein synthesis and lipogenesis by the liver. Although hepatic apolipoprotein synthesis is increased in the nephrotic syndrome, not all apolipoproteins are affected to the same degree, and mechanisms causing increased synthesis of the various apolipoproteins are also different. Secretion of apo A is increased approximately sixfold,³⁵⁷ whereas synthesis of apo B and E is only increased twofold. Synthesis of the apo C is not increased.

Apo A-I mRNA is increased transcriptionally in the livers of nephrotic and analbuminemic rats,³⁵⁸ suggesting that reduced plasma oncotic pressure or albumin concentration is responsible for the change in apo A-I gene expression. Although plasma apo B and E are both increased in nephrotic and analbuminemic rats, there is little or no change in transcription rates. Thus if increased synthesis is causing increased levels of these apolipoproteins in plasma, the mechanisms involved are most likely posttranscriptional, at the level of translational or protein processing, in contrast to apo A-I.

In addition to increased synthesis, studies in animals and humans have determined alterations in the catabolism of lipids in the nephrotic syndrome. The clearance of chylomicrons and VLDL is reduced following the onset of proteinuria but is normal in rats with hereditary analbuminemia,^{357,359} suggesting that urinary loss of a liporegulatory substance, and not reduced albumin concentration or oncotic pressure, may play a role in causing defective lipolysis.

One possible explanation for the defective removal of lipoproteins is a decrease in the activity of lipoprotein lipase (LPL), which hydrolyzes triglycerides in VLDL and chylomicrons, releasing free fatty acids. LPL activity is reduced in nephrotic rats, which provides a potential mechanism for delayed lipolysis.³⁵⁷ Chylomicron catabolism in hearts isolated from nephrotic rats was decreased *in vitro* and the LPL pool bound to vascular endothelium was reduced by approximately 90%. LPL activity not bound to the vascular endothelium, and hence unable to interact with large lipoproteins, was normal.³⁵⁹ Thus a specific reduction in LPL attached to the vascular endothelium may play a role in the reduced catabolism of chylomicrons and VLDL in the nephrotic syndrome.

The relationship between reduced endothelial-bound LPL activity and reduced catabolism of chylomicrons and VLDL is by no means clear. VLDL and chylomicron catabolism by analbuminemic rats is normal despite a marked reduction in heparin-releasable LPL activity.³⁶⁰ Moreover, it has been reported that HDL isolated from normal animals corrects defective lipolysis of VLDL isolated from nephrotic rats, whereas HDL isolated from nephrotic animals may be dysfunctional. Indeed, HDL isolated from nephrotic animals has been found to be structurally abnormal.³⁶¹ Thus multiple separate defects in the peripheral catabolism of triglyceride-rich lipoproteins may be responsible for delayed lipolysis.

Studies in patients with the nephrotic syndrome have not been as detailed as in the rat; however, when comparable studies are evaluated both species exhibit similar disturbances in lipid metabolism. The fractional turnover rate of triglycerides is reduced in nephrotic subjects compared with controls, and the half-life of triglycerides in VLDL is prolonged from 4 to 11 hours.³⁶² Not only is VLDL catabolism decreased, but the concentration curve over time has an unusual shape, presumably resulting from a delay in the conversion of VLDL into LDL.³⁶³

It has been suggested that the delay in lipolysis in humans, as in rats, is due to a decrease in LPL activity. Evidence

supporting this hypothesis is that LPL activity is reduced in children with the nephrotic syndrome and increases after remission. Furthermore, there is a strong inverse correlation between LPL activity and the concentration of triglycerides in the VLDL fraction,³⁶⁴ although not all investigators report decreased LPL activity in nephrotic patients.³⁶⁵

LDL catabolism has been shown to be either normal or reduced³⁶⁶ in patients with the nephrotic syndrome and only marginally reduced in nephrotic rats.³⁶⁷ Reduced receptor-mediated LDL clearance has been reported in some clinical studies,³⁶⁸ which may account in part for elevations in LDL. A defect in LDL receptor translation or enhanced receptor protein turnover has been hypothesized because normal LDL receptor mRNA was found in nephrotic rats despite marked reduction in LDL receptor protein expression in the liver.³⁶⁹

The nephrotic syndrome is also associated with abnormalities in the activity of enzymes required for effective function of HDL. Cholesterol ester transfer protein (CETP) catalyzes the transfer of the cholesterol ester-rich core of HDL₂ to VLDL remnant particles creating LDL, and thereby increasing LDL cholesterol at the expense of HDL cholesterol. CETP is increased in the plasma of nephrotic patients and correlates positively with VLDL cholesterol and negatively with HDL cholesterol.³⁷⁰

The enzyme lecithin-cholesterol acyl transferase (LCAT) catalyzes the esterification of cholesterol and its incorporation in HDL particles and promotes the conversion of HDL₃ to HDL₂. The observation that HDL₃ is preserved in plasma from nephrotic patients at the apparent expense of HDL₂ suggests that the LCAT activity is reduced in the nephrotic syndrome.³⁷¹ However, increased activity of CETP could also explain this pattern of HDL distribution by rapidly cycling the core of HDL₂ to VLDL remnant particles, thus increasing the flux of cholesterol from the surface of nascent HDL into the core of LDL by increased activity of this enzyme. Furthermore, mature HDL also transports a number of apolipoproteins that serve as cofactors. One of these apolipoproteins, Apo C-II, is an endogenous activator of LPL activity. Apo C-II is normally transported by HDL₂ to nascent VLDL and chylomicrons. Apo C-II may be lost in the urine of nephrotic patients, either as free protein or bound to HDL.³⁷² Additionally, an inhibitor of Apo C-II, Apo C-III, is increased in the nephrotic syndrome; with the resulting decreased Apo C-II-to-Apo C-III ratio, the activity of LPL is significantly decreased.

CLINICAL CONSEQUENCES OF HYPERLIPIDEMIA

The most important consequence of hyperlipidemia is its potential for inducing cardiovascular disease. The changes that occur in blood lipoprotein composition in the nephrotic syndrome,³⁵³ reduced HDL₂ cholesterol, a relative increase in HDL₃ cholesterol, and the massive increase in total cholesterol, mostly found in the LDL, IDL, and VLDL fractions, are likely to increase the risk of atherosclerotic disease. Nevertheless, the presence of additional risk factors for atherosclerosis in nephrotic patients, notably hypertension, hypercoagulability and chronic renal failure, makes it difficult to determine the individual contribution of hyperlipidemia to the increase in risk.

Given the natural history of atherosclerosis, one would predict that the patient with a protracted form of nephrotic syndrome has the highest risk of dying from premature

cardiovascular disease.³⁷³ Accelerated atherosclerosis has been reported in patients with proteinuria and hyperlipidemia and in some studies has been associated with a strongly increased incidence of cardiovascular disease and stroke.³⁷³ One study reported an 85-fold increase in the incidence of ischemic heart disease in nephrotic patients.³⁷⁴ In another retrospective analysis of 142 patients with proteinuria greater than 3.5 g/day, the relative risk of myocardial infarction was found to be 5.5 and the risk of cardiac death 2.8 compared with age- and sex-matched controls.³⁷³

A number of studies have indicated a potential role for hyperlipidemia in the progression of chronic kidney disease. It was proposed that filtered lipoproteins might accumulate in the mesangium and promote sclerosis.³⁷⁵ In animals, lipogenic diets have been shown to induce focal sclerosis, and the extent of glomerular damage correlates with the serum cholesterol; in the obese Zucker rat, focal sclerosis correlates with hyperlipidemia and can be ameliorated by lipid-lowering drugs.³⁷⁶ Similarly, free and esterified cholesterol was found in the glomeruli of nephrotic rats, and a close correlation was noted between plasma cholesterol levels and the number of sclerosing glomerular lesions.³⁷⁷

Whether hyperlipidemia also plays a role in the progression of chronic kidney disease in the human nephrotic syndrome has yet to be determined. Although there is no specific indication to treat the qualitative abnormalities that characterize the lipid disorder of the nephrotic syndrome, if it is anticipated that the duration of hyperlipidemia will be prolonged, it is wise to initiate therapy. Treatment of nephrotic patients with ACE inhibitors³⁷⁸ results in a decline in both proteinuria and blood lipid levels even if plasma albumin concentration does not increase. The decline in blood lipid levels includes a decrease in total cholesterol, lipoprotein(a), a decrease in VLDL and LDL cholesterol, and a decrease in the activities of CETP and LCAT.³⁷⁹

It is prudent to restrict dietary cholesterol and saturated lipids in patients with the nephrotic syndrome. The long-term effects of dietary supplementation with fish oil (rich in omega-3-polyunsaturated fatty acids) are yet unknown and it cannot be recommended as standard treatment except within the context of a controlled investigative trial. If reduction of

proteinuria and dietary fat restriction do not effectively reduce hyperlipidemia, a variety of lipid-lowering drugs, including the 3-hydroxy-3-methyl-glutaryl coenzyme A reductase inhibitors (statins), antioxidants, and fibric acid derivatives, may be useful.

HYPERCOAGULABILITY

Urinary loss of some of the proteins involved in the coagulation cascade and the adaptive increased synthesis of others can induce a hypercoagulable state.³⁸⁰ Although arterial thrombosis has been reported, it is venous thrombosis that occurs with a particularly high incidence in nephrotic subjects.³⁸¹

PATHOGENESIS OF HYPERCOAGULABILITY

In the nephrotic syndrome, there are widespread alterations in synthesis, turnover, and urinary losses of proteins involved both in coagulation and fibrinolysis. The numerous coagulation abnormalities that occur in the nephrotic syndrome are summarized in Fig. 30.14. Alterations in the concentrations of almost every coagulation factor, including zymogens (factors II, V, VII, IX, X, XI, and XII), cofactors (factors V and VIII), and fibrinogen, can occur.³⁸¹ Plasma proteins lost in the urine in patients with the nephrotic syndrome include factors IX, X and XII, which become deficient as there is no sufficient increase in synthetic rates.³⁸¹ In contrast, proteins of higher molecular weight, including factors V and VIII, and fibrinogen accumulate because of increased synthesis.³⁸⁰ Levels of factor VIII typically increase as much as two- to three-fold.³⁸⁰ However, because factor VIII is also an acute-phase reactant, high factor VIII levels may be an epiphenomenon rather than a causal factor in the development of venous thrombosis.

There is an inverse correlation between serum albumin and fibrinogen levels in the nephrotic syndrome.³⁸² The elevated plasma levels of fibrinogen likely result from increased hepatic synthesis, as catabolism is normal.³⁸³ Hyperfibrinogenemia may contribute to the procoagulant state by providing more substrate for fibrin formation and by promoting platelet hyperaggregability, increased blood viscosity, and red blood cell aggregation. Increased fibrin deposition, however, may

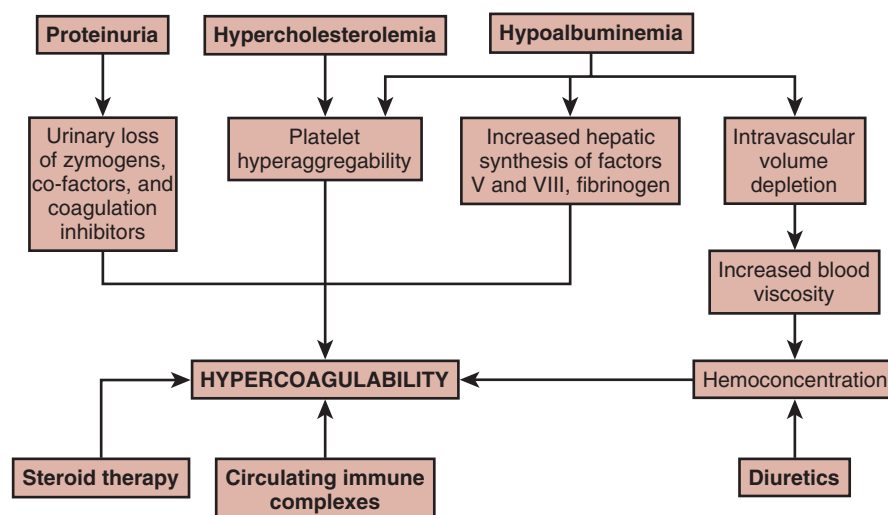


Fig. 30.14 Mechanisms in the pathophysiology of hypercoagulability in nephrotic syndrome. Altered levels and activity of factors in the intrinsic and extrinsic coagulation cascades, levels of antithrombotic and fibrinolytic components of plasma, platelet count and function, and other factors, such as steroids or diuretics, are the numerous abnormalities that contribute to hypercoagulability in the nephrotic syndrome.

also occur due to increased thrombin formation by the elevated levels of factors V and VIII.³⁸⁴

Nephrotic patients exhibit abnormalities in endogenous coagulation inhibitors including antithrombin III, which is deficient in 40% to 80% of patients.³⁸⁵ Plasma levels of antithrombin III correlate negatively with proteinuria and positively with serum albumin levels due to urinary losses of this factor.³⁸⁵ Antithrombin III deficiency has been associated with serum albumin levels of less than 2.0 g/dL³⁸⁶ and correlates with deep venous thrombosis and pulmonary embolism in some studies³⁸⁶ but not in others.³⁸⁷

Alterations in other endogenous anticoagulants may also occur in patients with nephrotic syndrome, but the findings are conflicting. Although plasma levels of total protein S are elevated, the active free fraction level is reduced as a consequence of urinary loss, which accounts for the reduction in the activity of this coagulation inhibitor.³⁸⁸ For protein C results have been contradictory.³⁸⁸ Levels of tissue factor pathway inhibitor (TFPI) were increased in patients with the nephrotic syndrome in one study despite being of relatively low molecular weight.³⁸⁹ Two additional factors that may predispose to thrombosis in nephrotic patients are elevations in serum levels of thrombin activable fibrinolysis inhibitor (TAFI), as well as reductions in levels of protein Z.³⁹⁰

A number of factors may lead to a reduction in plasmin-induced fibrinolysis in the nephrotic syndrome; much of the work has focused on plasminogen, the precursor for plasmin, and two major regulators of plasmin formation, plasminogen activator inhibitor (PAI-1) and tissue plasminogen activator (t-PA). Several studies noted decreased plasminogen levels in the nephrotic syndrome correlating with the magnitude of proteinuria.³⁹¹ Furthermore, hypoalbuminemia itself has been postulated to negatively affect fibrinolysis. Albumin is a cofactor for the binding of plasminogen to fibrin and their interaction with t-PA. One study demonstrated suppressed glomerular fibrinolytic activity in nephrotic syndrome, as there was a sixfold increase in PAI-1, but not t-PA, levels in patients with membranous glomerulopathy compared with controls.³⁹²

Maintenance of hemostasis also involves the formation of platelet plugs through platelet activation and aggregation. Studies examining platelet abnormalities have suggested a role for enhanced, platelet-vessel wall interaction and platelet aggregation in the development of thromboembolism in nephrotic syndrome. Thrombocytosis, decreased red blood cell deformability, and increased von Willebrand factor levels all favor platelet transport toward the vessel wall and increased platelet adhesion³⁹³ and are observed in the nephrotic syndrome.

In vitro studies have demonstrated increased platelet aggregation in nephrotic patients.³⁹³ In addition to platelet hyperaggregability, markers of platelet activation, including plasma P-selectin levels and circulating CD62P-positive platelets, were higher in nephrotic patients than in healthy controls. Increased CD62P expression was found in pediatric patients during the nephrotic but not during remission.³⁹⁴

Platelet hyperaggregability is associated with hypoalbuminemia, hypercholesterolemia, and hyperfibrinogenemia.³⁹⁴ Hypoalbuminemia results in increased availability of normally albumin-bound arachidonic acid, leading to increased formation of thromboxane A₂ in platelets, which promotes platelet aggregation.³⁹⁵ Elevated levels of LDL cholesterol may increase platelet aggregation as suggested by the observation that

lipid-lowering therapy reverses the spontaneous platelet hyperaggregability seen in such patients.³⁹⁶ Such an effect, however, has not been conclusively shown in the general population.³⁹⁷

To date, observations suggest that platelet activation and aggregation may play a role in the increased risk of thromboembolism in patients with nephrotic syndrome. However, attempts at correlating in vitro functional tests with clinically overt thromboembolic events have shown conflicting results.³⁸⁷ Other clinical features of the nephrotic state, such as intravascular volume depletion and exposure to steroids, also contribute to hypercoagulability. Increased blood viscosity is associated with hemoconcentration and enhanced using diuretics³⁹⁸ and by hyperfibrinogenemia.³⁸⁰ The nature of the underlying immunologic injury may play a role and account for the predilection of thrombosis for the renal vein and for the increased incidence of thrombotic complications in membranous glomerulopathy. The identification of circulating immune complexes in patients with membranous glomerulopathy with renal vein thrombosis, but not in those without thrombosis, supports this possibility.³⁸¹ The use of steroids has also been suggested to predispose patients to thromboembolic complications,³⁸¹ but other studies have reported a high incidence of thromboembolic complications in the absence of steroid therapy as well.³⁹⁹

Thus abnormalities in any of the steps that promote coagulation, including activation and termination of the coagulation cascade, fibrinolysis, and platelet activation and aggregation, may contribute to the hypercoagulable state seen in the nephrotic syndrome. The specific role of each of these alterations remains ill-defined.

CLINICAL CONSEQUENCE OF HYPERCOAGULABILITY

Thromboembolic events are serious complications of the nephrotic syndrome. The most frequent site of thrombosis is the renal vein. Retrospective and prospective studies have shown an incidence of renal vein thrombosis (RVT) in the nephrotic syndrome ranging from 5% to 62%.³⁸¹ Nephrotic syndrome is associated with RVT regardless of the underlying disease. Observational studies evaluated patients who underwent renal venography.^{400,401} These studies show that the prevalence of RVT is highest in patients with membranous nephropathy, on average 37%. However, the risk is still clinically important in other primary glomerular diseases, particularly in membranoproliferative glomerulonephritis and minimal change disease. Furthermore, the risk of developing RVT may have been underestimated in these largely cross-sectional studies because patients who were initially not found to have RVT may have subsequently developed it during the disease. In the largest prospective study assessing RVT in 151 patients with nephrotic syndrome, the cumulative incidence of RVT was 22%.⁴⁰⁰

RVT presents clinically in two ways,⁴⁰⁰ acute and chronic. Acute RVT is usually unilateral and characterized by acute flank pain, flank tenderness, macroscopic hematuria, and some deterioration of renal function. Chronic RVT is usually asymptomatic and occurs in the elderly. Selective renal venography is the gold standard for the diagnosis of RVT, and demonstration of venous collaterals establishes chronicity. However, renal venography is invasive and associated with complications that include pulmonary embolism due to clot dislodgment, inferior vena cava perforation, and contrast-induced acute kidney injury.⁴⁰² Consequently, noninvasive

diagnostic tests are preferred, including intravenous pyelography, computed tomography, and magnetic resonance imaging.⁴⁰² Nevertheless, there is need for further studies of the usefulness of these techniques in the diagnosis or exclusion of acute RVT remains unproven. Doppler ultrasonography appears to be inferior to renal venography in establishing the diagnosis of RVT and cannot be recommended based on the current data.⁴⁰³

Early data on the prognosis of nephrotic patients with RVT suggested a dismal outcome. More recently, it became clear that in the presence of anticoagulation therapy, symptomless chronic RVT is benign.⁴⁰⁰ Deep venous thrombosis of the lower extremities is also observed in the nephrotic syndrome and can occur in isolation in up to 15% of patients,⁴⁰¹ or in association with RVT.⁴⁰⁰

Pulmonary embolism may complicate deep vein thrombosis involving the lower extremities, inferior vena cava, or RVT. In a case series of 151 patients with the nephrotic syndrome,⁴⁰⁰ of whom 94 patients underwent ventilation–perfusion lung scanning, symptomatic pulmonary embolism was observed in 25% of patients with acute RVT and 20% of patients with chronic RVT. The incidence of asymptomatic pulmonary embolism was 12.8%. In both prospective and retrospective studies, the incidence of thromboembolic complications other than RVT ranges from 8% to 44%, with an average incidence of 20%.³⁸¹

Many clinical studies have demonstrated an association between hypoalbuminemia and venous thromboembolism, but serum albumin levels in patients with and without thromboembolic events were not significantly different.⁴⁰⁰ These data suggest that hypoalbuminemia is associated with, but not a prerequisite for, the development of thromboembolic complications in nephrotic patients.

The pattern of thrombosis is remarkably different between children and adults. Despite a lower incidence (1.8%–5%),⁴⁰⁴ thromboembolic complications in children tend to be more severe, and half of the children have arterial thrombosis that may cause clinical problems, such as persistent hemiplegia, mesenteric infarction, and peripheral occlusion, leading to amputation.⁴⁰⁴ In adults, arterial thrombosis is much less common than venous thrombosis, but it is a serious complication causing important morbidity. One case series described 43 patients with the nephrotic syndrome who had arterial thromboembolism at aortic, renal, femoral, mesenteric, cerebral, or brachial sites.³⁹⁹ An increased risk of coronary events in patients with the nephrotic syndrome has been documented in a retrospective study.³⁷³

The treatment of venous thromboembolism in nephrotic patients is similar to that in the general population. First-line treatment consists of conventional anticoagulation with low-molecular-weight heparin and oral vitamin K antagonists.³⁸¹ When therapy is initiated early in the course of acute RVT, renal function and other symptoms of RVT have been shown to improve significantly.⁴⁰⁰ Oral vitamin K antagonists are usually continued for the duration of nephrotic-range proteinuria,⁴⁰⁵ as RVT can recur in the setting of ongoing nephrosis after withdrawal of anticoagulation therapy. Low-molecular-weight heparins have a reasonable safety profile but should be used cautiously in patients with renal insufficiency because of excessive anticoagulant activity and an increased risk of bleeding due to drug accumulation.⁴⁰⁶

Controversy exists regarding the use of prophylactic anticoagulation therapy in patients with nephrotic syndrome who do not have RVT. The potential benefit of prophylactic

anticoagulant therapy for patients with membranous glomerulopathy has been documented using decision-analysis methodology.³⁸² Uncontrolled series show a high mortality from pulmonary embolism among patients not receiving anticoagulant therapy and very low rates of RVT and pulmonary embolism in patients given anticoagulant therapy. Prophylactic anticoagulation is warranted as long as the patient has nephrotic proteinuria, an albumin level below 2 g/dL, or both. In patients with other underlying diseases, a more cautious approach may be indicated, and prophylactic anticoagulant therapy should be initiated only if the risk of thromboembolic events is considered high.

SUSCEPTABILITY TO INFECTION

Loss of high filtered protein load through both urinary excretion and tubular catabolism,⁴⁰⁷ as well as a reduced rate of synthesis,⁴⁰⁸ may result in concurrent deficiency of IgG and components of the alternate complement pathway, including factor B. Indeed, patients with nephrotic syndrome have low serum levels of various IgG subclasses. Also, IgA levels are decreased in nephrotic syndrome, whereas IgM usually is increased, particularly in patients with minimal-change disease and normal renal function.⁴⁰⁹ Furthermore, defective cell-mediated immunity has been reported in nephrotic syndrome,^{410,411} including reduced number of total circulating T lymphocytes and blunted blastogenic response by lymphocytes to the mitogens concanavalin A and phytohemagglutinin.

The defects in the humoral and cellular-mediated immunity render the nephrotic patient highly susceptible to infection.⁴¹² The organisms most frequently encountered are *Streptococcus pneumoniae* and *Escherichia coli*. Although such susceptibility to infection is generalized, there seems to be a particular vulnerability to local infection at the sites of edema formation. Splits in the skin caused by edema and malnutrition may predispose nephrotic patients to cellulitis.³²⁰ Peritonitis has been reported in patients who have ascitis.⁴¹² It also occurred in approximately 6% of children with nephrotic syndrome who suffered one or more episodes of the infection.⁴¹²

The unusual susceptibility of children to infections with encapsulated microorganisms is associated with urinary losses of the alternate pathway complement components, particularly factor B, or C3 proactivator, and D, which are essential for the destruction of encapsulated bacteria in the absence of specific antibodies.⁴⁰⁷ The capacity to opsonize the encapsulated bacteria can be restored to normal by adding pure factor B to nephrotic serum.⁴⁰⁷ Fungemia due to *Candida lusitanae* has been also reported in low-birth-weight premature infants with congenital nephrotic syndrome.⁴¹³

New potent antibiotics have contributed to considerably decreasing the incidence of fatal infections in nephrotic syndrome. However, prophylactic measures, such as pneumococcal vaccine, are recommended in adults with severely depressed immunoglobulin levels and nephrotic children over 2 years of age, especially when early remission of nephrotic syndrome is not anticipated.⁴¹⁴

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Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

1. Haraldsson B, Nystrom J, Deen WM. Properties of the glomerular barrier and mechanisms of proteinuria. *Physiol Rev*. 2008;88:451–487.
3. Patrakka J, Tryggvason K. New insights into the role of podocytes in proteinuria. *Nat Rev Nephrol*. 2009;5:463–468.
6. Antiga L, Enc-Iordache B, Remuzzi G, et al. Automatic generation of glomerular capillary topological organization. *Microvasc Res*. 2001;62:346–354.
10. Deen WM, Bridges CR, Brenner BM. Biophysical basis of glomerular permselectivity. *J Membr Biol*. 1983;71:1–10.
22. Edwards A, Daniels BS, Deen WM. Ultrastructural model for size selectivity in glomerular filtration. *Am J Physiol*. 1999;276:F892–F902.
24. Hudson BG, Kalluri R, Tryggvason K. Pathology of glomerular basement membrane nephropathy. *Curr Opin Nephrol Hypertens*. 1994;3:334–339.
31. Winograd-Katz SE, Fassler R, Geiger B, et al. The integrin adhesome: from genes and proteins to human disease. *Nat Rev Mol Cell Biol*. 2014;15:273–288.
36. Gagliardini E, Conti S, Benigni A, et al. Imaging of the porous ultrastructure of the glomerular epithelial filtration slit. *J Am Soc Nephrol*. 2010;21:2081–2089.
40. Deen WM, Bridges CR, Brenner BM, et al. Heteroporous model of glomerular size selectivity: application to normal and nephrotic humans. *Am J Physiol*. 1985;249:F374–F389.
42. Remuzzi A, Remuzzi G. Assessment of glomerular size-selective function with fractional clearance of neutral dextran. *J Lab Clin Med*. 1998;132:360–362.
44. Deen WM, Lazzara MJ, Myers BD. Structural determinants of glomerular permeability. *Am J Physiol Renal Physiol*. 2001;281:F579–F596.
45. Guasch A, Deen WM, Myers BD. Charge selectivity of the glomerular filtration barrier in healthy and nephrotic humans. *J Clin Invest*. 1993;92:2274–2282.
48. Remuzzi A, Sangalli F, Fassi A, et al. Albumin concentration in the Bowman's capsule: multiphoton microscopy vs micropuncture technique. *Kidney Int*. 2007;72:1410–1411. author reply 1411.
51. Lazzara MJ, Deen WM. Model of albumin reabsorption in the proximal tubule. *Am J Physiol Renal Physiol*. 2007;292:F430–F439.
64. Benigni A, Gagliardini E, Remuzzi A, et al. Angiotensin-converting enzyme inhibition prevents glomerular-tubule disconnection and atrophy in passive Heymann nephritis, an effect not observed with a calcium antagonist. *Am J Pathol*. 2001;159:1743–1750.
66. Remuzzi G, Bertani T. Pathophysiology of progressive nephropathies. *N Engl J Med*. 1998;339:1448–1456.
69. Brinkkoetter PT, Ising C, Benzing T. The role of the podocyte in albumin filtration. *Nat Rev Nephrol*. 2013;9:328–336.
74. Durvasula RV, Petermann AT, Hiromura K, et al. Activation of a local tissue angiotensin system in podocytes by mechanical strain. *Kidney Int*. 2004;65:30–39.
80. Abbate M, Zoja C, Morigi M, et al. Transforming growth factor-beta1 is up-regulated by podocytes in response to excess intraglomerular passage of proteins: a central pathway in progressive glomerulosclerosis. *Am J Pathol*. 2002;161:2179–2193.
90. Fukuda A, Wickman LT, Venkatareddy MP, et al. Angiotensin II-dependent persistent podocyte loss from destabilized glomeruli causes progression of end stage kidney disease. *Kidney Int*. 2012;81:40–55.
96. Trionfini P, Benigni A. MicroRNAs as master regulators of glomerular function in health and disease. *J Am Soc Nephrol*. 2017;28:1686–1696.
103. Wu J, Zheng C, Wang X, et al. MicroRNA-30 family members regulate calcium/calineurin signaling in podocytes. *J Clin Invest*. 2015;125:4091–4106.
107. Hasegawa K, Wakino S, Simic P, et al. Renal tubular Sirt1 attenuates diabetic albuminuria by epigenetically suppressing Claudin-1 overexpression in podocytes. *Nat Med*. 2013;19:1496–1504.
111. Jeansson M, Haraldsson B. Morphological and functional evidence for an important role of the endothelial cell glycocalyx in the glomerular barrier. *Am J Physiol Renal Physiol*. 2006;290:F111–F116.
122. Wang B, Komers R, Carew R, et al. Suppression of microRNA-29 expression by TGF-beta1 promotes collagen expression and renal fibrosis. *J Am Soc Nephrol*. 2012;23:252–265.
128. Eremina V, Baelde HJ, Quaggin SE. Role of the VEGF—a signaling pathway in the glomerulus: evidence for crosstalk between components of the glomerular filtration barrier. *Nephron Physiol*. 2007;106:p32–p37.
135. Garsen M, Lenoir O, Rops AL, et al. Endothelin-1 induces proteinuria by heparanase-mediated disruption of the glomerular glycocalyx. *J Am Soc Nephrol*. 2016;27:3545–3551.
143. Smeets B, Angelotti ML, Rizzo P, et al. Renal progenitor cells contribute to hyperplastic lesions of podocytopathies and crescentic glomerulonephritis. *J Am Soc Nephrol*. 2009;20:2593–2603.
149. Morigi M, Locatelli M, Rota C, et al. A previously unrecognized role of C3a in proteinuric progressive nephropathy. *Sci Rep*. 2016;6:28445.
160. Kietzmann L, Guhr SS, Meyer TN, et al. MicroRNA-193a regulates the transdifferentiation of human parietal epithelial cells toward a podocyte phenotype. *J Am Soc Nephrol*. 2015;26:1389–1401.
164. Nangaku M. Chronic hypoxia and tubulointerstitial injury: a final common pathway to end-stage renal failure. *J Am Soc Nephrol*. 2006;17:17–25.
170. Erkan E, Garcia CD, Patterson LT, et al. Induction of renal tubular cell apoptosis in focal segmental glomerulosclerosis: roles of proteinuria and Fas-dependent pathways. *J Am Soc Nephrol*. 2005;16:398–407.
173. Caruso-Neves C, Pinheiro AA, Cai H, et al. PKB and megalin determine the survival or death of renal proximal tubule cells. *Proc Natl Acad Sci U. S. A*. 2006;103:18810–18815.
176. Khan S, AbuJawdeh BG, Goel M, et al. Lipotoxic disruption of NHE1 interaction with PI(4,5)P2 expedites proximal tubule apoptosis. *J Clin Invest*. 2014;124:1057–1068.
193. Rudnicki M, Eder S, Perco P, et al. Gene expression profiles of human proximal tubular epithelial cells in proteinuric nephropathies. *Kidney Int*. 2007;71:325–335.
194. Biemesderfer D. Regulated intramembrane proteolysis of megalin: linking urinary protein and gene regulation in proximal tubule? *Kidney Int*. 2006;69:1717–1721.
208. Nangaku M, Pippin J, Couser WG. Complement membrane attack complex (C5b-9) mediates interstitial disease in experimental nephrotic syndrome. *J Am Soc Nephrol*. 1999;10:2323–2331.
217. Heymann F, Meyer-Schwesinger C, Hamilton-Williams EE, et al. Kidney dendritic cell activation is required for progression of renal disease in a mouse model of glomerular injury. *J Clin Invest*. 2009;119:1286–1297.
237. Humphreys BD, Lin S-L, Kobayashi A, et al. Fate tracing reveals the stromal pericyte and not epithelial origin of myofibroblasts in the obstructive model of kidney fibrosis. *Am J Pathol*. 2010;176:85–95.
242. Kato M, Natarajan R. Diabetic nephropathy—emerging epigenetic mechanisms. *Nat Rev Nephrol*. 2014;10:517–530.
245. Zanchi C, Macconi D, Trionfini P, et al. MicroRNA-184 is a downstream effector of albuminuria driving renal fibrosis in rats with diabetic nephropathy. *Diabetologia*. 2017;60:1114–1125.
255. Khairoun M, van der Pol P, de Vries DK, et al. Renal ischemia-reperfusion induces a dysbalance of angiopoietins, accompanied by proliferation of pericytes and fibrosis. *Am J Physiol Renal Physiol*. 2013;305:F901–F910.
274. Ronconi E, Sagrinati C, Angelotti ML, et al. Regeneration of glomerular podocytes by human renal progenitors. *J Am Soc Nephrol*. 2009;20:322–332.
279. Zhang J, Pippin JW, Vaughan MR, et al. Retinoids augment the expression of podocyte proteins by glomerular parietal epithelial cells in experimental glomerular disease. *Nephron Exp Nephrol*. 2012;121:e23–e37.
282. Kaysen GA. Albumin metabolism in the nephrotic syndrome: the effect of dietary protein intake. *Am J Kidney Dis*. 1988;12:461–480.
321. Perico N, Remuzzi G. Edema of the nephrotic syndrome: the role of the atrial peptide system. *Am J Kidney Dis*. 1993;22:355–366.
328. Peterson C, Madsen B, Perlman A, et al. Atrial natriuretic peptide and the renal response to hypervolemia in nephrotic humans. *Kidney Int*. 1988;34:825–831.
335. Polzin D, Kaminski HJ, Kastner C, et al. Decreased renal corin expression contributes to sodium retention in proteinuric kidney diseases. *Kidney Int*. 2010;78:650–659.
351. Kaysen GA, de Sain-van der Velden MG. New insights into lipid metabolism in the nephrotic syndrome. *Kidney Int Suppl*. 1999;71:S18–S21.
380. Kanfer A. Coagulation factors in nephrotic syndrome. *Am J Nephrol*. 1990;10(suppl 1):63–68.

REFERENCES

- Haraldsson B, Nystrom J, Deen WM. Properties of the glomerular barrier and mechanisms of proteinuria. *Physiol Rev*. 2008;88:451–487.
- Ballermaun BJ, Stan RV. Resolved: capillary endothelium is a major contributor to the glomerular filtration barrier. *J Am Soc Nephrol*. 2007;18:2432–2438.
- Patrakka J, Tryggvason K. New insights into the role of podocytes in proteinuria. *Nat Rev Nephrol*. 2009;5:463–468.
- Abbate M, Zoja C, Remuzzi G. How does proteinuria cause progressive renal damage? *J Am Soc Nephrol*. 2006;17:2974–2984.
- Haraldsson B, Jeansson M. Glomerular filtration barrier. *Curr Opin Nephrol Hypertens*. 2009;18:331–335.
- Antiga L, Ene-Iordache B, Remuzzi G, et al. Automatic generation of glomerular capillary topological organization. *Microvasc Res*. 2001;62:346–354.
- Kaczmarek E, Becker RL Jr. Three-dimensional modeling of renal glomerular capillary networks. *Anal Quant Cytol Histol*. 1997;19:93–101.
- Iordache BE, Remuzzi A. Numerical analysis of blood flow in reconstructed glomerular capillary segments. *Microvasc Res*. 1995;49:1–11.
- Remuzzi A, Deen WM. Theoretical effects of a distribution of capillary dimensions on glomerular ultrafiltration. *Microvasc Res*. 1986;32:131–144.
- Deen WM, Bridges CR, Brenner BM. Biophysical basis of glomerular permselectivity. *J Membr Biol*. 1983;71:1–10.
- Neal CR, Crook H, Bell E, et al. Three-dimensional reconstruction of glomeruli by electron microscopy reveals a distinct restrictive urinary subpodocyte space. *J Am Soc Nephrol*. 2005;16:1223–1235.
- Neal CR, Muston PR, Njegovan D, et al. Glomerular filtration into the subpodocyte space is highly restricted under physiological perfusion conditions. *Am J Physiol Renal Physiol*. 2007;293:F1787–F1798.
- Salmon AH, Toma I, Sipos A, et al. Evidence for restriction of fluid and solute movement across the glomerular capillary wall by the subpodocyte space. *Am J Physiol Renal Physiol*. 2007;293:F1777–F1786.
- Salmon AH, Neal CR, Harper SJ. New aspects of glomerular filtration barrier structure and function: five layers (at least) not three. *Curr Opin Nephrol Hypertens*. 2009;18:197–205.
- Satchell SC, Braet F. Glomerular endothelial cell fenestrations: an integral component of the glomerular filtration barrier. *Am J Physiol Renal Physiol*. 2009;296:F947–F956.
- Hjalmarsson C, Johansson BR, Haraldsson B. Electron microscopic evaluation of the endothelial surface layer of glomerular capillaries. *Microvasc Res*. 2004;67:9–17.
- Yao Y, Rabodzey A, Dewey CF Jr. Glycocalyx modulates the motility and proliferative response of vascular endothelium to fluid shear stress. *Am J Physiol Heart Circ Physiol*. 2007;293:H1023–H1030.
- Jeansson M, Björck K, Tenstad O, et al. Adriamycin alters glomerular endothelium to induce proteinuria. *J Am Soc Nephrol*. 2009;20:114–122.
- Dane MJ, van den Berg BM, Avramut MC, et al. Glomerular endothelial surface layer acts as a barrier against albumin filtration. *Am J Pathol*. 2013;182:1532–1540.
- Henry CB, Duling BR. Permeation of the luminal capillary glycocalyx is determined by hyaluronan. *Am J Physiol*. 1999;277:H508–H514.
- Faas FG, Avramut MC, van den Berg BM, et al. Virtual nanoscopy: generation of ultra-large high resolution electron microscopy maps. *J Cell Biol*. 2012;198:457–469.
- Edwards A, Daniels BS, Deen WM. Ultrastructural model for size selectivity in glomerular filtration. *Am J Physiol*. 1999;276:F892–F902.
- Bolton GR, Deen WM, Daniels BS. Assessment of the charge selectivity of glomerular basement membrane using Ficoll sulfate. *Am J Physiol*. 1998;274:F889–F896.
- Hudson BG, Kalluri R, Tryggvason K. Pathology of glomerular basement membrane nephropathy. *Curr Opin Nephrol Hypertens*. 1994;3:334–339.
- Suh JH, Miner JH. The glomerular basement membrane as a barrier to albumin. *Nat Rev Nephrol*. 2013;9:470–477.
- Kruegel J, Rubel D, Gross O. Alport syndrome—insights from basic and clinical research. *Nat Rev Nephrol*. 2013;9:170–178.
- Pierson M, Cordier J, Hervouet F, et al. [An unusual congenital and familial congenital malformative combination involving the eye and kidney]. *J Genet Hum*. 1963;12:184–213 [in French].
- Abrahamson DR, Isom K, Roach E, et al. Laminin compensation in collagen alpha3(IV) knockout (Alport) glomeruli contributes to permeability defects. *J Am Soc Nephrol*. 2007;18:2465–2472.
- Noakes PG, Miner JH, Gautam M, et al. The renal glomerulus of mice lacking s-laminin/laminin beta 2: nephrosis despite molecular compensation by laminin beta 1. *Nat Genet*. 1995;10:400–406.
- Hynes RO. Integrins: versatility, modulation, and signaling in cell adhesion. *Cell*. 1992;69:11–25.
- Winograd-Katz SE, Fassler R, Geiger B, et al. The integrin adhesome: from genes and proteins to human disease. *Nat Rev Mol Cell Biol*. 2014;15:273–288.
- Schell C, Rogg M, Suhm M, et al. The FERM protein EPB41L5 regulates actomyosin contractility and focal adhesion formation to maintain the kidney filtration barrier. *Proc Natl Acad Sci U. S. A*. 2017;114:E4621–E4630.
- Fukasawa H, Bornheimer S, Kudlicka K, et al. Slit diaphragms contain tight junction proteins. *J Am Soc Nephrol*. 2009;20:1491–1503.
- Rodewald R, Karnovsky MJ. Porous substructure of the glomerular slit diaphragm in the rat and mouse. *J Cell Biol*. 1974;60:423–433.
- Wartiovaara J, Overstedt LG, Khoshnoodi J, et al. Nephrin strands contribute to a porous slit diaphragm scaffold as revealed by electron tomography. *J Clin Invest*. 2004;114:1475–1483.
- Gagliardini E, Conti S, Benigni A, et al. Imaging of the porous ultrastructure of the glomerular epithelial filtration slit. *J Am Soc Nephrol*. 2010;21:2081–2089.
- Conti S, Perico L, Grahammer F, et al. The long journey through renal filtration: new pieces in the puzzle of slit diaphragm architecture. *Curr Opin Nephrol Hypertens*. 2017;26:148–153.
- Srivastava T, Thiagarajan G, Alon US, et al. Role of biomechanical forces in hyperfiltration-mediated glomerular injury in congenital anomalies of the kidney and urinary tract. *Nephrol Dial Transplant*. 2017;32:759–765.
- Kriz W, Lemley KV. Potential relevance of shear stress for slit diaphragm and podocyte function. *Kidney Int*. 2017;91:1283–1286.
- Deen WM, Bridges CR, Brenner BM, et al. Heteroporous model of glomerular size selectivity: application to normal and nephrotic humans. *Am J Physiol*. 1985;249:F374–F389.
- Campbell R, Sangalli F, Perticucci E, et al. Effects of combined ACE inhibitor and angiotensin II antagonist treatment in human chronic nephropathies. *Kidney Int*. 2003;63:1094–1103.
- Remuzzi A, Remuzzi G. Assessment of glomerular size-selective function with fractional clearance of neutral dextran. *J Lab Clin Med*. 1998;132:360–362.
- Jeansson M, Haraldsson B. Glomerular size and charge selectivity in the mouse after exposure to glucosaminoglycan-degrading enzymes. *J Am Soc Nephrol*. 2003;14:1756–1765.
- Deen WM, Lazzara MJ, Myers BD. Structural determinants of glomerular permeability. *Am J Physiol Renal Physiol*. 2001;281:F579–F596.
- Guasch A, Deen WM, Myers BD. Charge selectivity of the glomerular filtration barrier in healthy and nephrotic humans. *J Clin Invest*. 1993;92:2274–2282.
- Christensen EI, Nielsen R. Role of megalin and cubilin in renal physiology and pathophysiology. *Rev Physiol Biochem Pharmacol*. 2007;158:1–22.
- Tojo A, Endou H. Intrarenal handling of proteins in rats using fractional micropuncture technique. *Am J Physiol*. 1992;263:F601–F606.
- Remuzzi A, Sangalli F, Fassi A, et al. Albumin concentration in the Bowman's capsule: multiphoton microscopy vs micropuncture technique. *Kidney Int*. 2007;72:1410–1411. author reply 1411.
- Yu W, Sandoval RM, Molitoris BA. Quantitative intravital microscopy using a Generalized Polarity concept for kidney studies. *Am J Physiol Cell Physiol*. 2005;289:C1197–C1208.
- Tanner GA. Glomerular sieving coefficient of serum albumin in the rat: a two-photon microscopy study. *Am J Physiol Renal Physiol*. 2009;296:F1258–F1265.
- Lazzara MJ, Deen WM. Model of albumin reabsorption in the proximal tubule. *Am J Physiol Renal Physiol*. 2007;292:F430–F439.
- Rovira-Halbach G, Alt JM, Brunkhorst R, et al. Single nephron hyperfiltration and proteinuria in a newly selected rat strain with superficial glomeruli. *Ren Physiol*. 1986;9:317–325.
- Parving HH. Microalbuminuria in essential hypertension and diabetes mellitus. *J Hypertens Suppl*. 1996;14:S89–S93, discussion S93–84.
- D'Amico G, Bazzi C. Pathophysiology of proteinuria. *Kidney Int*. 2003;63:809–825.
- Remuzzi A, Remuzzi G. Glomerular perm-selective function. *Kidney Int*. 1994;45:398–402.
- Pavenstadt H, Kriz W, Kretzler M. Cell biology of the glomerular podocyte. *Physiol Rev*. 2003;83:253–307.

57. Christiansen RE, Tenstad O, Leh S, et al. Glomerular charge selectivity is impaired in hypertensive nephropathy. *Nephrol Dial Transplant*. 2004;19:1083–1091.
58. Andersen S, Blouch K, Bialek J, et al. Glomerular permselectivity in early stages of overt diabetic nephropathy. *Kidney Int*. 2000;58:2129–2137.
59. Jalanko H, Patrakka J, Tryggvason K, et al. Genetic kidney diseases disclose the pathogenesis of proteinuria. *Ann Med*. 2001;33:526–533.
60. Obeidova H, Merta M, Reiterova J, et al. Genetic basis of nephrotic syndrome—review. *Prague Med Rep*. 2006;107:5–16.
61. Halimi J, Laouad I, Buchler M, et al. Early proteinuria is a strong indicator of donor renal lesions, ischemia-reperfusion injury and immunological aggression. *Transplant Proc*. 2006;38:2319–2320.
62. Andersson M, Nilsson U, Hjalmarsson C, et al. Mild renal ischemia-reperfusion reduces charge and size selectivity of the glomerular barrier. *Am J Physiol Renal Physiol*. 2007;292:F1802–F1809.
63. Ledingham JG. Tubular toxicity of filtered proteins. *Am J Nephrol*. 1990;10(suppl 1):52–57.
64. Benigni A, Gagliardini E, Remuzzi A, et al. Angiotensin-converting enzyme inhibition prevents glomerular-tubule disconnection and atrophy in passive Heymann nephritis, an effect not observed with a calcium antagonist. *Am J Pathol*. 2001;159:1743–1750.
65. Birn H, Christensen EI. Renal albumin absorption in physiology and pathology. *Kidney Int*. 2006;69:440–449.
66. Remuzzi G, Bertani T. Pathophysiology of progressive nephropathies. *N Engl J Med*. 1998;339:1448–1456.
67. Shankland SJ. The podocyte's response to injury: role in proteinuria and glomerulosclerosis. *Kidney Int*. 2006;69:2131–2147.
68. Kriz W, Shirato I, Nagata M, et al. The podocyte's response to stress: the enigma of foot process effacement. *Am J Physiol Renal Physiol*. 2012;304:F333–F347.
69. Brinkkoetter PT, Ising C, Benzing T. The role of the podocyte in albumin filtration. *Nat Rev Nephrol*. 2013;9:328–336.
70. Hohne M, Ising C, Hagmann H, et al. Light microscopic visualization of podocyte ultrastructure demonstrates oscillating glomerular contractions. *Am J Pathol*. 2012;182:332–338.
71. Grgic I, Brooks CR, Hofmeister AF, et al. Imaging of podocyte foot processes by fluorescence microscopy. *J Am Soc Nephrol*. 2012;23:785–791.
72. Peti-Peterdi J, Sipos A. A high-powered view of the filtration barrier. *J Am Soc Nephrol*. 2010;21:1835–1841.
73. Rennke HG, Klein PS. Pathogenesis and significance of nonprimary focal and segmental glomerulosclerosis. *Am J Kidney Dis*. 1989;13:443–456.
74. Durvasula RV, Petermann AT, Hiromura K, et al. Activation of a local tissue angiotensin system in podocytes by mechanical strain. *Kidney Int*. 2004;65:30–39.
75. Macconi D, Abbate M, Morigi M, et al. Permselective dysfunction of podocyte-podocyte contact upon angiotensin II unravels the molecular target for renoprotective intervention. *Am J Pathol*. 2006;168:1073–1085.
76. Yamazaki H, Saito A, Ooi H, et al. Differentiation-induced cultured podocytes express endocytically active megalin, a Heymann nephritis antigen. *Nephron Exp Nephrol*. 2004;96:e52–e58.
77. Benigni A, Corna D, Zoja C, et al. Targeted deletion of angiotensin II type 1A receptor does not protect mice from progressive nephropathy of overload proteinuria. *J Am Soc Nephrol*. 2004;15:2666–2674.
78. Davies DJ, Messina A, Thumwood CM, et al. Glomerular podocytic injury in protein overload proteinuria. *Pathology*. 1985;17:412–419.
79. Brewer DB, Filip O. The morphometry of the glomerular epithelial cell and its foot processes after the injection of bovine serum albumin or egg albumin. *J Pathol*. 1976;120:209–220.
80. Abbate M, Zoja C, Morigi M, et al. Transforming growth factor-beta1 is up-regulated by podocytes in response to excess intraglomerular passage of proteins: a central pathway in progressive glomerulosclerosis. *Am J Pathol*. 2002;161:2179–2193.
81. Morigi M, Buelli S, Angioletti S, et al. In response to protein load podocytes reorganize cytoskeleton and modulate endothelin-1 gene: implication for permselective dysfunction of chronic nephropathies. *Am J Pathol*. 2005;166:1309–1320.
82. Barisoni L, Kopp JB. Update in podocyte biology: putting one's best foot forward. *Curr Opin Nephrol Hypertens*. 2003;12:251–258.
83. Ichimura K, Kurihara H, Sakai T. Actin filament organization of foot processes in rat podocytes. *J Histochem Cytochem*. 2003;51:1589–1600.
84. Lemley KV, Lafayette RA, Safai M, et al. Podocytopenia and disease severity in IgA nephropathy. *Kidney Int*. 2002;61:1475–1485.
85. Steffes MW, Schmidt D, McCrery R, et al. Glomerular cell number in normal subjects and in type 1 diabetic patients. *Kidney Int*. 2001;59:2104–2113.
86. Schiffer M, Bitzer M, Roberts IS, et al. Apoptosis in podocytes induced by TGF-beta and Smad7. *J Clin Invest*. 2001;108:807–816.
87. Wada T, Pippin JW, Terada Y, et al. The cyclin-dependent kinase inhibitor p21 is required for TGF-beta1-induced podocyte apoptosis. *Kidney Int*. 2005;68:1618–1629.
88. Shankland SJ, Floege J, Thomas SE, et al. Cyclin kinase inhibitors are increased during experimental membranous nephropathy: potential role in limiting glomerular epithelial cell proliferation in vivo. *Kidney Int*. 1997;52:404–413.
89. Kuan CJ, Al-Douahji M, Shankland SJ. The cyclin kinase inhibitor p21WAF1, CIP1 is increased in experimental diabetic nephropathy: potential role in glomerular hypertrophy. *J Am Soc Nephrol*. 1998;9:986–993.
90. Fukuda A, Wickman LT, Venkatreddy MP, et al. Angiotensin II-dependent persistent podocyte loss from destabilized glomeruli causes progression of end stage kidney disease. *Kidney Int*. 2012;81:40–55.
91. Benigni A, Gagliardini E, Remuzzi G. Changes in glomerular permselectivity induced by angiotensin II imply podocyte dysfunction and slit diaphragm protein rearrangement. *Semin Nephrol*. 2004;24:131–140.
92. Doublier S, Salvidio G, Lupia E, et al. Nephron expression is reduced in human diabetic nephropathy: evidence for a distinct role for glycated albumin and angiotensin II. *Diabetes*. 2003;52:1023–1030.
93. Remuzzi A, Perico N, Amuchastegui CS, et al. Short- and long-term effect of angiotensin II receptor blockade in rats with experimental diabetes. *J Am Soc Nephrol*. 1993;4:40–49.
94. Gagliardini E, Corna D, Zoja C, et al. Unlike each drug alone, lisinopril if combined with avosentan promotes regression of renal lesions in experimental diabetes. *Am J Physiol Renal Physiol*. 2009;297:F1448–F1456.
95. Gagliardini E, Perico N, Rizzo P, et al. Angiotensin II contributes to diabetic renal dysfunction in rodents and humans via Notch1/Snai1 pathway. *Am J Pathol*. 2013;183:119–130.
96. Trionfini P, Benigni A. MicroRNAs as master regulators of glomerular function in health and disease. *J Am Soc Nephrol*. 2017;28:1686–1696.
97. Shi S, Yu L, Chiu C, et al. Podocyte-selective deletion of *dicer* induces proteinuria and glomerulosclerosis. *J Am Soc Nephrol*. 2008;19:2159–2169.
98. Harvey SJ, Jarad G, Cunningham J, et al. Podocyte-specific deletion of *dicer* alters cytoskeletal dynamics and causes glomerular disease. *J Am Soc Nephrol*. 2008;19:2150–2158.
99. Ho J, Ng KH, Rosen S, et al. Podocyte-specific loss of functional microRNAs leads to rapid glomerular and tubular injury. *J Am Soc Nephrol*. 2008;19:2069–2075.
100. Zhdanova O, Srivastava S, Di L, et al. The inducible deletion of *Drosha* and microRNAs in mature podocytes results in a collapsing glomerulopathy. *Kidney Int*. 2011;80:719–730.
101. Perico L, Conti S, Benigni A, et al. Podocyte-actin dynamics in health and disease. *Nat Rev Nephrol*. 2016;12:692–710.
102. Wu J, Zheng C, Fan Y, et al. Downregulation of microRNA-30 facilitates podocyte injury and is prevented by glucocorticoids. *J Am Soc Nephrol*. 2014;25:92–104.
103. Wu J, Zheng C, Wang X, et al. MicroRNA-30 family members regulate calcium/calceinurin signaling in podocytes. *J Clin Invest*. 2015;125:4091–4106.
104. Peng R, Zhou L, Zhou Y, et al. MiR-30a inhibits the epithelial-mesenchymal transition of podocytes through downregulation of NFATc3. *Int J Mol Sci*. 2015;16:24032–24047.
105. Shi S, Yu L, Zhang T, et al. Smad2-dependent downregulation of miR-30 is required for TGF-beta-induced apoptosis in podocytes. *PLoS ONE*. 2013;8:e75572.
106. Li M, Armelloni S, Zennaro C, et al. BDNF repairs podocyte damage by microRNA-mediated increase of actin polymerization. *J Pathol*. 2015;235:731–744.
107. Hasegawa K, Wakino S, Simic P, et al. Renal tubular Sirt1 attenuates diabetic albuminuria by epigenetically suppressing Claudin-1 overexpression in podocytes. *Nat Med*. 2013;19:1496–1504.
108. Hall JA, Dominy JE, Lee Y, et al. The sirtuin family's role in aging and age-associated pathologies. *J Clin Invest*. 2013;123:973–979.
109. Dhawan P, Singh AB, Deane NG, et al. Claudin-1 regulates cellular transformation and metastatic behavior in colon cancer. *J Clin Invest*. 2005;115:1765–1776.

110. Dai C, Stolz DB, Kiss LP, et al. Wnt/beta-catenin signaling promotes podocyte dysfunction and albuminuria. *J Am Soc Nephrol*. 2009;20:1997–2008.
111. Jeansson M, Haraldsson B. Morphological and functional evidence for an important role of the endothelial cell glycocalyx in the glomerular barrier. *Am J Physiol Renal Physiol*. 2006;290:F111–F116.
112. Grond J, Schilthuis MS, Koudstaal J, et al. Mesangial function and glomerular sclerosis in rats after unilateral nephrectomy. *Kidney Int*. 1982;22:338–343.
113. Grond J, Koudstaal J, Elema JD. Mesangial function and glomerular sclerosis in rats with aminonucleoside nephrosis. *Kidney Int*. 1985;27:405–410.
114. Schlondorff D. The glomerular mesangial cell: an expanding role for a specialized pericyte. *FASEB J*. 1987;1:272–281.
115. Gomez-Guerrero C, Suzuki Y, Egido J. The identification of IgA receptors in human mesangial cells: in the search for “Eldorado.” *Kidney Int*. 2002;62:715–717.
116. Schlondorff D, Banas B. The mesangial cell revisited: no cell is an island. *J Am Soc Nephrol*. 2009;20:1179–1187.
117. Abrera-Abeleda MA, Xu Y, Pickering MC, et al. Mesangial immune complex glomerulonephritis due to complement factor D deficiency. *Kidney Int*. 2007;71:1142–1147.
118. Schlondorff D. Roles of the mesangium in glomerular function. *Kidney Int*. 1996;49:1583–1585.
119. Ding G, Zhang A, Huang S, et al. ANG II induces c-Jun NH2-terminal kinase activation and proliferation of human mesangial cells via redox-sensitive transactivation of the EGFR. *Am J Physiol Renal Physiol*. 2007;293:F1889–F1897.
120. Isaka Y, Fujiwara Y, Ueda N, et al. Glomerulosclerosis induced by in vivo transfection of transforming growth factor-beta or platelet-derived growth factor gene into the rat kidney. *J Clin Invest*. 1993;92:2597–2601.
121. Wang Q, Wang Y, Minto AW, et al. MicroRNA-377 is up-regulated and can lead to increased fibronectin production in diabetic nephropathy. *FASEB J*. 2008;22:4126–4135.
122. Wang B, Komers R, Carew R, et al. Suppression of microRNA-29 expression by TGF-beta1 promotes collagen expression and renal fibrosis. *J Am Soc Nephrol*. 2012;23:252–265.
123. Park JT, Kato M, Lanting L, et al. Repression of let-7 by transforming growth factor-beta1-induced Lin28 upregulates collagen expression in glomerular mesangial cells under diabetic conditions. *Am J Physiol Renal Physiol*. 2014;307:F1390–F1403.
124. Lin CL, Wang JY, Ko JY, et al. Dickkopf-1 promotes hyperglycemia-induced accumulation of mesangial matrix and renal dysfunction. *J Am Soc Nephrol*. 2010;21:124–135.
125. Kato M, Arce L, Wang M, et al. A microRNA circuit mediates transforming growth factor-beta1 autoregulation in renal glomerular mesangial cells. *Kidney Int*. 2011;80:358–368.
126. Trionfini P, Benigni A, Remuzzi G. MicroRNAs in kidney physiology and disease. *Nat Rev Nephrol*. 2015;11:23–33.
127. Castro NE, Kato M, Park JT, et al. Transforming growth factor beta1 (TGF-beta1) enhances expression of profibrotic genes through a novel signaling cascade and microRNAs in renal mesangial cells. *J Biol Chem*. 2014;289:29001–29013.
128. Eremina V, Baelde HJ, Quaggin SE. Role of the VEGF—a signaling pathway in the glomerulus: evidence for crosstalk between components of the glomerular filtration barrier. *Nephron Physiol*. 2007;106:p32–p37.
129. Ballermann BJ. Contribution of the endothelium to the glomerular permselectivity barrier in health and disease. *Nephron Physiol*. 2007;106:p19–p25.
130. Kitamoto Y, Tokunaga H, Tomita K. Vascular endothelial growth factor is an essential molecule for mouse kidney development: glomerulogenesis and nephrogenesis. *J Clin Invest*. 1997;99:2351–2357.
131. Foster RR, Armstrong L, Baker S, et al. Glycosaminoglycan regulation by VEGFA and VEGFC of the glomerular microvascular endothelial cell glycocalyx in vitro. *Am J Pathol*. 2013;183:604–616.
132. Abrahamson DR, Robert B. Derivation and differentiation of glomerular endothelial cells. *Nephrol Dial Transplant*. 2003;18(suppl 6):vi2–vi7.
133. Segerer S, Kretzler M, Strutz F, et al. Mechanisms of tissue injury and repair in renal diseases. In: Schrier R, ed. *Diseases of the Kidney and Urinary Tract*. Philadelphia: Wolters Kluwer Health/Lippincott Williams & Wilkins; 2007.
134. Collino F, Bussolati B, Gerbaudo E, et al. Preeclamptic sera induce nephrin shedding from podocytes through endothelin-1 release by endothelial glomerular cells. *Am J Physiol Renal Physiol*. 2008;294:F1185–F1194.
135. Garsen M, Lenoir O, Rops AL, et al. Endothelin-1 induces proteinuria by heparanase-mediated disruption of the glomerular glycocalyx. *J Am Soc Nephrol*. 2016;27:3545–3551.
136. Gil N, Goldberg R, Neuman T, et al. Heparanase is essential for the development of diabetic nephropathy in mice. *Diabetes*. 2012;61:208–216.
137. Sangalli F, Carrara F, Gaspari F, et al. Effect of ACE inhibition on glomerular permselectivity and tubular albumin concentration in the renal ablation model. *Am J Physiol Renal Physiol*. 2011;300:F1291–F1300.
138. Falk R, Jennette J, Nachman P. Primary glomerular disease. In: Brenner BM, ed. *Brenner and Rector's the Kidney*. 8th ed. Philadelphia: Saunders Elsevier; 2008:987–1066:chap 30.
139. Boucher A, Droz D, Adafer E, et al. Relationship between the integrity of Bowman's capsule and the composition of cellular crescents in human crescentic glomerulonephritis. *Lab Invest*. 1987;56:526–533.
140. Nitta K, Horita S, Honda K, et al. Glomerular expression of cell-cycle-regulatory proteins in human crescentic glomerulonephritis. *Virchows Arch*. 1999;435:422–427.
141. Jennette JC, Hipp CG. The epithelial antigen phenotype of glomerular crescent cells. *Am J Clin Pathol*. 1986;86:274–280.
142. Sagrinati C, Netti GS, Mazzinghi B, et al. Isolation and characterization of multipotent progenitor cells from the Bowman's capsule of adult human kidneys. *J Am Soc Nephrol*. 2006;17:2443–2456.
143. Smeets B, Angelotti ML, Rizzo P, et al. Renal progenitor cells contribute to hyperplastic lesions of podocytopathies and crescentic glomerulonephritis. *J Am Soc Nephrol*. 2009;20:2593–2603.
144. Benigni A, Morigi M, Rizzo P, et al. Inhibiting angiotensin-converting enzyme promotes renal repair by limiting progenitor cell proliferation and restoring the glomerular architecture. *Am J Pathol*. 2011;179:628–638.
145. Peired A, Angelotti ML, Ronconi E, et al. Proteinuria impairs podocyte regeneration by sequestering retinoic acid. *J Am Soc Nephrol*. 2013;24:1756–1768.
146. Turnberg D, Lewis M, Moss J, et al. Complement activation contributes to both glomerular and tubulointerstitial damage in adriamycin nephropathy in mice. *J Immunol*. 2006;177:4094–4102.
147. Lenderink AM, Liegel K, Ljubicic D, et al. The alternative pathway of complement is activated in the glomeruli and tubulointerstitium of mice with adriamycin nephropathy. *Am J Physiol Renal Physiol*. 2007;293:F555–F564.
148. Abbate M, Zoja C, Corna D, et al. Complement-mediated dysfunction of glomerular filtration barrier accelerates progressive renal injury. *J Am Soc Nephrol*. 2008;19:1158–1167.
149. Morigi M, Locatelli M, Rota C, et al. A previously unrecognized role of C3a in proteinuric progressive nephropathy. *Sci Rep*. 2016;6:28445.
150. Rizzo P, Perico N, Gagliardini E, et al. Nature and mediators of parietal epithelial cell activation in glomerulonephritides of human and rat. *Am J Pathol*. 2013;183:1769–1778.
151. Fatima H, Moeller MJ, Smeets B, et al. Parietal epithelial cell activation marker in early recurrence of FSGS in the transplant. *Clin J Am Soc Nephrol*. 2012;7:1852–1858.
152. Smeets B, Kuppe C, Sicking EM, et al. Parietal epithelial cells participate in the formation of sclerotic lesions in focal segmental glomerulosclerosis. *J Am Soc Nephrol*. 2011;22:1262–1274.
153. Ueno T, Kobayashi N, Nakayama M, et al. Aberrant Notch1-dependent effects on glomerular parietal epithelial cells promotes collapsing focal segmental glomerulosclerosis with progressive podocyte loss. *Kidney Int*. 2013;83:1065–1075.
154. van Es JH, van Gijn ME, Riccio O, et al. Notch/gamma-secretase inhibition turns proliferative cells in intestinal crypts and adenomas into goblet cells. *Nature*. 2005;435:959–963.
155. Matsusaka T, Xin J, Niwa S, et al. Genetic engineering of glomerular sclerosis in the mouse via control of onset and severity of podocyte-specific injury. *J Am Soc Nephrol*. 2005;16:1013–1023.
156. Zhou H, Hasni SA, Perez P, et al. miR-150 promotes renal fibrosis in lupus nephritis by downregulating SOCS1. *J Am Soc Nephrol*. 2013;24:1073–1087.
157. Tomasoni S, Benigni A. Post-transcriptional gene regulation makes things clearer in renal fibrosis. *J Am Soc Nephrol*. 2013;24:1026–1028.
158. Macconi D, Tomasoni S, Romagnani P, et al. MicroRNA-324-3p promotes renal fibrosis and is a target of ACE inhibition. *J Am Soc Nephrol*. 2012;23:1496–1505.

159. Gebeshuber CA, Kornauth C, Dong L, et al. Focal segmental glomerulosclerosis is induced by microRNA-193a and its downregulation of WT1. *Nat Med*. 2013;19:481–487.
160. Kietzmann L, Guhr SS, Meyer TN, et al. MicroRNA-193a regulates the transdifferentiation of human parietal epithelial cells toward a podocyte phenotype. *J Am Soc Nephrol*. 2015;26:1389–1401.
161. Hohenstein P, Pritchard-Jones K, Charlton J. The yin and yang of kidney development and Wilms' tumors. *Genes Dev*. 2015;29:467–482.
162. Krebs CF, Kapfner S, Paust HJ, et al. MicroRNA-155 drives TH17 immune response and tissue injury in experimental crescentic GN. *J Am Soc Nephrol*. 2013;24:1955–1965.
163. Fine LG, Norman JT. Chronic hypoxia as a mechanism of progression of chronic kidney diseases: from hypothesis to novel therapeutics. *Kidney Int*. 2008;74:867–872.
164. Nangaku M. Chronic hypoxia and tubulointerstitial injury: a final common pathway to end-stage renal failure. *J Am Soc Nephrol*. 2006;17:17–25.
165. Norman JT, Fine LG. Intrarenal oxygenation in chronic renal failure. *Clin Exp Pharmacol Physiol*. 2006;33:989–996.
166. Nangaku M, Eckardt KU. Hypoxia and the HIF system in kidney disease. *J Mol Med*. 2007;85:1325–1330.
167. Rosenberger C, Pratschke J, Rudolph B, et al. Immunohistochemical detection of hypoxia-inducible factor-1 α in human renal allograft biopsies. *J Am Soc Nephrol*. 2007;18:343–351.
168. Erkan E, De Leon M, Devarajan P. Albumin overload induces apoptosis in LLC-PK(1) cells. *Am J Physiol Renal Physiol*. 2001;280:F1107–F1114.
169. Tejera N, Gomez-Garre D, Lazaro A, et al. Persistent proteinuria up-regulates angiotensin II type 2 receptor and induces apoptosis in proximal tubular cells. *Am J Pathol*. 2004;164:1817–1826.
170. Erkan E, Garcia CD, Patterson LT, et al. Induction of renal tubular cell apoptosis in focal segmental glomerulosclerosis: roles of proteinuria and Fas-dependent pathways. *J Am Soc Nephrol*. 2005;16:398–407.
171. Brunskill NJ, Stuart J, Tobin AB, et al. Receptor-mediated endocytosis of albumin by kidney proximal tubule cells is regulated by phosphatidylinositol 3-kinase. *J Clin Invest*. 1998;101:2140–2150.
172. Song G, Ouyang G, Bao S. The activation of Akt/PKB signaling pathway and cell survival. *J Cell Mol Med*. 2005;9:59–71.
173. Caruso-Neves C, Pinheiro AA, Cai H, et al. PKB and megalin determine the survival or death of renal proximal tubule cells. *Proc Natl Acad Sci U S A*. 2006;103:18810–18815.
174. Urahama Y, Ohsaki Y, Fujita Y, et al. Lipid droplet-associated proteins protect renal tubular cells from fatty acid-induced apoptosis. *Am J Pathol*. 2008;173:1286–1294.
175. Ruggiero C, Elks CM, Kruger C, et al. Albumin-bound fatty acids but not albumin itself alter redox balance in tubular epithelial cells and induce a peroxide-mediated redox-sensitive apoptosis. *Am J Physiol Renal Physiol*. 2014;306:F896–F906.
176. Khan S, Abu Jawdeh BG, Goel M, et al. Lipotoxic disruption of NHE1 interaction with PI(4,5)P2 expedites proximal tubule apoptosis. *J Clin Invest*. 2014;124:1057–1068.
177. Matzinger P. The danger model: a renewed sense of self. *Science*. 2002;296:301–305.
178. Kawai T, Akira S. TLR signaling. *Cell Death Differ*. 2006;13:816–825.
179. Ye Z, Ting JP. NLR, the nucleotide-binding domain leucine-rich repeat containing gene family. *Curr Opin Immunol*. 2008;20:3–9.
180. Bianchi ME. DAMPs, PAMPs and alarmins: all we need to know about danger. *J Leukoc Biol*. 2007;81:1–5.
181. Mariathasan S, Monack DM. Inflammasome adaptors and sensors: intracellular regulators of infection and inflammation. *Nat Rev Immunol*. 2007;7:31–40.
182. Zoja C, Donadelli R, Colleoni S, et al. Protein overload stimulates RANTES production by proximal tubular cells depending on NF-kappa B activation. *Kidney Int*. 1998;53:1608–1615.
183. Wang Y, Chen J, Chen L, et al. Induction of monocyte chemoattractant protein-1 in proximal tubule cells by urinary protein. *J Am Soc Nephrol*. 1997;8:1537–1545.
184. Tang S, Leung JC, Abe K, et al. Albumin stimulates interleukin-8 expression in proximal tubular epithelial cells in vitro and in vivo. *J Clin Invest*. 2003;111:515–527.
185. Donadelli R, Zanchi C, Morigi M, et al. Protein overload induces fractalkine upregulation in proximal tubular cells through nuclear factor kappaB- and p38 mitogen-activated protein kinase-dependent pathways. *J Am Soc Nephrol*. 2003;14:2436–2446.
186. Wolf G, Schroeder R, Ziyadeh FN, et al. Albumin up-regulates the type II transforming growth factor-beta receptor in cultured proximal tubular cells. *Kidney Int*. 2004;66:1849–1858.
187. Peruzzi L, Trusolino L, Amore A, et al. Tubulointerstitial responses in the progression of glomerular diseases: albuminuria modulates alpha v beta 5 integrin. *Kidney Int*. 1996;50:1310–1320.
188. Wang Y, Rangan GK, Tay YC, et al. Induction of monocyte chemoattractant protein-1 by albumin is mediated by nuclear factor kappaB in proximal tubule cells. *J Am Soc Nephrol*. 1999;10:1204–1213.
189. Drumm K, Lee E, Stanners S, et al. Albumin and glucose effects on cell growth parameters, albumin uptake and Na(+)/H(+)-exchanger isoform 3 in OK cells. *Cell Physiol Biochem*. 2003;13:199–206.
190. Morigi M, Macconi D, Zoja C, et al. Protein overload-induced NF-kappaB activation in proximal tubular cells requires H(2)O(2) through a PKC-dependent pathway. *J Am Soc Nephrol*. 2002;13:1179–1189.
191. Zandi-Nejad K, Eddy AA, Glassock RJ, et al. Why is proteinuria an ominous biomarker of progressive kidney disease? *Kidney Int*. 2004;66:S76–S89.
192. Nakajima H, Takenaka M, Kaimori JY, et al. Gene expression profile of renal proximal tubules regulated by proteinuria. *Kidney Int*. 2002;61:1577–1587.
193. Rudnicki M, Eder S, Perco P, et al. Gene expression profiles of human proximal tubular epithelial cells in proteinuric nephropathies. *Kidney Int*. 2007;71:325–335.
194. Biemesderfer D. Regulated intramembrane proteolysis of megalin: linking urinary protein and gene regulation in proximal tubule? *Kidney Int*. 2006;69:1717–1721.
195. Brown MS, Goldstein JL. The SREBP pathway: regulation of cholesterol metabolism by proteolysis of a membrane-bound transcription factor. *Cell*. 1997;89:331–340.
196. Motoyoshi Y, Matsusaka T, Saito A, et al. Megalin contributes to the early injury of proximal tubule cells during nonselective proteinuria. *Kidney Int*. 2008;74:1262–1269.
197. Schlondorff DO. Overview of factors contributing to the pathophysiology of progressive renal disease. *Kidney Int*. 2008;74:860–866.
198. Benigni A, Caroli C, Longaretti L, et al. Involvement of renal tubular Toll-like receptor 9 in the development of tubulointerstitial injury in systemic lupus. *Arthritis Rheum*. 2007;56:1569–1578.
199. Brentjens JR, Sepulveda M, Baliah T, et al. Interstitial immune complex nephritis in patients with systemic lupus erythematosus. *Kidney Int*. 1975;7:342–350.
200. Sano H, Morimoto C. Dna isolated from DNA/anti-DNA antibody immune complexes in systemic lupus erythematosus is rich in guanine-cytosine content. *J Immunol*. 1982;128:1341–1345.
201. Hirschberg R. Bioactivity of glomerular ultrafiltrate during heavy proteinuria may contribute to renal tubulo-interstitial lesions: evidence for a role for insulin-like growth factor I. *J Clin Invest*. 1996;98:116–124.
202. Wang SN, LaPage J, Hirschberg R. Role of glomerular ultrafiltration of growth factors in progressive interstitial fibrosis in diabetic nephropathy. *Kidney Int*. 2000;57:1002–1014.
203. Song JH, Lee SW, Suh JH, et al. The effects of dual blockade of the renin-angiotensin system on urinary protein and transforming growth factor-beta excretion in 2 groups of patients with IgA and diabetic nephropathy. *Clin Nephrol*. 2003;60:318–326.
204. Wang SN, Hirschberg R. Growth factor ultrafiltration in experimental diabetic nephropathy contributes to interstitial fibrosis. *Am J Physiol Renal Physiol*. 2000;278:F554–F560.
205. Hsu SI, Couser WG. Chronic progression of tubulointerstitial damage in proteinuric renal disease is mediated by complement activation: a therapeutic role for complement inhibitors? *J Am Soc Nephrol*. 2003;14:S186–S191.
206. Nangaku M. Complement regulatory proteins in glomerular diseases. *Kidney Int*. 1998;54:1419–1428.
207. Abbate M, Zoja C, Rottoli D, et al. Antiproteinuric therapy while preventing the abnormal protein traffic in proximal tubule abrogates protein- and complement-dependent interstitial inflammation in experimental renal disease. *J Am Soc Nephrol*. 1999;10:804–813.
208. Nangaku M, Pippin J, Couser WG. Complement membrane attack complex (C5b-9) mediates interstitial disease in experimental nephrotic syndrome. *J Am Soc Nephrol*. 1999;10:2323–2331.
209. Camussi G, Stratta P, Mazzucco G, et al. In vivo localization of C3 on the brush border of proximal tubules of kidneys from nephrotic patients. *Clin Nephrol*. 1985;23:134–141.
210. Pratt JR, Basheer SA, Sacks SH. Local synthesis of complement component C3 regulates acute renal transplant rejection. *Nat Med*. 2002;8:582–587.
211. Quigg RJ. Complement and the kidney. *J Immunol*. 2003;171:3319–3324.

212. Ferenbach D, Hughes J. Macrophages and dendritic cells: what is the difference? *Kidney Int.* 2008;74:5–7.
213. Soos TJ, Sims TN, Barisoni L, et al. CX3CR1+ interstitial dendritic cells form a contiguous network throughout the entire kidney. *Kidney Int.* 2006;70:591–596.
214. Rock KL, Shen L. Cross-presentation: underlying mechanisms and role in immune surveillance. *Immunol Rev.* 2005;207:166–183.
215. Pietrangelo A, Panduro A, Chowdhury JR, et al. Albumin gene expression is down-regulated by albumin or macromolecule infusion in the rat. *J Clin Invest.* 1992;89:1755–1760.
216. Lukacs-Kornek V, Burgdorf S, Diehl L, et al. The kidney-renal lymph node-system contributes to cross-tolerance against innocuous circulating antigen. *J Immunol.* 2008;180:706–715.
217. Heymann F, Meyer-Schwesinger C, Hamilton-Williams EE, et al. Kidney dendritic cell activation is required for progression of renal disease in a mouse model of glomerular injury. *J Clin Invest.* 2009;119:1286–1297.
218. Macconi D, Chiabrando C, Schiarea S, et al. Proteasomal processing of albumin by renal dendritic cells generates antigenic peptides. *J Am Soc Nephrol.* 2009;20:123–130.
219. Sean Eardley K, Cockwell P. Macrophages and progressive tubulointerstitial disease. *Kidney Int.* 2005;68:437–455.
220. Rodríguez-Iturbe B, Pons H, Herrera-Acosta J, et al. Role of immunocompetent cells in nonimmune renal diseases. *Kidney Int.* 2001;59:1626–1640.
221. Eddy AA. Interstitial nephritis induced by protein-overload proteinuria. *Am J Pathol.* 1989;135:719–733.
222. Romero F, Rodríguez-Iturbe B, Parra G, et al. Mycophenolate mofetil prevents the progressive renal failure induced by 5/6 renal ablation in rats. *Kidney Int.* 1999;55:945–955.
223. Boucher A, Droz D, Adafer E, et al. Characterization of mononuclear cell subsets in renal cellular interstitial infiltrates. *Kidney Int.* 1986;29:1043–1049.
224. Cohen CD, Calvaresi N, Armelloni S, et al. CD20-positive infiltrates in human membranous glomerulonephritis. *J Nephrol.* 2005;18:328–333.
225. Heller F, Lindenmeyer MT, Cohen CD, et al. The contribution of B cells to renal interstitial inflammation. *Am J Pathol.* 2007;170:457–468.
226. Martin F, Chan AC. B cell immunobiology in disease: evolving concepts from the clinic. *Annu Rev Immunol.* 2006;24:467–496.
227. Wada T, Sakai N, Matsushima K, et al. Fibrocytes: a new insight into kidney fibrosis. *Kidney Int.* 2007;72:269–273.
228. Sakai N, Wada T, Yokoyama H, et al. Secondary lymphoid tissue chemokine (SLC/CCL21)/CCR7 signaling regulates fibrocytes in renal fibrosis. *Proc Natl Acad Sci U. S. A.* 2006;103:14098–14103.
229. Eddy AA. Molecular insights into renal interstitial fibrosis. *J Am Soc Nephrol.* 1996;7:2495–2508.
230. Roberts IS, Burrows C, Shanks JH, et al. Interstitial myofibroblasts: predictors of progression in membranous nephropathy. *J Clin Pathol.* 1997;50:123–127.
231. Lan HY. Tubular epithelial-myofibroblast transdifferentiation mechanisms in proximal tubule cells. *Curr Opin Nephrol Hypertens.* 2003;12:25–29.
232. Abbate M, Zoja C, Rottoli D, et al. Proximal tubular cells promote fibrogenesis by TGF-beta1-mediated induction of peritubular myofibroblasts. *Kidney Int.* 2002;61:2066–2077.
233. Strutz FM. EMT and proteinuria as progression factors. *Kidney Int.* 2009;75:475–481.
234. Iwano M, Plieth D, Danoff TM, et al. Evidence that fibroblasts derive from epithelium during tissue fibrosis. *J Clin Invest.* 2002;110:341–350.
235. Eddy AA, Kim H, Lopez-Guisa J, et al. Interstitial fibrosis in mice with overload proteinuria: deficiency of TIMP-1 is not protective. *Kidney Int.* 2000;58:618–628.
236. Faulkner JL, Szykalski LM, Springer F, et al. Origin of interstitial fibroblasts in an accelerated model of angiotensin II-induced renal fibrosis. *Am J Pathol.* 2005;167:1193–1205.
237. Humphreys BD, Lin SL, Kobayashi A, et al. Fate tracing reveals the stromal pericyte and not epithelial origin of myofibroblasts in the obstructive model of kidney fibrosis. *Am J Pathol.* 2010;176:85–95.
238. Nuttall RK, Sampieri CL, Pennington CJ, et al. Expression analysis of the entire MMP and TIMP gene families during mouse tissue development. *FEBS Lett.* 2004;563:129–134.
239. Kassiri Z, Oudit GY, Kandam V, et al. Loss of TIMP3 enhances interstitial nephritis and fibrosis. *J Am Soc Nephrol.* 2009;20:1223–1235.
240. Kantharidis P, Wang B, Carew RM, et al. Diabetes complications: the microRNA perspective. *Diabetes.* 2011;60:1832–1837.
241. Kato M, Natarajan R. MicroRNAs in diabetic nephropathy: functions, biomarkers, and therapeutic targets. *Ann N Y Acad Sci.* 2015;1353:72–88.
242. Kato M, Natarajan R. Diabetic nephropathy—emerging epigenetic mechanisms. *Nat Rev Nephrol.* 2014;10:517–530.
243. McClelland A, Hagiwara S, Kantharidis P. Where are we in diabetic nephropathy: microRNAs and biomarkers? *Curr Opin Nephrol Hypertens.* 2014;23:80–86.
244. Rudnicki M, Beckers A, Neuwirt H, et al. RNA expression signatures and posttranscriptional regulation in diabetic nephropathy. *Nephrol Dial Transplant.* 2015;30(suppl 4):iv35–iv42.
245. Zanchi C, Macconi D, Trionfini P, et al. MicroRNA-184 is a downstream effector of albuminuria driving renal fibrosis in rats with diabetic nephropathy. *Diabetologia.* 2017;60:1114–1125.
246. Nangaku M. Mechanisms of tubulointerstitial injury in the kidney: final common pathways to end-stage renal failure. *Intern Med.* 2004;43:9–17.
247. Nakagawa T, Kang DH, Ohashi R, et al. Tubulointerstitial disease: role of ischemia and microvascular disease. *Curr Opin Nephrol Hypertens.* 2003;12:233–241.
248. Yuan HT, Li XZ, Pitera JE, et al. Peritubular capillary loss after mouse acute nephrotoxicity correlates with down-regulation of vascular endothelial growth factor-A and hypoxia-inducible factor-1 alpha. *Am J Pathol.* 2003;163:2289–2301.
249. Armulik A, Abramsson A, Betsholtz C. Endothelial/pericyte interactions. *Circ Res.* 2005;97:512–523.
250. Kida Y, Duffield JS. Pivotal role of pericytes in kidney fibrosis. *Clin Exp Pharmacol Physiol.* 2011;38:467–473.
251. Schrimpf C, Xin C, Campanholle G, et al. Pericyte TIMP3 and ADAMTS1 modulate vascular stability after kidney injury. *J Am Soc Nephrol.* 2012;23:868–883.
252. Cai J, Kehoe O, Smith GM, et al. The angiopoietin/Tie-2 system regulates pericyte survival and recruitment in diabetic retinopathy. *Invest Ophthalmol Vis Sci.* 2008;49:2163–2171.
253. Woolf AS, Gnudi L, Long DA. Roles of angiopoietins in kidney development and disease. *J Am Soc Nephrol.* 2009;20:239–244.
254. Rabelink TJ, Wijewickrama DC, de Koning EJ. Peritubular endothelium: the Achilles heel of the kidney? *Kidney Int.* 2007;72:926–930.
255. Khairoun M, van der Pol P, de Vries DK, et al. Renal ischemia-reperfusion induces a dysbalance of angiopoietins, accompanied by proliferation of pericytes and fibrosis. *Am J Physiol Renal Physiol.* 2013;305:F901–F910.
256. Norman JT, Clark IM, Garcia PL. Hypoxia promotes fibrogenesis in human renal fibroblasts. *Kidney Int.* 2000;58:2351–2366.
257. Wang Y, Wang YP, Tay YC, et al. Progressive adriamycin nephropathy in mice: sequence of histologic and immunohistochemical events. *Kidney Int.* 2000;58:1797–1804.
258. Zheng G, Wang Y, Mahajan D, et al. The role of tubulointerstitial inflammation. *Kidney Int Suppl.* 2005;S96–S100.
259. Sica A, Mantovani A. Macrophage plasticity and polarization: in vivo veritas. *J Clin Invest.* 2012;122:787–795.
260. Alikhan MA, Jones CV, Williams TM, et al. Colony-stimulating factor-1 promotes kidney growth and repair via alteration of macrophage responses. *Am J Pathol.* 2011;179:1243–1256.
261. Zhang MZ, Yao B, Yang S, et al. CSF-1 signaling mediates recovery from acute kidney injury. *J Clin Invest.* 2012;122:4519–4532.
262. Nishida M, Fujinaka H, Matsusaka T, et al. Absence of angiotensin II type 1 receptor in bone marrow-derived cells is detrimental in the evolution of renal fibrosis. *J Clin Invest.* 2002;110:1859–1868.
263. Rastaldi MP, Ferrario F, Crippa A, et al. Glomerular monocyte-macrophage features in ANCA-positive renal vasculitis and cryoglobulinemic nephritis. *J Am Soc Nephrol.* 2000;11:2036–2043.
264. Duffield JS, Forbes SJ, Constandinou CM, et al. Selective depletion of macrophages reveals distinct, opposing roles during liver injury and repair. *J Clin Invest.* 2005;115:56–65.
265. Asano M, Toda M, Sakaguchi N, et al. Autoimmune disease as a consequence of developmental abnormality of a T cell subpopulation. *J Exp Med.* 1996;184:387–396.
266. Thornton AM, Piccirillo CA, Shevach EM. Activation requirements for the induction of CD4+CD25+ T cell suppressor function. *Eur J Immunol.* 2004;34:366–376.
267. Fontenot JD, Rasmussen JP, Williams LM, et al. Regulatory T cell lineage specification by the forkhead transcription factor foxp3. *Immunity.* 2005;22:329–341.

268. Wang YM, Zhang GY, Wang Y, et al. Foxp3-transduced polyclonal regulatory T cells protect against chronic renal injury from adriamycin. *J Am Soc Nephrol*. 2006;17:697–706.
269. Remuzzi G, Benigni A, Remuzzi A. Mechanisms of progression and regression of renal lesions of chronic nephropathies and diabetes. *J Clin Invest*. 2006;116:288–296.
270. Marshall CB, Shankland SJ. Cell cycle and glomerular disease: a minireview. *Nephron Exp Nephrol*. 2006;102:e39–e48.
271. Prodromidi EI, Poulosom R, Jeffery R, et al. Bone marrow-derived cells contribute to podocyte regeneration and amelioration of renal disease in a mouse model of Alport syndrome. *Stem Cells*. 2006;24:2448–2455.
272. Becker JU, Hoerning A, Schmid KW, et al. Immigrating progenitor cells contribute to human podocyte turnover. *Kidney Int*. 2007;72:1468–1473.
273. Appel D, Kershaw DB, Smeets B, et al. Recruitment of podocytes from glomerular parietal epithelial cells. *J Am Soc Nephrol*. 2009;20:333–343.
274. Ronconi E, Sagrinati C, Angelotti ML, et al. Regeneration of glomerular podocytes by human renal progenitors. *J Am Soc Nephrol*. 2009;20:322–332.
275. Macconi D, Sangalli F, Bonomelli M, et al. Podocyte repopulation contributes to regression of glomerular injury induced by ACE inhibition. *Am J Pathol*. 2009;174:797–807.
276. Zhang J, Pippin JW, Kroff RD, et al. Podocyte repopulation by renal progenitor cells following glucocorticoids treatment in experimental FSGS. *Am J Physiol Renal Physiol*. 2013;304:F1375–F1389.
277. Lasagni L, Ballerini L, Angelotti ML, et al. Notch activation differentially regulates renal progenitors proliferation and differentiation toward the podocyte lineage in glomerular disorders. *Stem Cells*. 2010;28:1674–1685.
278. Darisipudi MN, Kulkarni OP, Sayyed SG, et al. Dual blockade of the homeostatic chemokine CXCL12 and the proinflammatory chemokine CCL2 has additive protective effects on diabetic kidney disease. *Am J Pathol*. 2011;179:116–124.
279. Zhang J, Pippin JW, Vaughan MR, et al. Retinoids augment the expression of podocyte proteins by glomerular parietal epithelial cells in experimental glomerular disease. *Nephron Exp Nephrol*. 2012;121:e23–e37.
280. Scheiner G. The nephrotic syndrome. In: Strauss M, Welt L, eds. *Diseases of the Kidney*. 2nd ed. Boston: Little Brown; 1961:503–636.
281. Rothschild MA, Oratz M, Schreiber SS. Albumin synthesis. 1. *N Engl J Med*. 1972;286:748–757.
282. Kaysen GA. Albumin metabolism in the nephrotic syndrome: the effect of dietary protein intake. *Am J Kidney Dis*. 1988;12:461–480.
283. Kaysen GA, Gambertoglio J, Jimenez I, et al. Effect of dietary protein intake on albumin homeostasis in nephrotic patients. *Kidney Int*. 1986;29:572–577.
284. Rothschild MA, Oratz M, Evans CD, et al. Role hepatic interstitial albumin in regulating albumin synthesis. *Am J Physiol*. 1966;210:57–68.
285. Kaysen GA, Jones H Jr, Martin V, et al. A low-protein diet restricts albumin synthesis in nephrotic rats. *J Clin Invest*. 1989;83:1623–1629.
286. Landwehr DM, Carvalho JS, Oken DE. Micropuncture studies of the filtration and absorption of albumin by nephrotic rats. *Kidney Int*. 1977;11:9–17.
287. Park CH, Maack T. Albumin absorption and catabolism by isolated perfused proximal convoluted tubules of the rabbit. *J Clin Invest*. 1984;73:767–777.
288. Katz J, Bonorris G, Sellers AL. Albumin metabolism in amino-nucleoside nephrotic rats. *J Lab Clin Med*. 1963;62:910–934.
289. Kaysen GA, Kirkpatrick WG, Couser WG. Albumin homeostasis in the nephrotic rat: nutritional considerations. *Am J Physiol*. 1984;247:F192–F202.
290. Baylis C, Ichikawa I, Willis WT, et al. Dynamics of glomerular ultrafiltration. IX. Effects of plasma protein concentration. *Am J Physiol*. 1977;232:F58–F71.
291. Drummond MC, Kristal B, Myers BD, et al. Structural basis for reduced glomerular filtration capacity in nephrotic humans. *J Clin Invest*. 1994;94:1187–1195.
292. Chagnac A, Kiberd BA, Farinas MC, et al. Outcome of the acute glomerular injury in proliferative lupus nephritis. *J Clin Invest*. 1989;84:922–930.
293. Austin SM, Lieberman JS, Newton LD, et al. Slope of serial glomerular filtration rate and the progression of diabetic glomerular disease. *J Am Soc Nephrol*. 1993;3:1358–1370.
294. Meyer TW, Rennke HG. Increased single-nephron protein excretion after renal ablation in nephrotic rats. *Am J Physiol*. 1988;255:F1243–F1248.
295. Lowenstein J, Schacht RG, Baldwin DS. Renal failure in minimal change nephrotic syndrome. *Am J Med*. 1981;70:227–233.
296. Gugler R, Shoeman DW, Huffman DH, et al. Pharmacokinetics of drugs in patients with the nephrotic syndrome. *J Clin Invest*. 1975;55:1182–1189.
297. Aronoff GR, Abel SR. Principles of administering drugs to patients with renal failure. In: Bennett WM, McCarron DA, Morrison G, et al, eds. *Drug Prescribing in Renal Failure: Dosing Guidelines for Adults*. Philadelphia: American College of Physicians; 1987.
298. Bergrem H. Pharmacokinetics and protein binding of prednisolone in patients with nephrotic syndrome and patients undergoing hemodialysis. *Kidney Int*. 1983;23:876–881.
299. Inoue M, Okajima K, Itoh K, et al. Mechanism of furosemide resistance in analbuminemic rats and hypoalbuminemic patients. *Kidney Int*. 1987;32:198–203.
300. Rane A, Villeneuve JP, Stone WJ, et al. Plasma binding and disposition of furosemide in the nephrotic syndrome and in uremia. *Clin Pharmacol Ther*. 1978;24:199–207.
301. Kirchner KA, Voelker JR, Brater DC. Intratubular albumin blunts the response to furosemide—a mechanism for diuretic resistance in the nephrotic syndrome. *J Pharmacol Exp Ther*. 1990;252:1097–1101.
302. Agarwal R, Gorski JC, Sundblad K, et al. Urinary protein binding does not affect response to furosemide in patients with nephrotic syndrome. *J Am Soc Nephrol*. 2000;11:1100–1105.
303. Kaysen GA. Plasma composition in the nephrotic syndrome. *Am J Nephrol*. 1993;13:347–359.
304. Alon U, Chan JC. Calcium and vitamin D homeostasis in the nephrotic syndrome: current status. *Nephron*. 1984;36:1–4.
305. Purkerson ML, Tollefsen DM, Klahr S. N-desulfated/acylated heparin ameliorates the progression of renal disease in rats with subtotal renal ablation. *J Clin Invest*. 1988;81:69–74.
306. Sato KA, Gray RW, Lemann J Jr. Urinary excretion of 25-hydroxyvitamin D in health and the nephrotic syndrome. *J Lab Clin Med*. 1982;99:325–330.
307. Barragry JM, France MW, Carter ND, et al. Vitamin-D metabolism in nephrotic syndrome. *Lancet*. 1977;2:629–632.
308. Goldstein DA, Haldimann B, Sherman D, et al. Vitamin D metabolites and calcium metabolism in patients with nephrotic syndrome and normal renal function. *J Clin Endocrinol Metab*. 1981;52:116–121.
309. Malluche HH, Goldstein DA, Massry SG. Osteomalacia and hyperparathyroid bone disease in patients with nephrotic syndrome. *J Clin Invest*. 1979;63:494–500.
310. Korkor A, Schwartz J, Bergfeld M, et al. Absence of metabolic bone disease in adult patients with the nephrotic syndrome and normal renal function. *J Clin Endocrinol Metab*. 1983;56:496–500.
311. Schieppati A, Dodesini P, Benigni A, et al. The metabolism of arachidonic acid by platelets in nephrotic syndrome. *Kidney Int*. 1984;25:671–676.
312. Koomans HA, Kortlandt W, Geers AB, et al. Lowered protein content of tissue fluid in patients with the nephrotic syndrome: observations during disease and recovery. *Nephron*. 1985;40:391–395.
313. Lewis DM, Tooke JE, Beaman M, et al. Peripheral microvascular parameters in the nephrotic syndrome. *Kidney Int*. 1998;54:1261–1266.
314. Schnittler HJ. Structural and functional aspects of intercellular junctions in vascular endothelium. *Basic Res Cardiol*. 1998;93(suppl 3):30–39.
315. Bernard DB, Alexander EA. Edema formation in the nephrotic syndrome: pathophysiologic mechanisms. *Cardiovasc Med*. 1979;4:605–625.
316. Berlyne GM, Brown C, Adler A, et al. Water immersion in nephrotic syndrome. *Arch Intern Med*. 1981;141:1275–1278.
317. Garnett ES, Webber CE. Changes in blood-volume produced by treatment in the nephrotic syndrome. *Lancet*. 1967;2:798–799.
318. Meltzer JI, Keim HJ, Laragh JH, et al. Nephrotic syndrome: vasoconstriction and hypervolemic types indicated by renin-sodium profiling. *Ann Intern Med*. 1979;91:688–696.
319. Kelsch RC, Light GS, Oliver WJ. The effect of albumin infusion upon plasma norepinephrine concentration in nephrotic children. *J Lab Clin Med*. 1972;79:516–525.
320. Harris RC, Ismail N. Extrarenal complications of the nephrotic syndrome. *Am J Kidney Dis*. 1994;23:477–497.

321. Perico N, Remuzzi G. Edema of the nephrotic syndrome: the role of the atrial peptide system. *Am J Kidney Dis.* 1993;22:355–366.
322. Dorhout EJ, Roos JC, Boer P, et al. Observations on edema formation in the nephrotic syndrome in adults with minimal lesions. *Am J Med.* 1979;67:378–384.
323. Brown EA, Markandu ND, Sagnella GA, et al. Evidence that some mechanism other than the renin system causes sodium retention in nephrotic syndrome. *Lancet.* 1982;2:1237–1240.
324. Tulassay T, Rascher W, Lang RE, et al. Atrial natriuretic peptide and other vasoactive hormones in nephrotic syndrome. *Kidney Int.* 1987;31:1391–1395.
325. Krishna GG, Danovitch GM. Effects of water immersion on renal function in the nephrotic syndrome. *Kidney Int.* 1982;21:395–401.
326. Ichikawa I, Rennke HG, Hoyer JR, et al. Role for intrarenal mechanisms in the impaired salt excretion of experimental nephrotic syndrome. *J Clin Invest.* 1983;71:91–103.
327. Perico N, Delaini F, Lupini C, et al. Blunted excretory response to atrial natriuretic peptide in experimental nephrosis. *Kidney Int.* 1989;36:57–64.
328. Peterson C, Madsen B, Perlman A, et al. Atrial natriuretic peptide and the renal response to hypervolemia in nephrotic humans. *Kidney Int.* 1988;34:825–831.
329. Healy DP, Fanestil DD. Localization of atrial natriuretic peptide binding sites within the rat kidney. *Am J Physiol.* 1986;250:F573–F578.
330. Perico N, Delaini F, Lupini C, et al. Renal response to atrial peptides is reduced in experimental nephrosis. *Am J Physiol.* 1987;252:F654–F660.
331. DiBona GF, Herman PJ, Sawin LL. Neural control of renal function in edema-forming states. *Am J Physiol.* 1988;254:R1017–R1024.
332. Valentin JP, Qiu C, Ldowney WP, et al. Cellular basis for blunted volume expansion natriuresis in experimental nephrotic syndrome. *J Clin Invest.* 1992;90:1302–1312.
333. Yan W, Sheng N, Seto M, et al. Corin, a mosaic transmembrane serine protease encoded by a novel cDNA from human heart. *J Biol Chem.* 1999;274:14926–14935.
334. Chan JC, Knudson O, Wu F, et al. Hypertension in mice lacking the proatrial natriuretic peptide convertase corin. *Proc Natl Acad Sci U S A.* 2005;102:785–790.
335. Polzin D, Kaminski HJ, Kastner C, et al. Decreased renal corin expression contributes to sodium retention in proteinuric kidney diseases. *Kidney Int.* 2010;78:650–659.
336. Klein JD. Corin: an ANP protease that may regulate sodium reabsorption in nephrotic syndrome. *Kidney Int.* 2010;78:635–637.
337. Armaly Z, Assady S, Abassi Z. Corin: a new player in the regulation of salt-water balance and blood pressure. *Curr Opin Nephrol Hypertens.* 2013;22:713–722.
338. Fang C, Shen L, Dong L, et al. Reduced urinary corin levels in patients with chronic kidney disease. *Clin Sci.* 2013;124:709–717.
339. Andreoli TE. An overview of salt absorption by the nephron. *J Nephrol.* 1999;12(suppl 2):S3–S15.
340. Passero CJ, Mueller GM, Rondon-Berrios H, et al. Plasmin activates epithelial Na⁺ channels by cleaving the gamma subunit. *J Biol Chem.* 2008;283:36586–36591.
341. Svenningsen P, Bistrup C, Friis UG, et al. Plasmin in nephrotic urine activates the epithelial sodium channel. *J Am Soc Nephrol.* 2009;20:299–310.
342. Kim SW, Wang W, Nielsen J, et al. Increased expression and apical targeting of renal ENaC subunits in puromycin aminonucleoside-induced nephrotic syndrome in rats. *Am J Physiol Renal Physiol.* 2004;286:F922–F935.
343. Lourdel S, Loffing J, Favre G, et al. Hyperaldosteronemia and activation of the epithelial sodium channel are not required for sodium retention in puromycin-induced nephrosis. *J Am Soc Nephrol.* 2005;16:3642–3650.
344. de Seigneux S, Kim SW, Hemmingsen SC, et al. Increased expression but not targeting of ENaC in adrenalectomized rats with PAN-induced nephrotic syndrome. *Am J Physiol Renal Physiol.* 2006;291:F208–F217.
345. Besse-Eschmann V, Klisic J, Nief V, et al. Regulation of the proximal tubular sodium/proton exchanger NHE3 in rats with puromycin aminonucleoside (PAN)-induced nephrotic syndrome. *J Am Soc Nephrol.* 2002;13:2199–2206.
346. Klisic J, Zhang J, Nief V, et al. Albumin regulates the Na⁺/H⁺ exchanger 3 in OKP cells. *J Am Soc Nephrol.* 2003;14:3008–3016.
347. Rodriguez-Iturbe B, Herrera-Acosta J, Johnson RJ. Interstitial inflammation, sodium retention, and the pathogenesis of nephrotic edema: a unifying hypothesis. *Kidney Int.* 2002;62:1379–1384.
348. Gur A, Adefuin PY, Siegel NJ, et al. A study of the renal handling of water in lipid nephrosis. *Pediatr Res.* 1976;10:197–201.
349. Fernandez-Llama P, Andrews P, Ecelbarger CA, et al. Concentrating defect in experimental nephrotic syndrome: altered expression of aquaporins and thick ascending limb Na⁺ transporters. *Kidney Int.* 1998;54:170–179.
350. Fernandez-Llama P, Andrews P, Nielsen S, et al. Impaired aquaporin and urea transporter expression in rats with adriamycin-induced nephrotic syndrome. *Kidney Int.* 1998;53:1244–1253.
351. Kaysen GA, de Sain-van der Velden MG. New insights into lipid metabolism in the nephrotic syndrome. *Kidney Int Suppl.* 1999;71:S18–S21.
352. Thomas EM, Rosenblum AH, Lander HB, et al. Relationships between blood lipid and blood protein levels in the nephrotic syndrome. *AMA Am J Dis Child.* 1951;81:207–214.
353. Muls E, Rosseneu M, Daneels R, et al. Lipoprotein distribution and composition in the human nephrotic syndrome. *Atherosclerosis.* 1985;54:225–237.
354. Deighan CJ, Caslake MJ, McConnell M, et al. Patients with nephrotic-range proteinuria have apolipoprotein C and E deficient VLDL1. *Kidney Int.* 2000;58:1238–1246.
355. Bernard DB. Extrarenal complications of the nephrotic syndrome. *Kidney Int.* 1988;33:1184–1202.
356. Marsh JB. Lipoprotein metabolism in experimental nephrosis. *J Lipid Res.* 1984;25:1619–1623.
357. Kaysen GA. Hyperlipidemia of the nephrotic syndrome. *Kidney Int Suppl.* 1991;31:S8–S15.
358. Sun X, Jones H Jr, Joles JA, et al. Apolipoprotein gene expression in albuminemic rats and in rats with Heymann nephritis. *Am J Physiol.* 1992;262:F755–F761.
359. Kaysen GA, Pan XM, Couser WG, et al. Defective lipolysis persists in hearts of rats with Heymann nephritis in the absence of nephrotic plasma. *Am J Kidney Dis.* 1993;22:128–134.
360. Davies RW, Staprans I, Hutchison FN, et al. Proteinuria, not altered albumin metabolism, affects hyperlipidemia in the nephrotic rat. *J Clin Invest.* 1990;86:600–605.
361. Furukawa S, Hirano T, Mamo JC, et al. Catabolic defect of triglyceride is associated with abnormal very-low-density lipoprotein in experimental nephrosis. *Metabolism.* 1990;39:101–107.
362. Kekki M, Nikkila EA. Plasma triglyceride metabolism in the adult nephrotic syndrome. *Eur J Clin Invest.* 1971;1:345–351.
363. Vega GL, Grundy SM. Lovastatin therapy in nephrotic hyperlipidemia: effects on lipoprotein metabolism. *Kidney Int.* 1988;33:1160–1168.
364. Yamada M, Matsuda I. Lipoprotein lipase in clinical and experimental nephrosis. *Clin Chim Acta.* 1970;30:787–794.
365. Warwick GL, Packard CJ, Stewart JP, et al. Post-prandial lipoprotein metabolism in nephrotic syndrome. *Eur J Clin Invest.* 1992;22:813–820.
366. McKenzie IF, Nestel PJ. Studies on the turnover of triglyceride and esterified cholesterol in subjects with the nephrotic syndrome. *J Clin Invest.* 1968;47:1685–1695.
367. Joven J, Masana L, Villabona C, et al. Low density lipoprotein metabolism in rats with puromycin aminonucleoside-induced nephrotic syndrome. *Metabolism.* 1989;38:491–495.
368. Warwick GL, Caslake MJ, Boulton-Jones JM, et al. Low-density lipoprotein metabolism in the nephrotic syndrome. *Metabolism.* 1990;39:187–192.
369. Vaziri ND, Sato T, Liang K. Molecular mechanisms of altered cholesterol metabolism in rats with spontaneous focal glomerulosclerosis. *Kidney Int.* 2003;63:1756–1763.
370. Dullaart RP, Gansevoort RT, Dikkeschei BD, et al. Role of elevated lecithin: cholesterol acyltransferase and cholesteryl ester transfer protein activities in abnormal lipoproteins from proteinuric patients. *Kidney Int.* 1993;44:91–97.
371. Vaziri ND, Liang K, Parks JS. Acquired lecithin-cholesterol acyltransferase deficiency in nephrotic syndrome. *Am J Physiol Renal Physiol.* 2001;280:F823–F828.
372. Kashyap ML, Srivastava LS, Hynd BA, et al. Apolipoprotein CII and lipoprotein lipase in human nephrotic syndrome. *Atherosclerosis.* 1980;35:29–40.
373. Ordóñez JD, Hiatt RA, Killebrew EJ, et al. The increased risk of coronary heart disease associated with nephrotic syndrome. *Kidney Int.* 1993;44:638–642.

374. Berlyne GM, Mallick NP. Ischaemic heart-disease as a complication of nephrotic syndrome. *Lancet*. 1969;2:399–400.
375. Moorhead JF, Chan MK, El-Nahas M, et al. Lipid nephrotoxicity in chronic progressive glomerular and tubulo-interstitial disease. *Lancet*. 1982;2:1309–1311.
376. Kasiske BL, O'Donnell MP, Cleary MP, et al. Treatment of hyperlipidemia reduces glomerular injury in obese Zucker rats. *Kidney Int*. 1988;33:667–672.
377. Grond J, van Goor H, Erkelens DW, et al. Glomerular sclerotic lesions in the rat. Histochemical analysis of their macromolecular and cellular composition. *Virchows Arch B Cell Pathol Incl Mol Pathol*. 1986;51:521–534.
378. Don BR, Kaysen GA, Hutchison FN, et al. The effect of angiotensin-converting enzyme inhibition and dietary protein restriction in the treatment of proteinuria. *Am J Kidney Dis*. 1991;17:10–17.
379. Wanner C, Rader D, Bartens W, et al. Elevated plasma lipoprotein(a) in patients with the nephrotic syndrome. *Ann Intern Med*. 1993;119:263–269.
380. Kanfer A. Coagulation factors in nephrotic syndrome. *Am J Nephrol*. 1990;10(suppl 1):63–68.
381. Llach F. Hypercoagulability, renal vein thrombosis, and other thrombotic complications of nephrotic syndrome. *Kidney Int*. 1985;28:429–439.
382. Bellomo R, Atkins RC. Membranous nephropathy and thromboembolism: is prophylactic anticoagulation warranted? *Nephron*. 1993;63:249–254.
383. Takeda Y, Chen AY. Fibrinogen metabolism and distribution in patients with the nephrotic syndrome. *J Lab Clin Med*. 1967;70:678–685.
384. Zwavinga JJ, Koomans HA, Sixma JJ, et al. Thrombus formation and platelet-vessel wall interaction in the nephrotic syndrome under flow conditions. *J Clin Invest*. 1994;93:204–211.
385. Vaziri ND, Paule P, Toohey J, et al. Acquired deficiency and urinary excretion of antithrombin III in nephrotic syndrome. *Arch Intern Med*. 1984;144:1802–1803.
386. Kauffmann RH, Veltkamp JJ, Van Tilburg NH, et al. Acquired antithrombin III deficiency and thrombosis in the nephrotic syndrome. *Am J Med*. 1978;65:607–613.
387. Robert A, Olmer M, Sampol J, et al. Clinical correlation between hypercoagulability and thrombo-embolic phenomena. *Kidney Int*. 1987;31:830–835.
388. Cosio FG, Harker C, Bataud MA, et al. Plasma concentrations of the natural anticoagulants protein C and protein S in patients with proteinuria. *J Lab Clin Med*. 1985;106:218–222.
389. Ariens RA, Moia M, Rivolta E, et al. High levels of tissue factor pathway inhibitor in patients with nephrotic proteinuria. *Thromb Haemost*. 1999;82:1020–1023.
390. Vigano-D'Angelo S, D'Angelo A, Kaufman CE Jr, et al. Protein S deficiency occurs in the nephrotic syndrome. *Ann Intern Med*. 1987;107:42–47.
391. Thomson C, Forbes CD, Prentice CR, et al. Changes in blood coagulation and fibrinolysis in the nephrotic syndrome. *Q J Med*. 1974;43:399–407.
392. Hamano K, Iwano M, Akai Y, et al. Expression of glomerular plasminogen activator inhibitor type 1 in glomerulonephritis. *Am J Kidney Dis*. 2002;39:695–705.
393. Rabelink TJ, Zwavinga JJ, Koomans HA, et al. Thrombosis and hemostasis in renal disease. *Kidney Int*. 1994;46:287–296.
394. Tkaczyk M, Baj Z. Surface markers of platelet function in idiopathic nephrotic syndrome in children. *Pediatr Nephrol*. 2002;17:673–677.
395. Jackson CA, Greaves M, Patterson AD, et al. Relationship between platelet aggregation, thromboxane synthesis and albumin concentration in nephrotic syndrome. *Br J Haematol*. 1982;52:69–77.
396. Yashiro M, Muso E, Shio H, et al. Amelioration of hypercholesterolaemia by HMG-CoA reductase inhibitor (Pravastatin) improved platelet hyperaggregability in nephrotic patients. *Nephrol Dial Transplant*. 1994;9:1842–1843.
397. Balk EM, Lau J, Goudas LC, et al. Effects of statins on nonlipid serum markers associated with cardiovascular disease: a systematic review. *Ann Intern Med*. 2003;139:670–682.
398. Fahal IH, McClelland P, Hay CR, et al. Arterial thrombosis in the nephrotic syndrome. *Postgrad Med J*. 1994;70:905–909.
399. Tarry WC, Moser AJ, Makhoul RG. Peripheral arterial thrombosis in the nephrotic syndrome. *Surgery*. 1993;114:618–623.
400. Llach F, Papper S, Massry SG. The clinical spectrum of renal vein thrombosis: acute and chronic. *Am J Med*. 1980;69:819–827.
401. Velasquez Forero F, Garcia Prugue N, Ruiz Morales N. Idiopathic nephrotic syndrome of the adult with asymptomatic thrombosis of the renal vein. *Am J Nephrol*. 1988;8:457–462.
402. Singhal R, Brimble KS. Thromboembolic complications in the nephrotic syndrome: pathophysiology and clinical management. *Thromb Res*. 2006;118:397–407.
403. Rostoker G, Texier JP, Jeandel B, et al. Asymptomatic renal-vein thrombosis in adult nephrotic syndrome ultrasonography and urinary fibrin-fibrinogen products: a prospective study. *Eur J Med*. 1992;1:19–22.
404. Mehls O, Andrassy K, Koderisch J, et al. Hemostasis and thromboembolism in children with nephrotic syndrome: differences from adults. *J Pediatr*. 1987;110:862–867.
405. Cameron JS. Coagulation and thromboembolic complications in the nephrotic syndrome. *Adv Nephrol Necker Hosp*. 1984;13:75–114.
406. Farooq V, Hegarty J, Chandrasekar T, et al. Serious adverse incidents with the usage of low molecular weight heparins in patients with chronic kidney disease. *Am J Kidney Dis*. 2004;43:531–537.
407. McLean RH, Forsgren A, Bjorksten B, et al. Decreased serum factor B concentration associated with decreased opsonization of *Escherichia coli* in the idiopathic nephrotic syndrome. *Pediatr Res*. 1977;11:910–916.
408. Heslan JM, Lautie JP, Intrator L, et al. Impaired IgG synthesis in patients with the nephrotic syndrome. *Clin Nephrol*. 1982;18:144–147.
409. Giangiacomo J, Cleary TG, Cole BR, et al. Serum immunoglobulins in the nephrotic syndrome. A possible cause of minimal-change nephrotic syndrome. *N Engl J Med*. 1975;293:8–12.
410. Fodor P, Saitua MT, Rodriguez E, et al. T-cell dysfunction in minimal-change nephrotic syndrome of childhood. *Am J Dis Child*. 1982;136:713–717.
411. Chapman S, Taube D, Brown Z, et al. Impaired lymphocyte transformation in minimal change nephropathy in remission. *Clin Nephrol*. 1982;18:34–38.
412. Krensky AM, Ingelfinger JR, Grupe WE. Peritonitis in childhood nephrotic syndrome: 1970–1980. *Am J Dis Child*. 1982;136:732–736.
413. Viudes A, Peman J, Canton E, et al. Two cases of fungemia due to *Candida lusitanae* and a literature review. *Eur J Clin Microbiol Infect Dis*. 2002;21:294–299.
414. Spika JS, Halsey NA, Le CT, et al. Decline of vaccine-induced antipneumococcal antibody in children with nephrotic syndrome. *Am J Kidney Dis*. 1986;7:466–470.

BOARD REVIEW QUESTIONS

1. Charge-selectivity in the glomerular basement membrane (GBM) functions in which of the following manners?
 - a. Positive charges on the GBM repel positively charged circulating proteins
 - b. Positive charges on the GBM attract negatively charged circulating proteins
 - c. Negative charges on the GBM attract positively charged circulating proteins
 - d. Negative charges on the GBM repel negatively charged circulating proteins

Answer: d

Rationale: The majority of circulating proteins including albumin are negatively charged. Evidence from theoretical models and experiments in animals suggests that negatively charged molecules in the glomerular capillary wall (in the glycocalyx of endothelial cells, the negatively charged heparansulfate of the glomerular basement membrane, and the glycoproteins of the cell membrane of podocytes) form an important component of glomerular permselective function.

2. The components of the glomerular capillary membrane wall are
 - a. Endothelial cells and GBM
 - b. Endothelial cells and basement membrane and podocyte foot process
 - c. Endothelial cells and podocyte foot process
 - d. Endothelial cells and basement membrane and parietal cells
 - e. Endothelial cells and parietal cells and podocyte foot processes

Answer: b

Rationale: The glomerular capillary membrane is a complex structure composed of the three layers stated in option b that allow a large amount of water and small solute filtration while efficiently restricting glomerular passage of protein macromolecules within blood circulation.

3. Which of the following molecules are thought to be important in the absorption of filtered proteins by proximal tubule cells? (Select all that apply.)
 - a. Uromodulin
 - b. Megalin

- c. C3 complement
 - d. Clathrin
 - e. Nephlin

Answer: b and d

Rationale: Renal proximal tubule cells have a remarkable ability to reabsorb large quantities of albumin through clathrin- and megalin receptor-mediated endocytosis. Uromodulin (Tamm-Horsfall protein) is secreted by tubule cells and is thought to play a role in preventing urinary infections. C3 is a component of the complement system that is thought to play an important role in the progressive renal injury resulting from proteinuria. Nephlin is a key component of the glomerular slit diaphragm.

4. Locally produced angiotensin II contributes to glomerular injury by
 - a. Sustaining glomerular hypertension-induced damage
 - b. Impairing the glomerular barrier sieving function through inhibition of nephrin expression by podocytes
 - c. Activating parietal epithelial progenitor cells in the setting of glomerular hyperplastic lesions
 - d. All the above mechanisms

Answer: d

Rationale: Angiotensin II has been identified as a key mediator of multiple mechanisms that contribute to proteinuria and progressive kidney damage including all those listed.

5. The main systemic consequences of nephrotic-range proteinuria include
 - a. Alteration in the composition of the body protein pool—hypoalbuminemia
 - b. A state of sodium retention
 - c. Dyslipidemia
 - d. Abnormalities of coagulation factors
 - e. Susceptibility to infection
 - f. All the above abnormalities

Answer: f

Rationale: The nephrotic syndrome is a complex pathophysiologic state characterized by sodium and water retention as well as multiple abnormalities resulting from excessive urinary loss of albumin as well as other proteins that are components of the coagulation and humoral immune systems.